

Exterior Hardy bounds for finite rank BFSS Hamiltonians

Global induction and a singlet decay improvement

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Technical review draft. This document presents a proposed analytical proof for specialist checking. The global implication is stated with its exact local inputs, and twelve appendices contain the baseline derivations and the added rotation argument. Those inputs are substantive parts of the proposed BFSS proof; their inclusion is not a substitute for verifying them. No independent human review or proof-assistant verification is represented as having occurred.

AI disclosure. OpenAI AI systems made substantial contributions to research, derivation, code, manuscript preparation, and mathematical checking of this draft. The author name identifies the person under whose direction the work was prepared; this version does not assert that the named author has personally checked every argument or approved a public submission. AI-generated checking is not independent expert peer review.

Abstract

For the trace-normalized sixteen-supercharge $SU(N)$ matrix Hamiltonian, we assemble a candidate exterior quadratic-form proof at every fixed finite rank. The baseline bound has every coefficient $c < 49/4$ on arbitrary physical exterior vectors. This revision adds $c < 121/4$ on physical $\text{Spin}(9)$ singlets, using a bounded-radius internal nonsinglet uncertainty estimate and rotation-equivariant cluster localization. Internal continua are retained; no internal spectral gap or ground-state projection is required. Combined with the existing Hasler-Hoppe theorem that every physical L^2 zero mode is rotation invariant, the proposed full argument gives ordinary radial moments of every order $0 \leq m < 9$ for each actual zero mode. These conclusions depend on the complete baseline local and global analytical inputs, whose candidate derivations remain subject to specialist review. Existence, uniqueness, the endpoint $m = 9$, uniformity in N , and full soft or scattering theorems are not established. Polchinski already stated the below-nine expectation through a Born-Oppenheimer tail argument; the rotation identities and weighted method also have substantial prior art. No priority or independent human or formal proof verification is claimed.

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1 Scope and prior work

The configuration space has nine traceless Hermitian matrices and noncompact commuting valleys. Separated-channel asymptotics are informative, but an exterior form inequality must hold for every physical test vector, including states whose internal cluster coordinates have unrestricted energy and configurations with several simultaneous or nested clusters. A local expansion with all eigenvalue gaps comparable does not by itself give that estimate.

The proposed statement is deliberately finite-rank. For each fixed N , an exterior radius may be chosen after all rank-dependent constants: $c < 49/4$ for arbitrary physical vectors, and $c < 121/4$ for rotation singlets. There is no bound on how that radius grows with N . A conclusion valid at each finite N cannot be exchanged with a large- N limit without further estimates.

Sethi and Stern derive weighted L^2 estimates from exterior supercharge coercivity and analyze the excited-oscillator correction and inverse-square cancellation in the two-particle problem [1, manuscript pp. 20–21 and 37–39]. The present bounded-weight multiplier argument is an instance of that established Hardy-Agmon method [4]. Konechny computes a free leading effective Hamiltonian in an all-rank large-separation regime [5]. Hasler and Hoppe establish an asymptotic factorization in a region where the selected eigenvalue gaps are all of the same large order [6]. These are essential precursors, with hypotheses different from a collision-uniform exterior form bound on arbitrary states.

The $SU(2)$ wavefunction analysis of Fröhlich, Graf, Hasler, Hoppe and Yau is explicitly formulated for a formal asymptotic series [7]. Lin and Yin propose a recursive cluster form of the leading asymptotic wavefunction [8]. These results motivate the below-nine tail benchmark. The new singlet comparison addresses the angular part of an actual-state estimate; the baseline errors and domains must still be controlled in every channel.

Polchinski already explicitly states convergence of the fourth moment and convergence for orders $L < 9$, using a Hamiltonian Born-Oppenheimer argument and an $l = 2$ two-block tail [9, printed p. 12, Eqs. (7.8)–(7.10)]. The $L < 9$ expectation, the $l = 2$ angular mechanism, and the associated constants are therefore not claimed as new. The distinction under examination is a global all-channel theorem for actual L^2 zero modes. The references inspected here do not provide a priority certificate or establish the absence of a stronger theorem elsewhere.

Sethi and Stern proved rotational invariance results for supersymmetric bound states [2]. Hasler and Hoppe gave explicit rotation primitives and proved $\text{Spin}(d)$ invariance for $SU(N)$ in dimensions $d = 3, 5, 9$ [3]. We use their kernel-invariance theorem and corrected primitive, rather than presenting them as new. The quantitative support-radius estimate and its insertion into the proposed collision-uniform ledger are the candidate addition under review.

How to read the proof status

Theorem 8.1 proves the baseline all-vector implication from the realization and local inputs stated below. Theorem 11.1 gives the added singlet implication, and Corollary 12.1 applies it to actual zero modes. Appendices A–J supply candidate derivations of the baseline inputs; Appendices K–L detail the added rotation and model-domain argument. This

is a review packet containing the proposed full argument, not a claim that its difficult local estimates have been independently certified. The finite tests check specified algebraic identities and constants; they do not verify an infinite-dimensional operator inequality.

The most important review points are the actual nonzero-internal source inventory, the colored complete-form reserve, differentiated orbit remainders, the singular-stratum incidence localization, and form-domain compatibility. A reviewer can inspect them directly without relying on a private report or an acceptance label. The relevant derivations, including the corrected $\mathcal{Q}(H)$ interpretation and explicit module identities, are reproduced here.

2 Operator and domain conventions

Let T_A , $1 \leq A \leq N^2 - 1$, be Hermitian generators with

$$\text{Tr}(T_A T_B) = \delta_{AB}, \quad [T_A, T_B] = if_{ABC} T_C, \quad X_i = X_i^A T_A \quad (1 \leq i \leq 9). \quad (2.1)$$

The Euclidean metric is $\sum_i \text{Tr}(\text{d}X_i^2)$, and

$$\rho_N^2 = \sum_i \text{Tr}(X_i^2) = \sum_{i,A} (X_i^A)^2. \quad (2.2)$$

Take Hermitian Majorana generators θ_α^A with $\{\theta_\alpha^A, \theta_\beta^B\} = \delta_{AB} \delta_{\alpha\beta}$, on a complex Clifford module $\mathcal{F}_N$ of dimension $2^{8(N^2-1)}$. The matrices Γ_i are real symmetric 16×16 matrices satisfying $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}$; set $\Gamma_{ij} = \frac{1}{2}[\Gamma_i, \Gamma_j]$.

For a fixed positive length ℓ , define

$$C_{Aij} = f_{ABC} X_i^B X_j^C, \quad (2.3)$$

$$(\text{ad } X_i)_{AB} = \text{Tr}(T_A [X_i, T_B]) = -if_{CAB} X_i^C, \quad (2.4)$$

$$D_{\text{ad}}(X) = \sum_i \Gamma_i \otimes \text{ad } X_i. \quad (2.5)$$

With the displayed ordering of spin and color indices, the kinetic-one expression is

$$h_N = -\Delta_X + \ell^{-6} \sum_{i < j, A} C_{Aij}^2 + \ell^{-3} \theta^T D_{\text{ad}}(X) \theta. \quad (2.6)$$

The Yukawa term is therefore $-i\ell^{-3} \sum f_{CAB} X_i^C \theta_\alpha^A (\Gamma_i)_{\alpha\beta} \theta_\beta^B$. This convention fixes the sign by defining $\text{ad } X$ explicitly. The matrix D_{ad} is Hermitian and purely imaginary antisymmetric in a Hermitian trace-orthonormal color basis.

The corresponding normalized real charges are

$$q_\alpha = \sqrt{2} \sum_{A,\beta} \left(\sum_i (\Gamma_i)_{\alpha\beta} p_i^A - \ell^{-3} \sum_{i < j} (\Gamma_{ij})_{\alpha\beta} C_{Aij} \right) \theta_\beta^A, \quad p_i^A = -i\partial_{X_i^A}. \quad (2.7)$$

Their spinor-averaged square gives (2.6) on the common smooth core. Individual squares have gauge terms off the physical sector. On physical vectors the usual gauge terms

vanish; on colored internal modules only the averaged identity is used. The physical Hamiltonian and charges in units with scalar kinetic coefficient $R_{\text{BFSS}}/2$ are

$$H_N = \frac{R_{\text{BFSS}}}{2} h_N, \quad Q_\alpha = \sqrt{R_{\text{BFSS}}/2} q_\alpha. \quad (2.8)$$

The rest of the proof uses $\ell = 1$, obtainable by a fixed dilation. Restoring ℓ changes the exterior radius, not the dimensionless Hardy coefficient in kinetic-one units.

Hypothesis 2.1 (Supersymmetric realization). Use the gauge-invariant subspace $\mathcal{H}_N = L^2(\mathbb{R}^{9(N^2-1)}; \mathcal{F}_N)^{SU(N)}$. On $\mathcal{C}_N = C_c^\infty(\mathbb{R}^{9(N^2-1)}; \mathcal{F}_N)^{SU(N)}$ take the supersymmetric form

$$\mathfrak{h}_N[u] = \frac{1}{16} \sum_{\alpha=1}^{16} \|q_\alpha u\|^2. \quad (2.9)$$

Assume its compatible closed supercharge-stack realization and the bounded Lipschitz, gauge-invariant scalar multiplier rule. The operator h_N is the nonnegative self-adjoint operator associated to this closed form. On unrestricted internal-color spaces use the corresponding closed averaged form. The differential expression and averaged identity agree on their common core.

This is the precise realization used throughout. It is not a Dirichlet realization in a fixed valley. The exterior domain $\mathcal{Q}_{N,R}$ is the closure of physical compactly supported smooth vectors in $\{\rho_N > R\}$ in the norm $(\|u\|^2 + \mathfrak{h}_N[u])^{1/2}$. The notation $\mathcal{Q}(h_N)$ means the quadratic-form domain; it is distinct from the operator domain $D(h_N)$. No moving projection is claimed to preserve $D(h_N)$. The standard physical realization carries the natural unitary Spin(9) action, which preserves its closed form and physical core. The kernel-invariance input below refers to this compatible realization, not an arbitrary self-adjoint extension. For real scalar multipliers with $\sum_j \chi_j^2 = 1$, the kinetic-one product rule gives the IMS identity

$$\mathfrak{h}_N[u] = \sum_j \mathfrak{h}_N[\chi_j u] - \int \sum_j |\nabla \chi_j|^2 |u|^2. \quad (2.10)$$

If $\mathfrak{h}_N[\Psi] = 0$, every closed charge annihilates Ψ , and for bounded real Lipschitz w ,

$$\mathfrak{h}_N[w\Psi] = \int |\nabla w|^2 |\Psi|^2. \quad (2.11)$$

The Clifford normalization in (2.7) makes the coefficient in both identities exactly one.

3 The local inputs and their proof dependencies

Fix a proper partition $\pi = (n_1, \dots, n_k)$ of N , $k \geq 2$. In an intrinsic stationary-flag branch write

$$X_i = Z_i + A_i + Y_i, \quad Z_i = \text{diag}(z_{1i} I_{n_1}, \dots, z_{ki} I_{n_k}), \quad \sum_a n_a z_a = 0. \quad (3.1)$$

The $A_i^{(a)}$ are traceless internal matrices, and Y is off-block with the intrinsic normal constraint $B_Z^* Y = 0$, where $B_Q \xi = (i[\xi, Q_i])_i$. Put

$$s^2 = \sum_a n_a |z_a|^2, \quad \alpha_a^2 = \sum_i \text{Tr}((A_i^{(a)})^2), \quad r_{ab} = |z_a - z_b|, \quad r_* = \min_{a < b} r_{ab}, \quad W_p = \sum_{a < b} r_{ab}^{-p}. \quad (3.2)$$

A local tube has $\alpha_a + \alpha_b \leq \kappa r_{ab}$ and $|Y| < \sigma r_*$. The first condition is pairwise relative; it does not bound an unrelated large block by the smallest gap elsewhere. All local constants below may depend on fixed N , but not on $r_{\max}/r_*$.

L1 Exact geometry and physical projection

The intrinsic slice has internal kinetic coefficient exactly one, exact physical density $J_{\text{phys}} = \det M$, and exact same-branch unitary transitions. Flattening by $J_{\text{phys}}^{1/2}$ includes its differentiated scalar term. A normalized even cutoff vacuum U_σ defines

$$u = U_\sigma f + v, \quad U_\sigma^* v = 0. \quad (3.3)$$

It preserves the branch form domain with padded slow and fast supports. For every slow scalar multiplier $m \geq 0$,

$$\int m |u|^2 = \int m (|f|^2 + \|v\|_{\text{fiber}}^2). \quad (3.4)$$

At $A = 0$, eight negative spin modes are occupied for each bifundamental color. The determinant character restricts trivially to each internal $SU(n_a)$; the normal-ordering off-set gives zero block-center charge, and the positive Gaussian is invariant. Equivariant internal-radial transport preserves this residual identification. Thus the low coefficient is physical for every internal $SU(n_a)$ factor. The complement need only obey total residual equivariance and can be colored internally. Appendix A derives the metric, density, projector and transition identities; Appendix I treats closed forms and the retained Clifford module. Spatial equivariance, the anchor connection, and the two-block reference model used for the singlet refinement are checked separately in Section 10 and Appendix L.

L2 Source and fast energy control

Let $H_{\text{exc}} = \sum_e (H_{\text{fast},e} - E_{0,e})$, with the frozen ground energy subtracted pair by pair. On the cutoff complement,

$$H_{\text{exc}}[v] \geq g r_* \|v\|_{\text{fiber}}^2 \quad (3.5)$$

for a fixed $g > 0$ beyond a sufficiently large radius. This is a fast-fiber inequality at each slow coordinate, integrated over slow coordinates when used as a full form. Set $H_s = T_C + H_I$, $H_I = \sum_a h_{n_a}$, omitting singleton internal factors. T_C keeps the normal connection on full fibers. On low coefficients it uses the internal-radial identification with the $A = 0$ line.

The actual leading source is $S = H_I U$, and the actual diagonal multiplier W obeys

$$\|W - S^* H_{\text{exc}}^{-1} S\| \leq C_N \kappa W_2. \quad (3.6)$$

Only a shifted fast excitation inverse occurs here. It is not an inverse of H_I or of the full interacting Hamiltonian. The frozen energy is paid separately:

$$|E_0[u]| \leq C_N R_*^{-3} H_I[u] + C_N \kappa W_2[u]. \quad (3.7)$$

Appendix B proves a commutator-square bound for the kinetic-aware frozen frequencies. Appendix C gives the centered cancellation, its first-jet condition, graph excitation denominators, and complete slow-form estimates. Appendix D computes the actual adapted H_1, H_2 coefficients and their $O(\kappa W_2)$ comparison.

At $A = 0$, one pair source has excitation energy $4r_e$, while a triangle source has energy $2(r_e + r_f + r_g)$. The summed triangle denominator is essential when an opposite pair is much closer than a distant block. Freezing A never permits discarding internal derivatives of the moving vacuum.

L3 Colored local reserve and mixed ledger

There is a genuine postcompletion nonnegative square $J[v]$, and $B[v] = H_s[v] + H_{\text{exc}}[v] + J[v]$ is nonnegative. For any prescribed small local losses, the actual mixed ledger is

$$\begin{aligned} \mathfrak{h}_N[U_\sigma f + v] &\geq \mathfrak{h}_N[U_\sigma f] + c_s H_s[v] + c_J J[v] + (1 - d - \eta) H_{\text{exc}}[v] + 2 \operatorname{Re} \langle Sf, v \rangle \\ &\quad - a_\eta H_s[f] - b_\eta W_2[f], \end{aligned} \quad (3.8)$$

where $c_s, c_J > 0$, and

$$a_\eta \leq C_\eta(\sigma^2 + R_*^{-3}), \quad (3.9)$$

$$b_\eta \leq C_\eta(\kappa + \kappa^2 + \sigma^2 + R_*^{-3}). \quad (3.10)$$

The diagonal part is $H_s^{\text{Berry}} + W + E_0$ plus errors bounded by

$$\theta H_s + C_\theta(\kappa + \sigma + \sigma^2 + R_*^{-3}) W_2 + C_N R_*^{-3} H_s. \quad (3.11)$$

The Berry comparison costs

$$|H_s^{\text{Berry}} - H_s| \leq \theta H_s + C_N(\theta^{-1} \kappa^4 + \kappa) W_2. \quad (3.12)$$

Normalized cutoff errors decay as a polynomial times $\exp(-c\sigma_0^2 R_*^3)$ with rootwise weights. Appendices E-I give the connection estimates, nonflat Berry comparison, averaged colored-charge identity, complementary reserve, mixed errors, cutoff bounds and form-domain argument. The square J is the actual retained postcompletion square, not a second copy of the original Gauss square.

L4 Finite incidence localization

Appendix J supplies a finite-sheet, gauge-invariant localization near the commuting cone. With fixed tree endpoint κ_{tree} , logarithmic width L , inner ratio $\delta = \kappa_{\text{tree}} e^{-L}$, and a positive resulting gap Δ_N , its leaf supports satisfy

$$\alpha_a + \alpha_b \leq \beta r_{ab}, \quad r_* \geq \Delta_N s, \quad s \leq \rho_N \leq D_N s. \quad (3.13)$$

The normalized incidence weights w_j obey $\sum_j w_j^2 = 1$. For any desired $e > 0$, a later sufficiently thin near-cone neighborhood has

$$E := \sum_j |\nabla_X w_j|^2 \leq \sum_j w_j^2 E_j + e \rho_N^{-2}. \quad (3.14)$$

This is a summed inequality, not a leafwise normalized identity. The path cost has a slow-only majorant

$$E_j \leq M_\pi = \gamma \left[\sum_{n_a > 1} \alpha_a^{-2} \mathbf{1}_{\{\alpha_a \geq T(s)\}} + \sum_I s_I^{-2} \right], \quad (3.15)$$

$$\gamma = C_N L^{-2}, \quad T(s) = d_N \delta \Delta_N s, \quad M_\pi \leq D_{\text{path}} s^{-2}. \quad (3.16)$$

The second sum runs over proper unions of at least two final blocks; s_I is their relative center radius. The constant in γ is independent of L, δ . The possibly large D_{path} is used only on complementary vectors and is paid at a later exterior radius.

L5 Scalar outer bounds

For $\mathcal{B}(X) = \sum_{i < j} \| [X_i, X_j] \|_F^2$ and $t = \mathcal{B}/\rho_N^4$, an outer cutoff has gradient cost bounded by $C\mathcal{B}/\rho_N^6$. On every padded proper branch,

$$\mathcal{B} \leq C(V_I + s^2|Y|^2), \quad V_I \leq H_I + C_N \sum_a \alpha_a, \quad (3.17)$$

as weighted forms, including on colored internal modules. Positive oscillator estimates give

$$\|Yg\|^2 \leq C_N \sum_e r_e^{-2} H_{\text{exc},e}[g] + C_N \sum_e r_e^{-1} \|g\|^2. \quad (3.18)$$

These elementary potential and oscillator bounds are used explicitly below. They require the full retained oscillator and colored internal forms, not merely their norm gaps.

4 Recovering the all vector local reserve

Lemma 4.1. *Under L1-L3, for every $\varepsilon > 0$ one can choose fixed positive tube widths and an exterior threshold such that*

$$\mathfrak{h}_N[U_\sigma f + v] \geq (1 - \varepsilon)H_s[f] - \varepsilon W_2[f] + bB[v] \quad (4.1)$$

for some $b > 0$, uniformly in unrelated gap ratios.

Proof. Choose $b < c_s, c_J$ and $d + \eta + b < 1/2$. Reserve $bH_{\text{exc}}[v]$ before completing the remaining source square. Since S is orthogonal to the uncut fast kernel,

$$(1 - d - \eta - b)H_{\text{exc}}[v] + 2 \operatorname{Re} \langle Sf, v \rangle \geq - \frac{\langle f, S^* H_{\text{exc}}^{-1} Sf \rangle}{1 - d - \eta - b}. \quad (4.2)$$

The virtual multiplier is at most $C_N W_2$, so replacing its coefficient by one costs at most $2C_N(d + \eta + b)W_2$. Combine this with (3.6), (3.7), and the diagonal, Berry and mixed errors in L3. First assign small θ, d, η, b , then choose κ , then fixed σ and a smaller cutoff width σ_0 , and finally enlarge R_* . Every loss fits $\varepsilon(H_s + W_2)$.

The complement was defined using U_σ , while source completion uses the uncut kernel. This does not invalidate completion: S lies in the uncut complementary range, and the nonnegative quadratic form can be completed there for any v . The cutoff-complement gap is obtained from the exponentially small angle between the cut and uncut vacuum lines. The result retains all of $bB[v]$. $\square$

For fixed N there are finitely many partition types, so take finite maxima of pre-parameter constants and minima of admitted widths and reserves. Their uniformity in unrelated gap ratios is required before any logarithmic hierarchy is chosen. Merely knowing a local Schur-infimum inequality would not retain the complementary mass needed globally.

5 Shape independent Hardy payments

Set $h_0 = 49/4$. In the mass metric $\sum_a n_a |\mathbf{d}z_a|^2$, the center dimension is $D_\pi = 9(k-1)$. Radial integration by parts, valid for Hilbert-valued sections with a metric connection, yields

$$T_C[f] \geq h_\pi \int s^{-2} |f|^2, \quad h_\pi = \frac{(D_\pi - 2)^2}{4} \geq h_0. \quad (5.1)$$

For compact Dirichlet sections there is no boundary term; form closure extends the inequality. One can prove the scalar constant by expanding $\int |\partial_r f + (D_\pi - 2)f/(2r)|^2 r^{D_\pi-1} \mathbf{d}r \geq 0$. Metric compatibility gives the same real cross term for a unitary connection.

The linear map $Z \mapsto z_a - z_b$ has squared norm $\nu_{ab} = 1/n_a + 1/n_b$. Orthogonal relative-coordinate Hardy and $T_{ab} \leq T_C$ give

$$\int r_{ab}^{-2} |f|^2 \leq (h_0 \nu_{ab})^{-1} T_C[f], \quad W_2[f] \leq K_\pi T_C[f], \quad K_\pi = h_0^{-1} \sum_{a < b} \nu_{ab}^{-1}. \quad (5.2)$$

Thus $K_N = \max_\pi K_\pi$ is finite and contains no inverse shape gap. For any proper union I of at least two blocks, the orthogonal relative-center subspace has dimension $9(|I| - 1)$, whence

$$\int s_I^{-2} |f|^2 \leq h_0^{-1} T_C[f]. \quad (5.3)$$

All other coordinates and fermionic variables remain spectators.

Lemma 5.1 (Internal soft Hardy). *Suppose a lower-rank physical form has the exterior coefficient 9. Then there are $A_n < \infty, T_n$ such that for $T \geq T_n$,*

$$\mathfrak{h}_n[g] \geq a_0 \int_{\alpha \geq T} \alpha^{-2} |g|^2 - A_n T^{-4} \|g\|^2, \quad a_0 = 5/2. \quad (5.4)$$

Proof. Start with $w_T^0(\alpha) = \alpha^2(T^4 + \alpha^4)^{-1/2}$, and multiply by a fixed exterior cutoff β that equals one beyond its annulus. Put $w_T = \beta w_T^0$ and $v_T = (1 - w_T^2)^{1/2}$. Exact IMS, nonnegativity of $\mathfrak{h}_n[v_T g]$, and exterior Hardy for $w_T g$ imply

$$\mathfrak{h}_n[g] \geq (9 - 4) \int \alpha^{-2} w_T^2 |g|^2 - O(T^{-4}) \|g\|^2. \quad (5.5)$$

Away from the fixed annulus, $|\nabla w_T|^2 + |\nabla v_T|^2 \leq 4\alpha^{-2} w_T^2$; the annular derivatives are $O(T^{-4})$. For large T , $w_T^2 \geq 1/2$ on $\alpha \geq T$, proving the claim. It remains valid through $\alpha = 0$ and does not require an internal gap. $\square$

The cost $\gamma \alpha^{-2} \mathbf{1}_{\alpha \geq T}$ is therefore paid by $(\gamma/a_0) \mathfrak{h}_n + (\gamma A_n/a_0) T^{-4}$. Apply this only to physical low coefficients, never to a colored complement.

6 Incidence localization and exact global IMS

For clarity, the geometry behind L4 is recorded here; its full derivation is Appendix J. A commuting configuration has cluster scales a_B and regularized separation scales

$$d_B^{-2} = \sum_{j \notin B} (|\lambda_j - c_B|^2 + a_B^2)^{-1}. \quad (6.1)$$

Use select/reject angles $\vartheta_B = \Theta(\log(a_B/(\delta d_B))/L)$ and their cosine/sine factors. Thin selected clusters are laminar. A final partition has the symmetric leaf weight given by the product of selected final-block cosines and rejected proper-union sines. Crossing omitted nodes have their sine identically one on the nonzero leaf support. Its derivative cost is $C_N L^{-2} a_B^{-2}$ on transition regions.

The decision tree orders proper subsets by decreasing cardinality and skips a subset if a previously selected block overlaps it; otherwise it splits into select and reject children. The elementary pointwise identity

$$|\nabla(a \cos \vartheta)|^2 + |\nabla(a \sin \vartheta)|^2 = |\nabla a|^2 + a^2 |\nabla \vartheta|^2 \quad (6.2)$$

shows recursively that the leaf square sum is one and that its total gradient cost is the leaf-weighted sum of visited angle costs. Laminarity identifies each effective visited node as either a final block or a proper union of final blocks; a crossing node has zero transition derivative on the nonzero leaf support. This proves the pathwise accounting used by the symmetric leaf formula.

Off the commuting cone, keep every admitted ordered stationary flag, rather than asserting a globally unique projector. For a sheet j of k blocks let

$$b_j = (k!)^{-1/2} \chi_Y q_\pi, \quad Q = \sum_j b_j^2, \quad w_j = b_j Q^{-1/2}. \quad (6.3)$$

The factor $k!$ accounts for ordering; there is no additional $N!$ factor. Padded supports, a nondegenerate flag Hessian, and a finite rank-only multiplicity bound give smooth sums even with monodromy. On the whole commuting cone the identities $Q = 1$, $dQ = 0$, and $E = \sum_j w_j^2 E_j$ hold for ambient derivatives, including singular strata. Finite-sheet continuity after fixing the hierarchy yields (3.14) in a sufficiently thin neighborhood.

One quantitative support choice is

$$\beta = \frac{2\kappa_{\text{tree}}}{\sqrt{1 - 2\kappa_{\text{tree}}^2}}, \quad M_N = \frac{N\sqrt{1 + \beta^2} + 1}{1 - \beta}, \quad \Delta_N = \frac{(\delta/M_N)^{N-2}}{\sqrt{N(1 + \beta^2)}}. \quad (6.4)$$

Padding decreases this positive gap if necessary. Since $\rho_N^2 = s^2 + \sum_a \alpha_a^2 + |Y|^2$ and $r_{ab} \leq \sqrt{2}s$, one can use $D_N^2 = 1 + 2N\beta^2 + 2\sigma^2$ before the harmless enlargement for padded supports.

Choose $0 < t_0 < t_1$ so that the near-cone cutoff lies inside that neighborhood and all contributing fast support cutoffs equal one. Write

$$\chi = \cos \vartheta_{\text{out}}(t), \quad \chi_O = \sin \vartheta_{\text{out}}(t), \quad u_j = \chi w_j u, \quad u_O = \chi_O u, \quad E_{\text{out}} = |\nabla \vartheta_{\text{out}}(t)|^2. \quad (6.5)$$

The exact scalar IMS identity and L4 give

$$\begin{aligned} \mathfrak{h}_N[u] &\geq \sum_j \left\{ \mathfrak{h}_N[u_j] - \int (E_{\text{out}} + E_j + e\rho_N^{-2}) |u_j|^2 \right\} \\ &\quad + \mathfrak{h}_N[u_O] - \int E_{\text{out}} |u_O|^2. \end{aligned} \quad (6.6)$$

Every error is assigned on both sides of each split. No agreement between projections on different sheets is required.

After flattening, project each u_j as in (3.3). The normalized weights and χ can depend on Y . The projected coefficient has only the leaf's slow support; $U_\sigma f$ may leave the original

near-cone set, but stays in its padded branch. This is why all majorants and local estimates are required on the entire padded fast fiber. Slow support has strict margins, so zero extension to canonical center variables is legitimate for the Hardy inequalities. The projection bound is in the form norm only.

7 Paying the path and outer errors

Assume the theorem at all lower ranks. Fix a small $a > 0$. Equations (5.4) and (5.3) applied to (3.15) give, after choosing L large enough,

$$M_\pi[f] \leq aH_I[f] + aT_C[f] + C_{\text{soft}} \int s^{-4} |f|^2. \quad (7.1)$$

For example $\gamma \leq aa_0$ and $\gamma 2^N/h_0 \leq a$ suffice. The soft constant contains the now fixed inverse powers of $d_N \delta \Delta_N$ and lower-rank constants; its large size affects only a later radius. Enforce $T(s) \geq \max_{n < N} T_n$ at that radius.

For v , use the other part of L4:

$$M_\pi[v] \leq D_{\text{path}} \int s^{-2} \|v\|_{\text{fiber}}^2. \quad (7.2)$$

This is absorbed by any assigned fraction of $H_{\text{exc}}[v]$ once s is large, since $r_* \geq \Delta_N s$. For both low and complementary pieces, $e\rho_N^{-2} \leq es^{-2}$; the latter is a slow multiplier, so (3.4) applies exactly.

Differentiating the quartic polynomial gives

$$|\nabla \mathcal{B}| \leq C_N \rho_N \sqrt{\mathcal{B}}, \quad |\nabla t| \leq C_N \sqrt{\mathcal{B}}/\rho_N^3, \quad E_{\text{out}} \leq C_{\text{out}} \mathcal{B}/\rho_N^6. \quad (7.3)$$

On each branch, L5 bounds this by $C'_{\text{out}}(s^{-6}V_I + s^{-4}|Y|^2)$ on the full padded support. Because the s^{-6} weight commutes with internal differentiation and $\sum \alpha_a \leq C_N s$, the internal contribution costs

$$C'_{\text{out}} s^{-6} H_I + C''_{\text{out}} s^{-5} \quad (7.4)$$

as fiberwise integrated forms, for low or colored complementary inputs. The oscillator estimate (3.18) yields

$$\int s^{-4} |Yv|^2 \leq C_D \int s^{-6} H_{\text{exc}}[v \text{ at fixed slow variables}] + C_D \int s^{-5} \|v\|_{\text{fiber}}^2, \quad (7.5)$$

$$\int s^{-4} |YU_\sigma f|^2 \leq C_D \int s^{-5} |f|^2. \quad (7.6)$$

The V_I multiplier has zero low/complement cross term. The positive $|Y|^2$ cross is bounded by the two diagonal pieces, with a factor two. At sufficiently large radius these estimates imply

$$\int E_{\text{out}} |u_j|^2 \leq aH_I[f] + C_{\text{outer}} \int s^{-5} |f|^2 + \frac{b}{4} B[v]. \quad (7.7)$$

A norm gap alone would not control the unbounded $|Y|^2$ multiplier; the retained full excitation form is indispensable.

On the outside branch $t \geq t_0$, the bosonic potential is at least $v_N t_0 \rho_N^4$, whereas the fermionic norm is at most $F_N \rho_N$. The outer gradient cost is lower order. Hence, beyond a larger radius,

$$\mathfrak{h}_N[u_O] - \int E_{\text{out}} |u_O|^2 \geq c \int \rho_N^{-2} |u_O|^2. \quad (7.8)$$

There is no branch-weight cost on u_O in (6.6).

8 Baseline all vector theorem and rank induction

Theorem 8.1 (Baseline all-vector global implication). *Assume Hypothesis 2.1 and L1–L5 for every proper partition at each fixed finite rank, with the parameter uniformities specified above. Then for every integer $N \geq 2$ and every $0 < c < 49/4$, some finite $R_{N,c}$ satisfies*

$$\mathfrak{h}_N[u] \geq c \int \rho_N^{-2} |u|^2, \quad u \in \mathcal{Q}_{N,R_{N,c}}. \quad (8.1)$$

In physical units the right side is multiplied by $R_{\text{BFSS}}/2$.

Proof. Fix N, c and take

$$0 < a < \min \left\{ \frac{1}{12}, \frac{h_0 - c}{8(h_0 + 1)} \right\}, \quad \varepsilon(1 + K_N) \leq a. \quad (8.2)$$

Lemma 4.1 and (5.2) give

$$\mathfrak{h}_N[u_j] \geq (1 - a)(H_I[f] + T_C[f]) + bB[v]. \quad (8.3)$$

Choose $e \leq a$ and L for (7.1). Enlarge the final radius until $C_{\text{soft}} s^{-4} + C_{\text{outer}} s^{-5} \leq a s^{-2}$, the complementary path and defect costs each use at most $(b/8)H_{\text{exc}}[v]$, and (7.7) holds. Subtracting all assigned errors in (6.6) gives the single ledger

$$\begin{aligned} \mathfrak{h}_N[u_j] - \int (E_{\text{out}} + E_j + e \rho_N^{-2}) |u_j|^2 \\ \geq (1 - 3a)H_I[f] + (1 - 2a)T_C[f] - 2a \int s^{-2} |f|^2 + \frac{b}{2} B[v]. \end{aligned} \quad (8.4)$$

The complementary payments are exactly $b/4 + b/8 + b/8 = b/2$. There is no second Gauss reserve. Positivity of H_I and (5.1) give low coefficient $[(1 - 2a)h_0 - 2a] > c$. Increase the radius so that $(b/2)g\Delta_N s \geq c s^{-2}$. Then the last term controls $c \int s^{-2} \|v\|_{\text{fiber}}^2$. Thus the whole right side of (8.4) is at least

$$c \int s^{-2} (|f|^2 + \|v\|_{\text{fiber}}^2) = c \int s^{-2} |u_j|^2 \geq c \int \rho_N^{-2} |u_j|^2. \quad (8.5)$$

This exact slow-multiplier orthogonality prevents replacing full localized mass by low mass. Sum over leaves and use (7.8), (6.6), and the square-sum identity. Gauge averaging, compact exhaustion and exterior form closure extend the result from the physical core to the stated domain.

At $N = 2$ the only partition is $1 + 1$; there are no internal forms or selected-block costs. The same proof is a base case with center dimension nine and no inductive input. If all ranks below N are proved, each nontrivial block of a proper partition has smaller rank, giving exactly Lemma 5.1. This is ordinary finite-rank induction. $\square$

The fully split leaf has center Hardy coefficient $[9(N-1)-2]^2/4$, but the common bound is limited by possible two-block channels. No sharpness theorem is inferred from this bottleneck. Simultaneous $2+2$, $2+1+1$, and $2+2+1$ clusters are covered by the symmetric leaf construction; no relative ordering of disjoint internal sizes is imposed.

Acyclic parameter order

The proof requires the following order, rather than mutually dependent smallness choices.

1. Fix N, c, a , lower-rank soft constants, and the rank-only K_N .
2. Fix $\varepsilon \leq a/(1+K_N)$ and the local Young allocations, completion loss, and retained b .
3. Choose the local endpoint κ , a smaller tree endpoint, then fixed fast width σ and vacuum cutoff width σ_0 .
4. Choose L for the L^{-2} payment, then δ , the positive Δ_N , padded supports and their finite continuity constants.
5. Choose the near-cone thickness for the summed defect, then t_1, t_0 and the finite outer-cutoff constant.
6. Finally enlarge the physical exterior radius to meet every gap, lower-rank threshold, Gaussian tail, s^{-4} , s^{-5} and complementary absorption requirement.

Inverse powers of the small hierarchy gaps occur only in constants already fixed before the final radius. The local εW_2 cost was paid by center Hardy before the hierarchy was chosen, avoiding a circular $\varepsilon \Delta_N^{-2}$ requirement.

9 Rotation invariance and internal uncertainty

The extra input for the singlet improvement is rotational representation theory on complete physical internal forms. It does not replace them by internal ground states. We first record an older theorem and then a quantitative consequence of its rotation identity.

Theorem 9.1 (Kernel invariance of Hasler and Hoppe). *For the standard physical realization in Hypothesis 2.1, every existing L^2 zero mode of h_N is invariant under $\text{Spin}(9)$, at every fixed finite N .*

This is [3, Theorem 1(a), Section 2], building on [2]; it is not a new assertion of this draft. Its realization and domain are important. Appendix K recalls the cutoff argument: linear growth of the rotation primitive cancels the inverse-radius cutoff derivative, and L^2 shell tails suffice. No positive moment, uniqueness, or internal bound state is assumed.

For an arbitrary finite collection of physical internal blocks, write $H_I = \sum_a \mathfrak{h}_{n_a}$ and $\alpha^2 = \sum_a \alpha_a^2$. Let P_I be Haar averaging for their simultaneous internal $\text{Spin}(9)$ action; it fixes all external spectators.

Lemma 9.2 (Internal nonsinglet uncertainty on a ball). *For physical form vectors supported in $\alpha \leq R$,*

$$H_I[g] \geq \frac{1}{4R^2} \|(1 - P_I)g\|^2. \quad (9.1)$$

The estimate also holds for Hilbert-valued spectators and in the Dirichlet form closure of the internal ball. It is a radius-of-support estimate, not a positive spectral gap on the full internal space and not the operator inequality $H_I \geq \frac{1}{4}\alpha^{-2}(1 - P_I)$.

Proof. Use charges $\tilde{q}_\alpha = q_\alpha/\sqrt{2}$, where q_α are the coefficient-16 charges of Section 2. On physical vectors, $\sum_\alpha \|\tilde{q}_\alpha g\|^2 = 8\mathfrak{h}[g]$. For one fixed generator $j = (u, v)$, the corrected Hasler-Hoppe primitive is

$$\tilde{a}_{j,\alpha} = \frac{5b_{j,\alpha} + 2c_{j,\alpha}}{7}, \quad J_j = \sum_\alpha \{\tilde{q}_\alpha, \tilde{a}_{j,\alpha}\}, \quad (9.2)$$

$$b_{j,\alpha} = \frac{1}{16}(X_u \Gamma_v - X_v \Gamma_u)_{\alpha\epsilon} \theta_\epsilon, \quad c_{j,\alpha} = \frac{1}{16}X_w(\Gamma_w \Gamma_{uv})_{\alpha\epsilon} \theta_\epsilon. \quad (9.3)$$

Color contractions and the sum over w are understood. The bare b alone has the wrong orbital-to-spin ratio in nine dimensions. The CAR and Clifford identities give, for this one j ,

$$\sum_\alpha \tilde{a}_{j,\alpha}^2 = \left(\frac{|X_u|^2 + |X_v|^2}{32} + \frac{\sum_{w \notin \{u,v\}} |X_w|^2}{392} \right) I \leq \frac{\alpha^2}{32} I. \quad (9.4)$$

The expression summed over all 36 generators instead equals $(9/28)\alpha^2 I$. These are different summation scopes in the same convention. In the baseline convention the primitive must also be rescaled: $a_{j,\alpha} = \tilde{a}_{j,\alpha}/\sqrt{2}$. Then $J_j = \sum_\alpha \{q_\alpha, a_{j,\alpha}\}$; the fixed-generator bound is $\alpha^2/64$ and the all-generator identity is $9\alpha^2/56$. Appendix K gives both computations and a stronger Casimir variant.

Sum the blockwise fixed- j identities. On the physical compact core, Cauchy-Schwarz gives

$$|\langle g, J_j^I g \rangle| \leq 2\sqrt{8H_I[g]} \sqrt{R^2 \|g\|^2 / 32} = R\sqrt{H_I[g]} \|g\|. \quad (9.5)$$

The coefficient is unchanged in the baseline convention because $2\sqrt{16/64} = 1$. Only a fixed-generator norm has been used.

The ball and the closed form are rotation invariant. Decompose into compact-group isotypes. On each irreducible representation factor the invariant form is the identity tensored with a multiplicity-space form. Every nontrivial $\text{Spin}(9)$ irrep has a highest-weight vector with J_{12} eigenvalue $m \geq 1/2$. Apply (9.5) to that vector tensored with any multiplicity-space form vector; it gives $H_I \geq m^2/R^2 \geq 1/(4R^2)$. Invariance then gives the same inequality on the full isotype. Sum the nontrivial isotypes and use nonnegativity on the invariant part. Compact-group smoothing, finite isotypic truncation, and form closure justify the passage from the core; no unbounded rotation generator is applied to an uncontrolled form vector. Singleton blocks add no internal variables. $\square$

10 The singlet slow model and its domain

For a two-block partition $n_1 + n_2 = N$, let $\mathcal{H}_a$ be its physical internal Hilbert space, with $\mathcal{H}_a = \mathbb{C}$ for a singleton. In the mass-normalized center variable $x \in \mathbb{R}^9$ put

$$s = |x|, \quad r_{12} = \nu s, \quad \nu^2 = \frac{1}{n_1} + \frac{1}{n_2} \leq 2. \quad (10.1)$$

The reference slow space and closed form are

$$\mathcal{K} = L^2(\mathbb{R}_x^9; \mathcal{F}_{\text{rel}} \otimes \mathcal{H}_1 \otimes \mathcal{H}_2), \quad (10.2)$$

$$q_s = T_x + H_I, \quad \mathcal{Q}(q_s) = \mathcal{Q}(T_x) \cap \mathcal{Q}(H_I). \quad (10.3)$$

Here T_x is the ordinary Hilbert-valued center Dirichlet form. The internal spaces contain their complete bosonic and fermionic variables. Residual gauge and equal-block exchange constraints restrict the admissible subspace and do not add angular states.

Why the model has the ordinary diagonal action

The stationary-flag functional obeys

$$\Phi_{RX}(P) = \sum_{a,i} n_a^{-1} (\text{Tr}(P_a(RX)_i))^2 = \Phi_X(P) \quad (10.4)$$

for the same ordered flag P . This equality holds as a function on the entire flag manifold; its flag gradient and Hessian are equal as well. Hence spatial rotations preserve stationary flags, Hessian admission and incidence sheets. All scalar weights in L4 are rotation invariant. Color frames can be chosen from the flag alone and are constant along spatial rotation orbits; gauge transitions commute with spatial rotations. A global singlet therefore localizes to invariant sections without an angular partition.

At $A = 0$ the two-block fast ground line has stabilizer $\text{Spin}(8)$. Its genuine one-dimensional unitary representation is trivial because that group is connected and semisimple. This includes its central element -1 ; the filled determinant has $8n_1n_2$ occupied modes, an even number. The resulting global equivariant normalized anchor has zero angular Berry one-form, since $S^8 = \text{Spin}(9)/\text{Spin}(8)$ has no invariant tangent covector. Its radial Berry form also vanishes. This gives the ordinary T_x in (10.3).

The negative fast spectral projector along $A \mapsto tA$ is covariant. Its Kato generator $[\dot{P}_t, P_t]$ is covariant, so uniqueness of the transport equation makes the radial transport equivariant. The positive Gaussian and normalized scalar cutoff have the same property. The transported injection U_σ thus intertwines the actual action with

$$J_{\text{slow}} = L_x + S_{\text{rel}} + J_I. \quad (10.5)$$

Finite- A Berry curvature is not asserted to vanish. Its comparison and the differentiated vacuum sources remain exactly the terms already paid in L2–L3. Appendix L supplies the global anchor and transition details.

Zero extension precedes internal averaging

The bounded projection in L1 places the actual coefficient f in the slow Dirichlet form closure of its padded branch. Its zero extension consequently belongs to (10.3); this is

the same extension needed for the baseline Hardy payments. Arbitrary distributional sections are not being extended across a boundary. The extended f is supported in the larger radial tube

$$\mathcal{C}_\beta = \{(x, A) : \alpha_1 + \alpha_2 \leq \beta_{\text{pad}} \nu |x|\}. \quad (10.6)$$

The detailed leaf may have additional direction-dependent tests and need not be invariant under internal-only rotations.

Haar averaging P_I acts only on the simultaneous internal variables and fermions. It commutes with T_x , the complete H_I , center radial weights, and the diagonal action after averaging. It is a contraction in the model form norm and preserves (10.6). Thus it gives orthogonality separately in T_x , H_I , and the s^{-2} norm.

The order of application is essential:

1. Apply the paid local ledger (8.4) to the actual localized vector and its actual coefficient f .
2. Regard that same f as its zero extension in $\mathcal{K}$.
3. Split $f = P_I f + (1 - P_I)f$ in this reference space.
4. Estimate the full-space model form q_s on these summands.

Averaging may leave the detailed leaf but cannot leave the larger radial tube. No local Hamiltonian comparison, detailed-leaf multiplier bound, or projection theorem is reapplied to an averaged component outside its original leaf. Both components remain diagonal singlets when f is one. Fubini, compact slow exhaustion, and closed-form approximation justify the fiberwise estimates below.

Proposition 10.1 (Two-block singlet comparator). *Choose the padded endpoint width initially so that $\beta_{\text{pad}} \leq 1/12$. A diagonal singlet f in (10.3) supported in (10.6) satisfies*

$$(T_x + H_I)[f] \geq h_* \|s^{-1} f\|^2, \quad h_* := \frac{121}{4}. \quad (10.7)$$

Proof. Set $f_0 = P_I f$ and $f_1 = (1 - P_I)f$. The relative sixteen-Majorana module is $\mathcal{F}_{\text{rel}} = \mathbf{44} \oplus \mathbf{84} \oplus \mathbf{128}$, as recalled in [8, Section 2.4]. Scalar harmonics on S^8 have highest weight $(l, 0, 0, 0)$. Among those three summands only $\mathbf{44}$ matches a harmonic, at $l = 2$. Thus the internal-invariant diagonal singlet f_0 has angular eigenvalue $2(2 + 7) = 18$. This applies to its arbitrary internal invariant multiplicity space, not only an internal ground state. Radial Hardy gives

$$T_x[f_0] \geq (49/4 + 18) \|s^{-1} f_0\|^2. \quad (10.8)$$

For each fixed x , f_1 has internal support $\alpha \leq \beta_{\text{pad}} \nu s$. Lemma 9.2, followed by Fubini and form closure, gives

$$H_I[f_1] \geq \frac{1}{4\beta_{\text{pad}}^2 \nu^2} \|s^{-1} f_1\|^2 \geq 18 \|s^{-1} f_1\|^2. \quad (10.9)$$

Ordinary center Hardy supplies $49/4$ on f_1 as well. Orthogonal additivity proves (10.7). Half-integral internal representations, gapless continua, and entanglement between blocks are included in Lemma 9.2. Equal-block exchange constraints only restrict this space. At $N = 2$ both internal factors are scalar and $f_1 = 0$. $\square$

For $k \geq 3$, the center dimension is at least 18, and (5.1) alone gives $h_\pi \geq 64 > h_*$. No angular classification or internal averaging is needed on those leaves.

11 Singlet exterior estimate

Theorem 11.1 (Singlet global implication). *Assume the same realization and local analytical inputs as in Theorem 8.1, with the equivariant constructions and Dirichlet model identification detailed in Sections 9–10 and Appendices K–L. For every fixed $N \geq 2$ and every $0 < c < 121/4$, there is a finite $R_{N,c}^{\text{sing}}$ such that*

$$\mathfrak{h}_N[u] \geq c \|\rho_N^{-1}u\|^2, \quad u \in \mathcal{Q}_{N,R_{N,c}^{\text{sing}}}^{\text{Spin}(9)}. \quad (11.1)$$

The superscript restricts the physical exterior form domain to rotation singlets. No coefficient above $49/4$ is asserted for arbitrary nonsinglet vectors.

Proof. Keep the lower-rank all-vector Theorem 8.1 for all internal soft-Hardy payments. There is no induction using the stronger conclusion. Given $c < h_*$, choose

$$0 < a < \min \left\{ \frac{1}{12}, \frac{h_* - c}{2(3h_* + 2)} \right\}. \quad (11.2)$$

At the initial local-width selection, also require $\beta_{\text{pad}} \leq 1/12$ for two-block leaves. Choose the tree endpoint small enough to retain this padded restriction. This precedes L, δ, Δ_N and introduces no new inverse-gap dependency; follow the acyclic parameter order of the baseline thereafter.

Apply (8.4) before internal averaging. Its low part has the smaller internal coefficient $1 - 3a$. Discarding only the nonnegative extra aT_C gives

$$\begin{aligned} (1 - 3a)H_I[f] + (1 - 2a)T_C[f] - 2a \|s^{-1}f\|^2 &\geq [(1 - 3a)h_* - 2a] \|s^{-1}f\|^2 \\ &> c \|s^{-1}f\|^2. \end{aligned} \quad (11.3)$$

For two blocks this uses the model comparator after zero extension. For three or more blocks it uses $h_\pi \geq 64$ and $H_I \geq 0$. Replacing h_0 mechanically while keeping the coefficient $1 - 2a$ would be incorrect, since H_I now contributes to coercivity.

The colored complement receives no rotational improvement. Increase the final radius until its retained $(b/2)H_{\text{exc}}$ controls $c \|s^{-1}v\|^2$, exactly as before. The slow mass identity, $\rho_N \geq s$, outside-cone estimate, and incidence square sum then prove (11.1). Gauge averaging, rotation averaging, compact exhaustion and exterior form closure preserve the stated domain. Every IMS, Berry and Schur loss remains charged once, and no internal spectral gap or bound-state projection was introduced. $\square$

This theorem inherits all substantive analytical dependencies of Appendices A–J. The rotation argument closes an additional implication if those baseline inputs hold; it does not independently certify them. The number $121/4$ is the threshold supplied by this comparator, not a proved sharpness result for the full Hamiltonian.

12 Moments of actual zero modes

Corollary 12.1. *Under Theorem 11.1, every $\Psi \in \mathcal{H}_N$ with $\mathfrak{h}_N[\Psi] = 0$ satisfies*

$$\int \rho_N^m |\Psi|^2 < \infty \quad (0 \leq m < 9). \quad (12.1)$$

No existence of such a Ψ is asserted.

Proof. By Theorem 9.1, Ψ is a rotation singlet. Bounded radial multipliers preserve that property. Choose a fixed smooth radial inner cutoff η supported in the exterior of (11.1), equal to one farther out. For $a_p \geq 1$, use bounded truncated powers $g_M(\rho) = \min(\rho^{a_p}, M^{a_p})$, or smooth Lipschitz approximants, so that $|\nabla g_M| \leq a_p g_M / \rho$. Apply (11.1) and (2.11) to $\eta g_M \Psi$. For any $\zeta > 0$, Young's inequality gives

$$[c - (1 + \zeta)a_p^2] \|\rho_N^{-1} \eta g_M \Psi\|^2 \leq C_{\eta, \zeta, a_p} \|\Psi\|^2, \quad (12.2)$$

with the right side independent of large M : all derivatives of η lie in one fixed annulus. Choose $a_p^2 < c < h_*$ and ζ small, then let $M \rightarrow \infty$ by monotone convergence. This proves exterior integrability with exponent $m = 2a_p - 2$. The bounded interior is harmless since $m \geq 0$ and $\Psi \in L^2$. Every $0 \leq m < 9$ is obtained by taking $a_p = (m + 2)/2 < 11/2$ and choosing c between a_p^2 and $121/4$. $\square$

The endpoint $m = 9$ is not obtained because this coercive margin vanishes there. No point-wise asymptotic expansion or nonzero leading tail is proved. Polchinski's expected free $l = 2$ two-cluster behavior r^{-9} , with radial density $r^8 dr$, already predicts the same below-nine threshold [9]. The candidate contribution is the actual-state implication through the stated all-channel form estimates, not that asymptotic prediction.

At fixed rank the corollary includes the eighth ordinary radial moment. Every coordinate polynomial of degree at most four, with bounded fermion-matrix coefficients, times Ψ belongs to L^2 ; polynomial expectations through degree eight are absolutely finite. A gauge-invariant such observable therefore has a well-defined centered physical L^2 source $(O - \langle O \rangle)\Psi$. This gives no rank-uniform source norm, reduced-resolvent estimate at zero energy, or inverse spectral moment. Momentum insertions, all Ward identities, all bootstrap levels, numerical radius bounds, large- N tightness, scattering completeness and full soft theorems require additional arguments.

13 Reproducibility and review boundaries

The accompanying bundle contains this editable $\text{\LaTeX}$ source, twelve mathematical source notes, eleven executable regression programs, a runner, dependency versions, finite-check outputs, a separate Chinese summary, and SHA-256 manifests. It excludes private coordination and review transcripts, unrelated research, email, and downloaded third-party papers. The source notes retain the mathematical equations and local scope boundaries; procedural review labels and superseded status commentary are removed. Provenance records the exact input hashes and included sections.

The regression suite covers finite-matrix geometry, adapted Gauss Taylor coefficients, frozen-energy identities, source excitation bookkeeping, square completion, regularized tree/IMS examples, and the scalar global budget. The global budget uses exact rational arithmetic over 2,693 proper integer partition types through $N = 20$, six target coefficients including $49/4 - 10^{-6}$, the retained complementary fraction, and moment arithmetic. Floating-point checks include hierarchical gap stress cases. Two added programs check exact Clifford coefficients for all 36 rotation generators, the reciprocal normalization conversion, the two-block width payment, and the revised singlet ledger and moment arithmetic. Successful execution is evidence for those finite calculations only.

A specialist review should check, in order:

1. the trace/CAR normalization and realization assumptions in Section 2;
2. exact geometry and density, Appendix A; the frozen energy bound, Appendix B;
3. centered first jets and actual adapted source cancellation, Appendices C-D;
4. rootwise differential estimates, full colored-charge comparison, and the non-double-counted reserve, Appendices E-I;
5. normalized incidence localization at singular commuting strata and its padded domains, Appendix J;
6. the common ledger and all-vector induction;
7. the corrected rotation primitive, bounded-radius isotypic argument, equivariant vacuum line and zero-extension order, Appendices K-L;
8. the smaller retained internal coefficient in the singlet ledger and the bounded-weight passage.

The most recent domain correction is implemented consistently: the Schur infimum and any minimizing complement belong to $\mathcal{Q}(h_N)$, and projection boundedness is in the form norm, equivalently the graph norm of the closed charge stack. Operator-domain preservation is not inferred. The global lower bound does not depend on minimizer attainment. This draft is ready to be examined, not certified as correct or submission-ready by the existence of the bundle. A future public version should reflect the named author's own review and a specialist check of the full analytical dependencies. No priority, affiliation, human endorsement, formal verification, or journal outcome is asserted.

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Technical appendices

The first ten appendices reproduce the substantive mathematical derivations on which the BFSS specialization depends. Their subsection and displayed equation numbers are local to each source note. Historical source labels are retained for traceability, including gaps and nonsequential labels; these do not indicate omitted PDF pages. These are included for direct review, not as external certified lemmas. Their local scope boundaries should be read together with the dependency map in Section 3; no single appendix purports to prove the global theorem by itself. The last two appendices add the corrected rotation algebra and equivariant reference-domain construction for the singlet refinement.

The main text and appendices use typeset mathematics. The longer derivations in Appendices A–J have been transcribed from the source-level formulas, with their coefficient inventory, operator order, and local labels retained. The original editable source notes and an expression-by-expression transcription record accompany this edition for comparison. This production revision does not provide an additional mathematical certification.

A Intrinsic multiblock geometry

Technical derivation for review. Equation and subsection numbers in this appendix are local to the source derivation. Historical local labels are retained, including numbering gaps or nonsequential labels. The accompanying source notes retain the original expressions for comparison.

1. Setting and precise scope

Fix N and a partition $N = n_1 + \dots + n_k, k \geq 2$. Hermitian matrix tuples have the real trace inner product. Retain the residual group $K = S(U(n_1) \times \dots \times U(n_k))$, including its action on the fermions; do not divide by any internal nonabelian orbit. Write

$$\begin{aligned} X_i &= Z_i + A_i + Y_i, \\ Z_i &= \text{diag}(z_{1i}I_{n_1}, \dots, z_{ki}I_{n_k}), \\ \sum_a n_a z_a &= 0, \quad \text{Tr } A_i^{(a)} = 0, \\ b_{ab} &= z_a - z_b, \quad r_{ab} = |b_{ab}|, \quad r_* = \min_{a < b} r_{ab} > 0. \end{aligned}$$

A is block diagonal and Y is off block. Put $\alpha_a^2 = \sum_i \|A_i^{(a)}\|_F^2$. Assume only

$$\alpha_a + \alpha_b \leq \kappa r_{ab} \quad \text{for every } a < b. \quad (1)$$

There is no hypothesis $\alpha_a \ll r_*$ for a block unrelated to a shortest pair. Set $\kappa \leq 1/100$. For the fast tube use

$$|Y| \leq \sigma r_*, \quad \sigma \leq 1/100. \quad (2)$$

Condition (2) is deliberately a fast-coordinate condition. It is compatible with the Gaussian widths at sufficiently large r_* and does not impose the forbidden common scale on A . A smooth smaller cutoff may use the harmonic minimum of the r_{ab} instead of the non-smooth literal minimum.

Let C be the Euclidean block-center tuple space, I the space of blockwise traceless internal tuples, and m the off-block Hermitian gauge-parameter space. Its real dimension is

$$q = 2 \sum_{a < b} n_a n_b.$$

For a Hermitian tuple Q let $B_Q \xi = (i[\xi, Q_i])_i$. Define

$$\begin{aligned} R &= (B_Z^* B_Z)^{1/2} = \bigoplus_{a < b} r_{ab} I_{2n_a n_b}, \\ E &= B_Z R^{-1}, \quad E^* E = I_m, \\ N_Z &= \ker B_Z^* \text{ in the off-block tuple space.} \end{aligned}$$

The intrinsic normal constraints are $B_Z^*Y = 0$, equivalently $b_{ab} \cdot Y_{ab} = 0$ for every pair. The slice S consists of the triples (Z, A, Y) satisfying these constraints. Statements below are tensor identities on this slice and its local gauge-saturated image. A simultaneous smooth projector branch exists locally by the nonsingular Hessian proved below. This does not assert a globally unique branch across incompatible cluster partitions.

2. Horizontal differential and the exact internal coefficient

Use the normal connection obtained by orthogonal projection onto N_Z . For $h = (\delta Z, \delta A, \delta Y_N)$ in $H = C \oplus I \oplus N_Z$, horizontal differentiation of $B_Z^*Y = 0$ gives

$$\begin{aligned}\delta Y &= \delta Y_N + EK h, \\ Kh &= -R^{-1}B_{\delta Z}^*Y.\end{aligned}\tag{3}$$

K vanishes identically on I and N_Z . The slice embedding differential, in the orthogonal ambient splitting $H \oplus \text{range } E$, is

$$\begin{aligned}Sh &= (h, Kh), \\ g_S &= S^*S = I_H + K^*K \\ &= g_C \oplus I_I \oplus I_{N_Z}.\end{aligned}\tag{4}$$

Thus the induced slice has internal kinetic coefficients **exactly one**, and there are no internal/center or internal/fast induced-metric cross terms. The full reduced co-metric additionally contains the positive orbital contribution from the Gauss square; it is not claimed that its total AA block equals I . The coefficient-one t_A summand is retained before that square is estimated. This is an exact identity for every A and Y in the coordinate domain, not an oscillator expectation.

Skew-adjointness of the gauge action gives a useful equivalent identity. For $O = B_X, O_C = P_C O$ and $O_H = P_H O$,

$$O_C = P_C B_Y, \quad K = R^{-1}O_C^*.\tag{5}$$

The elementary commutator bound $\|B_Y\| \leq 2|Y|$ proves

$$\begin{aligned}\|K\| &\leq 2s, \quad s = |Y|/r_*, \\ (1 + 4s^2)^{-1}I_C &\leq g_C^{-1} \leq I_C.\end{aligned}\tag{6}$$

All normal-frame connections remain inside the covariant center momentum. Their unbounded $Y\partial_Y$ generators are never estimated by oscillator energy on arbitrary vectors.

3. Faddeev-Popov matrix, explicit noncircular invertibility, and density

The differential of the normalized slice constraint is $N^* = (-K, I_m)$; thus

$$N = (-K^*, I_m), \quad W = N^* N = I_m + K K^*.$$

The full transverse-orbit matrix is

$$\begin{aligned} M &= N^* O = E^* B_X - K O_C \\ &= E^* B_{Z+A} + E^* B_Y - R^{-1} O_C^* O_C. \end{aligned} \tag{7}$$

Put $F = E^* B_{Z+A}$. F is pair-block diagonal. Under the canonical complex identification of a pair, its block is unitarily equivalent to

$$\begin{aligned} &r_{ab} I + T_{ab,n}, \\ &T_{ab,n} V = A_n^{(a)} V - V A_n^{(b)}, \quad n = b_{ab}/r_{ab}. \end{aligned}$$

Consequently

$$(1 - \kappa)R \leq F \leq (1 + \kappa)R, \quad FR = RF. \tag{8}$$

The transverse-frame convention $E = B_Z R^{-1}$ has absorbed the fixed real root-plane complex structure; that is why F is positive here. In a fixed real root frame the corresponding matrix carries that orthogonal complex structure.

Equations (5)-(8) give the explicit bound

$$\|MR^{-1} - I\| \leq \kappa + 2s + 4s^2 \leq 0.0304. \tag{9}$$

No $r_{\max}/r_*$ or unrelated internal radius enters (9). The full differential from (h, ξ) to ambient coordinates has block matrix

$$T = \begin{bmatrix} I_H & O_H \\ K & E^* O \end{bmatrix}.$$

Its determinant is $\det M$. Equivalently the induced slice volume is $\sqrt{\det W}$, while the normalized-normal orbit contraction is $|\det(W^{-1/2}M)|$. They cancel. The physical measure after the compact off-block gauge integration is therefore

$$J dZ dAdY_N, \quad J = \det M > 0, \tag{10}$$

up to one constant orbit normalization. This is physical measure including orbit volume, not bare quotient Riemannian volume.

At $Y = 0$,

$$\begin{aligned}
J_0 &= \det F = \prod_{a < b} \det_C (r_{ab}I + T_{ab,n})^2, \\
d_0 &= \sqrt{J_0} = \prod_{a < b} \det_C (r_{ab}I + T_{ab,n}).
\end{aligned} \tag{11}$$

The determinant is positive by (1). Internal coincidences introduce no singularity.

There is no linear- Y density term, even for nonzero internal A . Indeed $L = E^*B_Y$ has zero diagonal pair blocks: acting on an ab gauge parameter, an off-block Y produces block-diagonal output or a different pair. Since F^{-1} is pair-block diagonal,

$$\text{tr}(F^{-1}L) = 0. \tag{12}$$

The remaining correction $-K O_C$ is quadratic in Y . In fact the absolutely convergent logarithm series gives, for the stated κ and σ ,

$$|\log(J/J_0)| \leq 8qs^2. \tag{13}$$

For verification: $\|F^{-1}L\| \leq 2s/(1-\kappa)$, $\|F^{-1}KO_C\| \leq 4s^2/(1-\kappa)$. The linear trace vanishes, and the sum of powers ≥ 2 is bounded by $qe^2/[2(1-e)]$, where $e = (2s+4s^2)/(1-\kappa)$. Together with the quadratic trace this is $< 8qs^2$.

4. Intrinsic projectors and transition law

For an ordered orthogonal family of projectors P_a of ranks n_a define the gauge-invariant function

$$\Phi_X(P) = \sum_{a,i} n_a^{-1} (\text{Tr}(P_a X_i))^2.$$

Stationarity under off-block variations is precisely $B_Z^*Y = 0$. At such a point its Hessian in a gauge parameter ξ is

$$\begin{aligned}
&\text{Hess } \Phi_X[\xi, \xi] \\
&= 2\|O_C \xi\|^2 - 2\langle B_Z \xi, B_X \xi \rangle \\
&= -2\langle \xi, RM \xi \rangle.
\end{aligned} \tag{14}$$

Thus RM is symmetric. In the normalized orbit variable $\eta = R\xi$, the negative half-Hessian is MR^{-1} , which by (9) lies between $0.9696I$ and $1.0304I$. Every such stationary decomposition is a strict local maximum with a nonsingular transverse Hessian. The implicit function theorem therefore gives a unique smooth nearby projector branch, equivariant under $SU(N)$, with no internal eigenvalue gap assumption.

On overlap of charts describing the same branch, the ordered P_a agree. Their block frames differ by K and, when the labels are changed consistently, by a permutation of

equal-rank blocks. Normal frames differ by orthogonal maps on N_Z . These are exact unitary bundle changes on bosonic fibers and the fermionic module. J is invariant; no extra nonunit half-density amplitude appears. All expressions (3)-(14) and the kinetic identity below intertwine under these changes. One may prove them in local frames without introducing scalar angular IMS cutoffs.

This is a **same-branch** transition theorem. Comparing different nested partitions, proving that their selectors agree with all commuting-tree choices, and uniformly transporting every decision-tree cutoff off the commuting cone remain separate obligations. Local nonsingularity alone does not prove any of those global claims.

4.1 Smooth fast projection on the same branch

The frozen pair matrices G_{ab} , K_{ab} and D_{ab} of the general fast-energy lemma are obtained at $Y = 0$ from (15)-(16) and the exact quadratic potential. Their bosonic frequencies and fermionic spectral gaps are bounded below by fixed positive multiples of r_{ab} under (1). Functional calculus therefore gives a smooth positive Gaussian and a smooth negative-energy occupied subspace, including at every internal eigenvalue collision. The resulting rank-one fast-vacuum projection P is gauge equivariant. Local vacuum frames may have a nonflat scalar Berry connection; none is declared parallel here.

A same-branch chart change acts on every finite pair matrix by its actual orthogonal/color/spin unitary representation. Spectral functional calculus and the positive Gaussian construction commute with that action. Hence $P' = V P V^*$ exactly. The same is true after a smooth invariant radial cutoff, for example with $h = (\sum_{a < b} r_{ab}^{-2})^{-1/2}$ and $\chi(|Y|/(\sigma h))$, followed by fiberwise normalization. There is no angular scalar partition in this construction. Since $r_*/\sqrt{k(k-1)/2} \leq h \leq r_*$, the cutoff tail bounds differ only by fixed N -dependent constants.

For the uncut product U , an internal derivative in A_a differentiates only the pair factors incident to a . For U_σ the derivative also differentiates its global scalar normalization; this term is bounded by the same derivative norms, and its difference from the uncut normalization is exponentially small in the indicated tail scale. The resolvent formula for their spectral projections and the Gaussian square-root formula give norm bounds $C_{N,j} r_{ab}^{-j}$ for each fixed number j of such derivatives, after the natural fast dilation. These follow from external pair gaps, never from an internal Hamiltonian resolvent. Mixed derivatives have the corresponding products of inverse incident scales. Global phase choices are unnecessary because the projection and its induced connection are intrinsic.

5. Exact all-vector kinetic plus Gauss form

Let p_H be the slice coordinate momentum stack, with the normal connection in its center component and ordinary internal and fast components. Let S_f denote the off-block fermionic Gauss generator, in the same sign convention as $O^*p = S_f$. Residual K-equivariance remains imposed. The orthogonal tangential projection of the orbit is

$$O_T = S g_S^{-1} S^* O.$$

For a raw equivariant section ϕ define

$$D\phi = W^{1/2} M^{-*} [S_f \phi - O^* S g_S^{-1} p_H \phi]. \quad (15)$$

Then the exact ambient reduced kinetic form, in coefficient-one units, is

$$t_{\text{raw}}[\phi] = \int J \{ \langle p_H \phi, g_S^{-1} p_H \phi \rangle + \|D\phi\|^2 \}. \quad (16)$$

Proof: write an ambient covector as its slice-tangent projection plus $N\beta$. Its tangential restriction is p_H ; the Gauss equation fixes $M^*\beta = S_f - O^* S g_S^{-1} p_H$. Tangent and normal parts are orthogonal, and $\|N\beta\|^2 = \langle \beta, W\beta \rangle$. This proves (16) for arbitrary covectors, arbitrary spinor values, and unrestricted momenta. It is a first-order identity, not substitution into a squared differential operator.

For the flat measure put $d = \sqrt{J}$ and $\psi = d\phi$. The exact flattened formula is obtained from (15)-(16) by

$$p_H \rightarrow p_H + id_H \log d. \quad (17)$$

This fixes every ordering. Equivalently integration of the real density cross terms gives the unshifted quadratic stack plus the scalar multiplication

$$V_d = d^{-1} \text{div}_H(ad_H d), \quad (18)$$

where a is the complete scalar principal co-metric, including the orbital part of D . Divergence is that of the flat bundle measure. The finite Hermitian spin terms have zero real cross pairing with the purely imaginary scalar density shift. No separate spin connection is appended. Formula (17) remains the definitive expression if a convention for momenta is changed.

On a fixed-shape regular region $r_* \geq \delta \rho_C$ with $\rho_C^2 = \sum_a n_a |z_a|^2$, and a closed smaller tube, V_d and all coefficient derivatives needed at any fixed order are bounded by constants times their homogeneous powers. In particular $|V_d| \leq C_{N,\delta,\kappa,\sigma} \rho_C^{-2}$. These are bounds from explicit smooth finite matrices and compact shape sets; they do not use a full Hamiltonian inequality.

Combining the exact internal kinetic coefficient with the exact internal potential and Yukawa pieces retains

$$\sum_a [t_{A_a} + V_a + \text{Yukawa}_a] = \sum_a h_{n_a}$$

with coefficient one, also on colored internal sections. The averaged-supercharge non-negativity of these internal forms is the usual algebraic statement, not a physical-sector Hardy inequality or an internal spectral gap.

Equation (16) does **not** justify adding a separately extracted adapted fast kinetic matrix and the whole same Gauss square. The frozen adapted kinetic matrix is already part of that square.

6. Incident-edge reserve for unrestricted low internal derivatives

Let $U(Z,A)$ be the normalized product of the pairwise kinetic-aware Gaussian and occupied fermionic determinant from the frozen-pair lemma. Interpret it as an isometric injection of its vacuum line; no flat Berry connection is assumed. On each pair its covariance obeys

$$E_U |Y_{ab}|^2 \leq C n_a n_b \ell^3 / r_{ab},$$

with an absolute C on the stated κ tube. Use a smooth invariant cutoff strictly inside (2), normalize fiberwise, and call the resulting injection U_σ . At fixed finite N and sufficiently large r_* , the same covariance bound holds with twice C ; cutoff errors are bounded by polynomial factors times $\exp(-c\sigma^2 r_*^3 / \ell^3)$, uniformly in unrelated A and $r_{\max}$. Derivatives can have fixed shape-dependent prefactors if expressed in ρ_C units.

The internal orbital map in (15) is exactly

$$O_{T,A} = O_A = P_I B_Y. \quad (19)$$

There is no Y -independent internal orbital derivative and no contribution from an unrelated pair. For an internal covector $p = (p_a)$, the ab component satisfies

$$|(O_A^* p)_{ab}| \leq C |Y_{ab}| (|p_a| + |p_b|). \quad (20)$$

Write $M = F + P$. The elementary bound

$$\|PF^{-1}\| \leq (2s + 4s^2)/(1 - \kappa) < 0.021$$

shows that $M^{-*} = (I + F^{-*}P^*)^{-1}F^{-*}$ has a uniformly bounded left factor. $W^{1/2}$ is uniformly bounded too. Thus (20) gives the pointwise estimate

$$\begin{aligned} & \|W^{1/2}M^{-*}O_A^*p\|^2 \\ & \leq C \sum_{a < b} |Y_{ab}|^2 / r_{ab}^2 (|p_a|^2 + |p_b|^2). \end{aligned} \quad (21)$$

This is valid for every internal covector, not just bounded internal momenta. Compressing only its coefficients in U_σ yields

$$\begin{aligned} & \|D_A(U_\sigma \nabla_A f)\|^2 \\ & \leq \sum_a w_a |\nabla_A f|^2, \\ & w_a = C \sum_{b \neq a} n_a n_b \ell^3 r_{ab}^{-3}. \end{aligned} \quad (22)$$

The derivatives here are the covariant derivatives on the possibly nonflat vacuum line. The expression in (22) denotes the part where the derivative acts on f ; derivatives acting on U_σ are separate coefficient sources. Estimate (22) records the incident-weighted low-leg coefficient bound. Its use in the mixed form must follow the complete-charge and postcompletion estimates of Appendix I, Sections 4 and 5.2-5.4; it does not authorize retaining or spending a second copy of the original Gauss square.

Most importantly, the loss is incident-edge weighted. For an ordinary untwisted internal derivative, both $t_{A_a} \leq h_a + C_{n_a} \ell^{-3} \alpha_a$ and $V_a \leq h_a + C_{n_a} \ell^{-3} \alpha_a$ follow from the finite-Clifford Yukawa bound. Independently of that form comparison, (1) gives

$$w_a \alpha_a \leq C \kappa \sum_{b \neq a} n_a n_b \ell^3 r_{ab}^{-2}. \quad (23)$$

The weighted derivative error has a direct payment that does not require a new weighted full-supercharge theorem. Use the internal-radial gauge constructed in source Section 4.2 in Appendix F: each pair connection obeys $|a_{ab}| \leq C_N \kappa^2 / r_{ab}$, and its A_a component vanishes unless that pair is incident to a . In the resulting identification with the flat $A = 0$ center line,

$$\begin{aligned} & \sum_a w_a \|\nabla_{A_a} f\|^2 \\ & \leq 2 \sum_a w_a t_{A_a}[f] \\ & \quad + C_N \kappa^4 \ell^3 r_*^{-3} \\ & \quad \int \sum_{a < b} r_{ab}^{-2} |f|^2 \\ & \leq 2 \sum_a w_a h_a[f] \\ & \quad + C_N [\kappa + \kappa^4 \ell^3 r_*^{-3}] \\ & \quad \int \sum_{a < b} r_{ab}^{-2} |f|^2. \end{aligned} \quad (26)$$

All rank-dependent Clifford and endpoint factors are included in C_N . The first inequality is the elementary $|\partial f + i a f|^2 \leq 2|\partial f|^2 + 2|a|^2 |f|^2$; the second uses (23). The weights depend only on centers, so multiplying the internal form creates no omitted internal weight derivative. At large r_* their h_a coefficients vanish. The inverse-square term is arbitrarily small in κ and has a shape-independent pair-center Hardy payment. This argument pays only the small derivative error (22); it does not replace the full-supercharge comparison needed for the principal Berry-coupled slow form. A cruder replacement $w_a \leq C r_*^{-3}$, followed by $r_*^{-3} \alpha_a$, would destroy this conclusion for a far extended block and is not used.

The variable induced center metric is also paid as a genuine derivative-form pairing against a retained covariant center kinetic square. Its correction is exactly

$$g_C^{-1} - I_C = -K^*(I + K K^*)^{-1} K.$$

Its Gaussian coefficient norm is $O_N(\ell^3 r_*^{-3})$, so the squared cost in that direct pairing is $O_N(\ell^6 r_*^{-6})$ times the low center derivative form. This step involves only center derivatives; it cannot create an internal kinetic loss. Derivatives of U_σ supply further coefficient sources, which still require their own full source inventory.

B Frozen fast energy defect

Technical derivation for review. Equation and subsection numbers in this appendix are local to the source derivation. Historical local labels are retained, including numbering gaps or nonsequential labels. The accompanying source notes retain the original expressions for comparison.

This appendix writes the frozen energy in physical units. Multiply E_0 by $2/R_{\text{BFSS}}$ to use the kinetic-one form of the main text; set $\ell = 1$ after the stated fixed dilation.

1. Statement and normalization

Let n, m be positive integers, $d = nm$, and let A_i and $B_i, i = 1, \dots, 9$, be Hermitian internal matrices of the two blocks. Rotate the spatial coordinates so that their center difference is $b = re_9, r > 0$. On $\text{Hom}(C^m, C^n)$, with its Hilbert-Schmidt metric, put

$$\begin{aligned} T_i Z &= A_i Z - Z B_i, \\ Q_i &= r \delta_{i9} I_d + T_i, \\ H &= \sum_i Q_i^2, \quad C_{ij} = [Q_i, Q_j] = [T_i, T_j]. \end{aligned}$$

Assume each $\|T_i\|_{\text{op}} \leq r/100$. A coordinate-invariant sufficient condition is

$$\left(\sum_i \|A_i\|_F^2\right)^{1/2} + \left(\sum_i \|B_i\|_F^2\right)^{1/2} \leq r/100.$$

Thus it suffices that each endpoint radius is at most $r/200$. No relation to any unrelated pair's separation is imposed. Each pair may use its own orthonormal spatial frame; a common global axial direction for all pairs is unnecessary. All traces below are finite ordinary, unnormalized traces.

On the slice $Z_9 = 0$ define 8-by-8 matrices with d -by- d blocks:

$$\begin{aligned} G_{ij} &= \delta_{ij} I_d + Q_i Q_9^{-2} Q_j, \\ K_{ij} &= \delta_{ij} H - Q_i Q_j + 2C_{ij}, \quad i, j \leq 8. \end{aligned}$$

The kinetic-aware bosonic frequency matrix is $B = G^{1/2} K G^{1/2}$. For Hermitian $\text{Spin}(9)$ matrices satisfying $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij} I_{16}$ put

$$D = \sum_i \Gamma_i \otimes Q_i.$$

Both K and B are positive, and D has precisely 8d negative and 8d positive eigenvalues. The frozen ground energy, in physical units, is

$$E_0 = (R/\ell^3) [\text{Tr}_{8d} \sqrt{B} - (1/2) \text{Tr}_{16d} |D|].$$

The following explicit, deliberately non-sharp constant is valid for every n, m :

$$|E_0| \leq 150002(R/\ell^3)r^{-3} \sum_{i < j} \|[T_i, T_j]\|_F^2. \quad (1)$$

The constant in (1) is independent of d ; endpoint weights below carry the finite-rank dependence. The proof neither diagonalizes internal matrices nor invokes an internal Hamiltonian gap, internal eigenvalue gap, regularity of the commuting variety, or a nearest-commuting-tuple estimate.

2. Why these are the correct frozen matrices

The eliminated off-block Gauss principal equation is

$$Q_9 p_9 + \sum_{i \leq 8} Q_i p_i = 0.$$

Its positive kinetic square gives G exactly. It must not be dropped or treated as a separate small bounded perturbation.

For $V(X) = (R/(2\ell^6)) \sum_{i < j} \|[X_i, X_j]\|_F^2$, the quadratic part in one off-block pair is $(R/\ell^6)Z^*KZ$. The square of the linear off-block commutator gives $\delta_{ij}H - Q_jQ_i$. The internal-background commutator paired with $[Y_i, Y_j]$ contributes C_{ij} . Their sum is

$$\delta_{ij}H - Q_jQ_i + C_{ij} = \delta_{ij}H - Q_iQ_j + 2C_{ij}.$$

This works for two arbitrary internal blocks. The second block contributes with the minus sign in T_i , including in its induced commutator. At quadratic order other block pairs do not couple into this pair's Hessian. Interactions involving three distinct blocks begin in the higher-degree cross terms and are not estimated here.

The 8d complex bosonic coordinates are 16d real oscillators. Their ground energy is $(R/\ell^3)\text{Tr}\sqrt{B}$. The fast fermionic sector has 16d complex annihilators; filling the 8d negative eigenmodes gives $-(R/(2\ell^3))\text{Tr}|D|$. The reverse block is the adjoint, not an independent second copy.

3. Uniform cross gaps

Scale r to one and write $\kappa = 1/100$. Then $Q_9 = I + T_9, Q_i = T_i$ for $i \leq 8$. Block norm row sums give

$$\begin{aligned} \|K - I\| &\leq p(\kappa), \quad p(t) = 2t + 29t^2, \\ 0 \leq G - I, \quad \|G - I\| &\leq g(\kappa), \quad g(t) = 8t^2/(1-t)^2. \end{aligned}$$

For K , a diagonal block has $2T_9$ plus eight squares, and each of its seven off-diagonal blocks is $T_i T_j - 2T_j T_i$. This explains the coefficient 29. Consequently

$$K \geq 0.9771I, \quad G \geq I, \quad B \geq 0.9771I.$$

Also $\|D - \Gamma_9 \otimes I\| \leq \sum_i \|T_i\| \leq 9\kappa$, so $\min |D| \geq 0.91$. Continuous scaling of T to zero preserves its signature. $H \geq Q_9^2 \geq 0.9801 I$. Restoring r multiplies K, B, H, D^2 by r^2 and D by r .

4. The real-trace two-commutator lemma

Let $X_1, \dots, X_s$ be Hermitian matrices with $\|X_i\| \leq \kappa$. Set $F^2 = \sum_{i < j} \|[X_i, X_j]\|_F^2$. For any words w and w' of length q having the same multiset of letters,

$$\begin{aligned} & |\operatorname{Re} \operatorname{Tr} w - \operatorname{Re} \operatorname{Tr} w'| \\ & \leq ((q)_4/8)\kappa^{q-4}F^2, \quad q \geq 4, \end{aligned} \tag{2}$$

where $(q)_4 = q(q-1)(q-2)(q-3)$. The left side is zero for $q \leq 3$.

Proof. At most $q(q-1)/2$ adjacent swaps transform one ordering into another. The trace difference from one swap is $\operatorname{Tr} U[X_i, X_j]V$. By cyclicity this is $\operatorname{Tr} CW$, with $C = [X_i, X_j]$ anti-Hermitian and $W = VU$ a word of length $q-2$. Exactly,

$$\operatorname{Re} \operatorname{Tr}(CW) = (1/2) \operatorname{Tr}(C(W - W^*)). \tag{3}$$

Reversing W takes at most $(q-2)(q-3)/2$ adjacent swaps. Each resulting term contains a second commutator and $q-4$ other letters. Hilbert-Schmidt Cauchy-Schwarz bounds its trace by $\kappa^{q-4}F^2$. Equation (3), followed by the first sequence of swaps, proves (2). If $q \leq 3$, W is constant or Hermitian and (3) is zero.

In particular, take a real-coefficient homogeneous noncommutative trace polynomial whose commutative polynomial is zero. Sort every word to the same canonical word for its multiset. The sorted contributions cancel coefficient by coefficient; (2) bounds the original real trace by the sum of absolute coefficients times $(q)_4\kappa^{q-4}F^2/8$. This is an explicit algebraic estimate; it does not divide by an ideal on a singular algebraic variety.

5. Bosonic functional calculus and an explicit coefficient majorant

Continue at $r = 1$. Put $L = GK - I$. Since GK is similar to B ,

$$\begin{aligned} \operatorname{Tr} \sqrt{B} &= \operatorname{Tr} \sqrt{I + L} \\ &= 8d + \sum_{j \geq 1} \binom{1/2}{j} \operatorname{Tr} L^j. \end{aligned} \tag{4}$$

The binomial series converges in operator norm: $\|L\| \leq l(\kappa) < 1$, where

$$l(t) = p(t) + g(t)(1 + p(t)).$$

The expansion $Q_9^{-2} = \sum_{j \geq 0} (-1)^j (j+1) T_9^j$ converges as well. Thus (4), and the corresponding expansion of $8\text{Tr } \sqrt{H}$, define absolutely convergent noncommutative power series with real coefficients.

Here is a coefficientwise majorant, not just a bound on evaluations. Replace every letter T_i by a common nonnegative variable t , every scalar coefficient by its absolute value, and take maximum block row sums before taking the eight spatial diagonal traces. The series for $K - I$ is majorized by $p(t)$, for $G - I$ by $g(t)$, for L by $l(t)$, and for $H - I$ by

$$h(t) = 2t + 9t^2.$$

Since $\sum_{j \geq 1} \binom{1/2}{j} |z^j| = 1 - \sqrt{1 - z}$, the sum of absolute trace-word coefficients of

$$F_B(T) = \text{Tr } \sqrt{B} - 8 \text{Tr } \sqrt{H}$$

is majorized degree by degree by

$$A(t) = 8[2 - \sqrt{1 - l(t)} - \sqrt{1 - h(t)}]. \quad (5)$$

The eight is the spatial trace multiplicity; there is no hidden factor d , because the color trace is retained inside each word and estimated by (2).

For scalar commuting variables, $GK = HI_8$ exactly, by the rank-one identity

$$(I + vv^*/q_9^2)(HI - vv^*) = HI, \quad H = q_9^2 + v^*v.$$

It follows that the commutative specialization of F_B vanishes identically as a convergent power series. Each homogeneous polynomial therefore has zero commutative polynomial. Apply (2) termwise, which is allowed by the absolute majorant.

For a wholly explicit bound use $t_0 = 1/10$. Direct arithmetic gives

$$\begin{aligned} p(t_0) &= 49/100, \\ g(t_0) &= 8/81, \\ l(t_0) &= 5161/8100 < 1, \\ h(t_0) &= 29/100, \\ A(t_0) &< 5. \end{aligned}$$

If $A(t) = \sum a_q t^q$, all a_q are nonnegative and

$$\begin{aligned} & \sum_{q \geq 4} a_q(q)_4 \kappa^{q-4} \\ & \leq 24t_0^{-4} A(t_0), \quad \kappa \leq 1/100. \end{aligned}$$

Indeed $(q)_4(\kappa/t_0)^{q-4}$ is maximized at $q = 4$: the consecutive-term ratio is at most $1/2$. Combining with (2) gives

$$\begin{aligned} & |\mathrm{Tr} \sqrt{B} - 8 \mathrm{Tr} \sqrt{H}| \\ & \leq 150000 \sum_{i < j} \|C_{ij}\|_F^2. \end{aligned} \tag{6}$$

Restoring r adds r^{-3} . This establishes the needed bosonic quadratic-commutator estimate without a claim that vanishing on commuting matrices alone implies it.

6. Fermionic resolvent cancellation

Exactly,

$$\begin{aligned} D^2 &= I_{16} \otimes H + S, \\ S &= \sum_{i < j} \Gamma_i \Gamma_j \otimes C_{ij}. \end{aligned}$$

The Clifford trace gives

$$\begin{aligned} \mathrm{Tr}[(I \otimes H)^{-1/2} S] &= 0, \\ \|S\|_F^2 &= 16 \sum_{i < j} \|C_{ij}\|_F^2. \end{aligned} \tag{7}$$

For positive matrices A , $A + S \geq \lambda I$, the square-root resolvent identity gives

$$\begin{aligned} & \mathrm{Tr} \sqrt{A + S} - \mathrm{Tr} \sqrt{A} - (1/2) \mathrm{Tr}(A^{-1/2} S) \\ &= -(1/\pi) \int_0^\infty t^{1/2} \\ & \mathrm{Tr}[(A + t)^{-1} S (A + t)^{-1} S (A + S + t)^{-1}] dt. \end{aligned}$$

The absolute value is bounded by

$$\|S\|_F^2 / (8\lambda^{3/2}),$$

using $\int_0^\infty t^{1/2}(\lambda + t)^{-3} dt = \pi/(8\lambda^{3/2})$. Take $A = I \otimes H$ and $\lambda = (0.91r)^2$. Equations (7) prove

$$\begin{aligned}
& |(1/2) \operatorname{Tr} |D| - 8 \operatorname{Tr} \sqrt{H}| \\
& \leq (0.91)^{-3} r^{-3} \sum_{i < j} \|C_{ij}\|_F^2 \\
& < 2r^{-3} \sum_{i < j} \|C_{ij}\|_F^2.
\end{aligned} \tag{8}$$

Combining (6) and (8) proves (1). The first-order fermionic term cancels exactly under the Clifford trace; the bosonic first-order-in-commutators term cancels by (3) and commutative specialization.

7. Endpoint commutators, pair weights, and absorption

In a matrix representation of $\operatorname{Hom}(C^m, C^m)$,

$$\begin{aligned}
T_i &= A_i \otimes I_m - I_n \otimes B_i^T, \\
[T_i, T_j] &= [A_i, A_j] \otimes I_m - I_n \otimes [B_i, B_j]^T.
\end{aligned}$$

There is no mixed term $[A_i, B_j]$: the two endpoint actions commute. Since traces of commutators vanish, exactly

$$\begin{aligned}
& \| [T_i, T_j] \|_F^2 \\
& = m \| [A_i, A_j] \|_F^2 + n \| [B_i, B_j] \|_F^2.
\end{aligned} \tag{9}$$

For a partition with block sizes n_a and separations r_{ab} , summing (1) gives the pairwise-relative statement

$$\begin{aligned}
& \left| \sum_{a < b} E_{0,ab} \right| \\
& \leq C(R/\ell^3) \sum_a F_a^2 \sum_{b \neq a} n_b r_{ab}^{-3}, \\
& F_a^2 = \sum_{i < j} \| [A_i^{(a)}, A_j^{(a)}] \|_F^2, \quad C = 150002.
\end{aligned} \tag{10}$$

The sum of absolute pair defects obeys the same bound. Defining the internal bosonic potential $V_a = RF_a^2/(2\ell^6)$, its multiplier is

$$2C\ell^3 \sum_{b \neq a} n_b r_{ab}^{-3}. \tag{11}$$

This tends to zero once all retained pair separations tend to ∞ . Its constants have no dependence on a ratio involving the smallest unrelated center gap. At fixed finite N , $r_{ab} \geq r_*$ gives the explicit upper bound $2C \ell^3 (N - n_a) r_*^{-3}$.

For clarity about full-form absorption, V_a by itself is not bounded by H_a with coefficient one without a Yukawa remainder. If the elementary finite-Clifford estimate is written as

$$\begin{aligned}\|Y_a(A)\| &\leq c_{n_a}(R/\ell^3)\alpha_a, \\ \alpha_a^2 &= \sum_i \|A_i^{(a)}\|_F^2,\end{aligned}$$

then $V_a \leq H_a + c_{n_a}(R/\ell^3)\alpha_a$, since the internal kinetic form is nonnegative. One explicit choice in these conventions is $c_n = 24(n^2 - 1)$, with $c_1 = 0$. Indeed, on the $16(n^2 - 1)$ real Majoranas, the internal mass is $D_{\text{ad}} = \sum_i \Gamma_i \otimes \text{ad}(A_i)$, and $Y_a = (R/(2\ell^3))\theta^T D_{\text{ad}}\theta$. In a real Hermitian generator basis D_{ad} is purely imaginary antisymmetric. Pairing its $\pm\lambda$ eigenvalues gives $\|Y_a\| = (R/(4\ell^3)) \text{Tr}|D_{\text{ad}}|$, while

$$\|D_{\text{ad}}\| \leq 2 \sum_i \|A_i\|_{\text{op}} \leq 6\alpha_a.$$

This proves the stated c_n without an internal spectral gap. Hence (10) is bounded by an arbitrarily small multiple of $\sum_a H_a$, after taking r_* large, plus

$$2CR \sum_a c_{n_a} \alpha_a \sum_{b \neq a} n_b r_{ab}^{-3}.$$

Under the endpoint condition $\alpha_a \leq \kappa r_{ab}$, this last expression is at most

$$2CR\kappa \sum_{a < b} (c_{n_a} n_b + c_{n_b} n_a) r_{ab}^{-2}.$$

It is an arbitrarily small center-Hardy cost by reducing κ before choosing the final exterior radius. This is a precise way to use the defect; replacing V_a by H_a with no remainder would be incorrect.

For an explicit finite-N allocation, write $T_{\text{cent}} = (R/2)t_{\text{cent}}$ in the mass metric $\sum_a n_a |dz_a|^2$. The nine-dimensional relative Hardy inequality gives

$$\begin{aligned}R \int r_{ab}^{-2} |f|^2 \\ \leq (8/49)(1/n_a + 1/n_b)^{-1} T_{\text{cent}}[f].\end{aligned}$$

The preceding residual multiplier is therefore bounded by

$$\begin{aligned}(16C\kappa/49) \sum_{a < b} \\ (c_{n_a} n_b + c_{n_b} n_a)(1/n_a + 1/n_b)^{-1} T_{\text{cent}}[f] \\ \leq (192C/49) N^4 \kappa T_{\text{cent}}[f].\end{aligned}$$

The last bound uses $c_n = 24(n^2 - 1)$ and $\sum_{a < b} n_a^2 n_b^2 \leq N^4/2$. Thus, for any $\epsilon > 0$, the choices

$$\begin{aligned}\kappa &\leq \min(1/200, 49\epsilon/(192CN^4)), \\ r_* &\geq [2C\ell^3 N/\epsilon]^{1/3}, \\ \alpha_a &\leq \kappa r_{ab} \text{ for every incident pair ,} \\ r_{ab} &\geq r_* \text{ for every pair ,}\end{aligned}$$

give the form bound

$$\begin{aligned}&\int \sum_{a < b} |E_{0,ab}| |f|^2 \\ &\leq \epsilon \sum_a H_a[f] + \epsilon T_{\text{cent}}[f].\end{aligned}\tag{12}$$

Here the standard nonnegative internal supersymmetric forms are used on their common core, and the center Hardy estimate is valid also for Hilbert-valued coefficients and unitary connections. The multiplier of each H_a depends only on the centers, so no derivative of an internal-dependent weight has been omitted. Equation (12) is a bound on this defect multiplier, not a claim that the full Hamiltonian has already been compared with the slow forms on its right.

C Centered sources and graph estimates

Technical derivation for review. Equation and subsection numbers in this appendix are local to the source derivation. Historical local labels are retained, including numbering gaps or nonsequential labels. The accompanying source notes retain the original expressions for comparison.

0. Result and scope

For any fixed partition $(n_1, \dots, n_k)$, the centered coefficient calculation has an exact root-graph resolution:

- Each two-block charge source occupies two quanta of one root pair and has fast energy $4r_{ab}$.
- Each genuine three-block charge source occupies one quantum on every edge of a triangle and has fast energy $2(r_{ab} + r_{bc} + r_{ca})$.
- On residual block-center-neutral states, the leading inverse-square Schur multiplication cancels as an operator on the entire internal Clifford module. Internal derivatives of the adapted vacuum are included in this statement; freezing the internal coordinates is not permission to discard those derivatives.
- The associated finite dressing has pair weight $C_N r_{ab}^{-3}$ and triangle weight $C_N [r_{ab}^{-2} + r_{bc}^{-2} + r_{ca}^{-2}] / (r_{ab} + r_{bc} + r_{ca})$. The latter summed denominator is essential beside a far large block and a close opposite pair.
- A graph-local multiplier lemma turns these weights into a small multiple of the complete slow supersymmetric form plus a small pairwise center-Hardy multiplier, without expanding an inverse around an internal Hamiltonian and without applying physical internal Hardy to colored vectors.

The nonzero internal coefficient continuation is supplied in Appendix *D*; analyticity of the oscillator matrices alone would not supply that continuation.

1. Fibers, neutral sector, and the correct fast inverse

Write $b_{ab} = z_a - z_b$, $r_{ab} = |b_{ab}|$, and $d_{ab} = n_a n_b$. Centers have mass metric $\sum_a n_a |dz_a|^2$. Internal matrices $A^{(a)}$ are traceless Hermitian, with radii

$$\alpha_a^2 = \sum_i \|A_i^{(a)}\|_F^2.$$

The hypotheses used in graph estimates are genuinely pairwise:

$$\begin{aligned} \alpha_a &\leq \kappa r_{ab} \quad \text{for every } b \neq a, \\ r_{ab} &\geq r_* > 0. \end{aligned}$$

No comparison of one gap with an unrelated gap is made.

At $A = 0$ the fast vacuum is a product $U_0 = \otimes_{a < b} U_{ab}$. Each pair has $16d_{ab}$ real transverse bosonic coordinates and $16d_{ab}$ complex fermionic modes, with $8d_{ab}$ occupied modes. The

reverse block is the adjoint, not a second independent pair. The vacuum is invariant under the residual $S(\prod U(n_a))$ group and is even in boson parity and fast fermion parity. For any one normalized real supercharge let q_0 be its centered fast part. Before imposing the block-center Gauss constraints, its square contains the center-weighted gauge term:

$$q_0^2 = H_f + \sum_i (\Gamma_i)_{\alpha\alpha} \sum_a z_a^i G_{a,\text{center,fast}},$$

up to the common gauge-generator normalization. The background $z_a I_{n_a}$ annihilates all traceless internal generators in this term. Consequently

$$q_0^2 = H_f$$

on the residual block-center-neutral fast space. This includes fast states carrying non-trivial internal $SU(n_a)$ representations. It does not assert the identity on arbitrary block-center-charged vectors. Physical total residual equivariance implies this center neutrality because traceless internal coordinates and internal adjoint fermions carry no block-center charge.

Every source below is center neutral: a pair contracts opposite root charges, while a triangle follows an oriented root cycle. On this space $P = U_0 U_0^*$ is the kernel of H_f and of q_0 , and

$$q_0^{-1} = q_0 H_f^{-1}$$

is a genuine inverse on $1-P$. This is not the inverse of an interacting internal Hamiltonian.

2. The internal vacuum jet cannot be omitted

Let $U(z, A)$ be the normalized adapted product vacuum from the gapped quadratic matrices in Appendix B. At one pair, rotate $n = b/r$ to direction 9 and write

$$\begin{aligned} T_i &= A_i^{(a)} \otimes I - I \otimes (A_i^{(b)})^T, \\ F &= rI + T_9, \\ D &= \Gamma_9 r + \sum_i \Gamma_i T_i. \end{aligned}$$

At $A = 0$, the bosonic kinetic correction $G - I$ has no linear term and the bosonic frequency has first variation $\delta\Omega_{ij} = \delta_{ij}\delta T_9$. The fermionic negative projector has first variation

$$\delta P_- = -(1/(2r)) \sum_{i \perp n} \Gamma_i \otimes \delta T_i.$$

In a local unitary frame with zero scalar Berry connection at this point, the vacuum derivative is orthogonal to U_0 and obeys the exact norm identity

$$\|\delta U_{ab}\|^2 = (2/r_{ab}^2) \sum_i \text{Tr}_{d_{ab}}(\delta Q_i^2), \quad (2.1)$$

where $\delta Q_i = \delta b_i I + \delta T_i$. The longitudinal contribution is the Gaussian variation; the transverse contributions are the occupied-determinant variation. Formula (2.1) is a norm in the fast fiber, leaving the entire slow Clifford module untouched.

For example, summing over all canonical center and endpoint-internal coordinate directions gives

$$\begin{aligned} \sum_{\mu} \|\partial_{\mu} U_{ab}\|^2 \\ = 18n_a n_b (n_a + n_b) / r_{ab}^2. \end{aligned} \quad (2.2)$$

Indeed the center contribution is $18(n_a + n_b)/r^2$, while the internal contributions are $18[n_b(n_a^2 - 1) + n_a(n_b^2 - 1)]/r^2$. Thus the adapted vacuum has nonzero internal derivatives even at $A = 0$.

2.1 Zero first jet of the frozen-charge defect

Let $q_f(z, A)$ be the frozen linear fast charge, with the exact frozen eliminated-Gauss kinetic coefficient. It need not annihilate the adapted vacuum at nonzero internal commutators. Nevertheless

$$\begin{aligned} R(z, A) &= q_f(z, A)U(z, A), \\ R(z, 0) &= 0, \quad \partial_A R(z, 0) = 0. \end{aligned} \quad (2.3)$$

Here is the algebra behind the first derivative. In the frozen superalgebra the discrepancy between q_f^2 and the quadratic Hamiltonian consists of the residual internal gauge term and background-commutator terms. The latter are quadratic in A . The derivative of the former annihilates U_0 , since U_0 is internal-gauge invariant. Differentiating the adapted ground equation and using $E'_0(0) = 0$ therefore gives

$$q_0[q'_f(0)U_0 + q_0U'(0)] = 0.$$

The vector in brackets is center neutral and odd in boson parity: a linear fast charge maps an even Gaussian to an odd Gaussian-polynomial vector. The kernel of q_0 is the even vacuum line, so the bracket is zero. This parity step is required; the differentiated square alone does not rule out a vacuum component.

There is also a direct commuting-axis proof. Vary one canonical internal coordinate $A_i^{(a)} = tT$, with all other internal matrices zero. This whole one-parameter background is commuting. A unitary diagonalizing the single Hermitian T splits the adapted fast system

into scalar color-pair oscillators at shifted relative vectors. The kinetic-aware Gaussian is the scalar transverse vacuum written in the chosen oblique slice, and the filled fermion vacuum is neutral for each color-pair phase. The frozen charge annihilates this product exactly. Thus every canonical internal partial derivative of R at zero vanishes, and linearity gives the full first jet. This argument is only along commuting coordinate axes; it does not incorrectly extend annihilation to noncommuting internal tuples.

Center derivatives of R vanish as well because $R(z, 0) = 0$ identically in z . Equation (2.3), rather than a false assertion that the adapted fast charge has an exact kernel for all A , is what the centered homological calculation uses.

3. Exact centered coefficient identity, including internal derivatives

Under the simultaneous coefficient dilation

$$z = L\zeta, \quad A = La, \quad Y = L^{-1/2}\xi,$$

the exact reduced charge has the formal homogeneous inventory

$$\begin{aligned} q(L) &= L^2 Q_I^V(a) + L^{1/2} q_0(\zeta, a) \\ &+ L^{-1} q_1 + L^{-5/2} q_2 + L^{-4} q_3 + \dots \end{aligned} \tag{3.1}$$

This is a coefficient expansion near the zero section, not a claim that the internal interacting charge is bounded or small. In an intrinsic multiblock chart the inverse Gauss matrix can create further terms; one must not copy the finite three-term axial expansion without checking this. All charge coefficients are first order in slow momenta. Since $Q_I^V(a)$ and its first derivative vanish at $a = 0$, its anticommutators with q_2 and q_3 vanish there. Thus the relevant centered Hamiltonian coefficients are exactly

$$\begin{aligned} H_1 &= \{q_0, q_1\}, \\ H_2 &= q_1^2 + \{q_0, q_2\}, \end{aligned} \tag{3.2}$$

when evaluated at the centered internal jet. Terms from the quadratic density of the intrinsic Gauss determinant belong to this inventory. The fact that its linear-in- Y trace vanishes does not permit omitting its quadratic triangle terms.

Put

$$\begin{aligned} D_s &= U^* q_1 U, \\ B &= (1 - P) q_1 U. \end{aligned}$$

The derivative of a coefficient section f in $q_1(Uf)$ lies in the vacuum line. Therefore B is a multiplication map, but it includes the internal derivatives in (2.1):

$$q_1(Uf) = U D_s f + B f.$$

Equation (2.3) gives the centered off-diagonal source and diagonal compression

$$\begin{aligned}(1 - P)H_1U &= q_0B, \\ U^*H_2U &= D_s^2 + B^*B.\end{aligned}\tag{3.3}$$

These are differential-operator/form identities on the common smooth coefficient core. There is no projection onto internal bound states or restriction of the spectator Clifford module.

The exact neutral fast inverse then gives

$$B^*q_0H_f^{-1}q_0B = B^*B.\tag{3.4}$$

Consequently the centered effective coefficient is D_s^2 , and the entire additional multiplication coefficient cancels. It is important to identify D_s with its first jet before replacing D_s^2 by a free slow Laplacian.

3.1 Compression and its first jet

The following checks supply that last identification at $A = 0$.

1. The scalar Berry curvature is zero at $A = 0$, including pure internal and mixed center/internal components. A local gauge therefore has zero connection and zero first jet there.
2. The expectation of the quadratic-Y commutator charge is zero to first order in A . The centered covariance and its first variation are spatial-diagonal, so contraction with Γ_{ij} vanishes.
3. The triangle orbital term and the fast-fast spin-Gauss term have an odd number of fast fermions and have zero vacuum compression. The same remains true after one internal derivative.
4. The δ part of the mixed slow-fast fermion covariance cancels the half-density derivative. The remaining sign-covariance term is, up to the fixed charge sign and factors of i , a linear combination of

$$\Gamma_n \operatorname{Im} \operatorname{Tr}_{\text{color}}[T_c F^{-1} \operatorname{sign}(D)],\tag{3.5}$$

where T_c is the Hermitian bifundamental action of a slow color generator. The trace leaves the spin matrix untraced. At the centered point it vanishes. Its first variation is a linear combination of

$$\begin{aligned}r^{-2} \operatorname{Im} \operatorname{Tr}(T_c \delta T_n) \Gamma_n, \\ r^{-2} \operatorname{Im} \operatorname{Tr}(T_c \delta T_i) \Gamma_i.\end{aligned}$$

Both are zero because the trace of a product of two Hermitian matrices is real. This is why the compression must be checked as a jet, not only by its value at $A = 0$.

In a realification of the complex pair space, the covariance is $I/2$ plus the complex structure times the realification of $\text{sign}(D)$, multiplied by $i/2$. The contracted mixed Gauss coefficient is the realification of $F^{-1}T_c$. Its δ trace gives the density derivative, and its complex-structure trace is exactly the imaginary color trace in (3.5). This verifies the color mechanism without assuming that F commutes with the internal generators.

It follows that D_s has the centered free slow Dirac value and first jet, so the coefficient in (3.3)-(3.4) is the full free center-plus-internal kinetic operator. At nonzero internal coordinates, the complete internal potential charge must of course be retained; (3.1) never authorizes discarding it from an all-vector estimate.

4. Root graph grading and exact virtual denominators

Use a root edge $e = \{a, b\}$ and a triangle $t = \{a, b, c\}$. For a root oscillator let N_e count bosonic excitations plus fermionic particles and holes relative to its vacuum. At the centered point

$$H_f = 2 \sum_e r_e N_e.$$

The leading complementary charge map has the decomposition

$$B = \sum_e B_e + \sum_t B_t. \quad (4.1)$$

Every B_e has $N_e = 2$ and every other $N_f = 0$. Its terms are the Gaussian and determinant derivatives, the two-fast-variable commutator with slow fermion output, and the complementary mixed spin-Gauss contraction. Removing P removes their only possible zero-quantum part.

Every B_t has one excitation on each of its three edges. Its possible charge monomials are exactly of these types, with color contractions along an oriented cycle:

$$\begin{aligned} & Y_e Y_f \theta_g, \\ & r_g^{-1} Y_e p_f \theta_g, \\ & r_h^{-1} \theta_e \theta_f \theta_g. \end{aligned} \quad (4.2)$$

Here e, f, g are the three distinct triangle edges, while h is one of them. A fast Majorana applied once to the filled vacuum creates one particle or hole. Different edges have independent Gaussian and Fock factors, so no within-edge vacuum contraction occurs in a triangle monomial.

Thus

$$\begin{aligned} H_f B_e &= 4r_e B_e, \\ H_f B_t &= 2s_t B_t, \quad s_t = \sum_{e \in t} r_e. \end{aligned} \quad (4.3)$$

Distinct graph summands are orthogonal as fast-space maps. Pair sources are nonvacuum on only their own edge. Triangle sources are odd in combined excitation parity on exactly their three edges. The centered charge preserves each combined excitation parity and annihilates vacuum factors, so the orthogonality survives application of q_0 and H_f^{-1} .

The finite dressing has the explicit graph resolution

$$\begin{aligned} Z_e &= -(4r_e)^{-1} q_0 B_e, \\ Z_t &= -(2s_t)^{-1} q_0 B_t. \end{aligned} \tag{4.4}$$

In particular, as positive operators on the full slow Clifford module,

$$\begin{aligned} Z_e^* Z_e &= (4r_e)^{-1} B_e^* B_e, \\ Z_t^* Z_t &= (2s_t)^{-1} B_t^* B_t. \end{aligned} \tag{4.5}$$

Replacing the second denominator by one arbitrarily selected root frequency is incorrect. Replacing every denominator by the unrelated global minimum throws away exactly the information needed for large far-away blocks.

5. Explicit coefficient seminorms and shape-independent graph estimates

All constants below are finite at fixed N . They can be calculated from the normalized Lie structure constants, γ matrices, and the displayed source monomials; no compact global gap-ratio chart occurs in their definition.

One precise conservative convention is the following. Write each pair source as a sum of elementary maps after factoring $1/r_e$; evaluate the elementary Hermite/Fock monomials on unit-frequency normalized vacua and include their operator norms and absolute scalar coefficients in a sum K_e . Include the derivative contribution using (2.1). For a triangle, group the monomials (4.2) into the three types, factoring respectively

$$\begin{aligned} &(r_e r_f)^{-1/2}, \\ &r_g^{-1} (r_f / r_e)^{1/2}, \\ &r_h^{-1}. \end{aligned}$$

Let K_t be the sum of their unit-frequency Hermite/Fock norms times absolute coefficients. Γ matrices have norm one, a real Majorana has norm $1/\sqrt{2}$, and degree-two normalized Gaussian monomials have norms at most $\sqrt{3}/2$. The remaining coefficients are normalized color brackets and unit spatial-direction contractions. This is a finite, explicit coefficient seminorm, not an unspecified supremum over gap ratios. For example (2.2) and Cauchy-Schwarz give a derivative-source contribution no larger than

$$K_{e,\text{derivative}}^2 \leq 162N^2 n_a n_b (n_a + n_b).$$

Enlarge K_t , if necessary, by a factor two to account for the following weight reduction. Triangle geometry gives $r_f \leq r_e + r_g$, and hence

$$\begin{aligned} r_g^{-1} \sqrt{r_f/r_e} &\leq r_g^{-1} + (r_e r_g)^{-1/2} \\ &\leq (3/2)r_g^{-1} + (1/2)r_e^{-1}. \end{aligned}$$

Also $(r_e r_f)^{-1/2} \leq (r_e^{-1} + r_f^{-1})/2$. Therefore

$$\begin{aligned} \|B_e\| &\leq K_e/r_e, \\ \|B_t\| &\leq K_t \sum_{e \in t} r_e^{-1}. \end{aligned} \tag{5.1}$$

Set $w_t = \sum_{e \in t} r_e^{-2}$. Equations (4.5) and (5.1) imply

$$\begin{aligned} \|Z_e\|^2 &\leq K_e^2/(4r_e^3), \\ \|Z_t\|^2 &\leq 3K_t^2 w_t/(2s_t). \end{aligned} \tag{5.2}$$

The centered map by itself has only center derivatives; an internal derivative requires an extension. For the following internal derivative bounds, choose smooth gapped pairwise Gaussian/Fock transport of the centered graph coefficient from $A = 0$, for example transport along internal radial rays. This gives a concrete graph-factorized extension, but it is not asserted to equal the actual nonzero-internal homological source. The center derivative statement is unconditional. For these derivatives, the same graph coefficients and one differentiated vacuum factor give, with another computable coefficient seminorm K'_g ,

$$\begin{aligned} \|\nabla Z_e\|^2 &\leq K_e'^2 r_e^{-5}, \\ \|\nabla Z_t\|^2 &\leq K_t'^2 \left(\sum_{e \in t} r_e^{-1} \right)^2 w_t/s_t. \end{aligned} \tag{5.3}$$

In (5.3), ∇ differentiates the active graph coefficient and its active oscillator factors. The full map also contains a vacuum factor on every spectator edge. Its full derivative has the corrected bound

$$\begin{aligned} \|\nabla Z_g\|^2 &\leq C_N[d_{g,\text{active}} + m_g W_2], \\ W_p &= \sum_e r_e^{-p}, \end{aligned} \tag{5.4}$$

where m_g is the right side of (5.2). The extra term is necessary when a spectator edge is much shorter. For adapted vacua it includes internal derivatives of spectator factors as well as center derivatives. The active triangle right side in (5.3) is at most a fixed numerical multiple of $K_t'^2 \sum_{e \in t} r_e^{-5}$, by Holder and $s_t \geq \max r_e$. Moreover

$$W_3 W_2 \leq (\#\text{edges}) W_5.$$

Thus the full derivatives still have a summed $C_N W_5$ majorant. Graph-resolved weights must nevertheless be kept when an internal Yukawa multiplier is to be paid.

5.1 Perturbed finite source spaces

The Gaussian/Fock statement also has a useful conditional continuation. On the adapted pairwise-relative tube, unitary/symplectic transport identifies the finite degree-at-most-three source spaces with their centered spaces. Frequencies have fixed positive bounds $cr_e \leq \omega_{e,j} \leq Cr_e$. A source odd in each triangle edge therefore has complementary excitation energy at least cs_t , even when modes within one pair split or coincide.

If the exact leading Hamiltonian graph source $S_g(A)$ has the coefficient bounds

$$\begin{aligned} \|S_e(A)\|^2 &\leq C_N/r_e, \\ \|S_t(A)\|^2 &\leq C_N s_t w_t, \end{aligned}$$

then the inverse of the shifted excitation operator

$$H_{\text{exc}} = H_{\text{fast}} - E_{0,\text{total}}$$

on the relevant excited sector satisfies precisely the graph dressing bounds (5.2), with enlarged constants. The vacuum-energy defect is handled separately by the general frozen-energy lemma. An unshifted inverse of the full fast Hamiltonian need not have these denominators, because unrelated pair vacuum-energy defects would enter it. This is solely the finite-source fast excitation inverse. No internal momentum appears in its denominator.

To deduce a small adapted inverse-square remainder from the centered identity, one additionally needs the exact diagonal H_2 inventory, including its density, ordering, and internal-potential cross terms. Under coefficientwise perturbation bounds of order κ , the finite resolvent identity gives a defect at most $C_N \kappa W_2$. Establishing all those hypotheses from the full reduced Hamiltonian is separate from the graph calculation.

6. A graph-local complete slow form lemma

This is an all-vector statement, not a finite-Gaussian approximation to an internal wavefunction.

Let

$$\begin{aligned} H_s &= T_{\text{cent}} + \sum_a H_a, \\ H_a &= T_a + V_a + Y_a, \\ V_a &\geq 0, \quad \|Y_a(A)\| \leq c_{n_a} \alpha_a. \end{aligned}$$

The averaged sixteen-charge identity makes H_a nonnegative also on colored internal vectors. The ordinary physical lower-rank Hardy estimate is not being invoked. For a graph-local multiplication map Z_g , suppose

- it is an active-graph coefficient, depending on internal coordinates at graph vertices only, tensored with the vacua on spectator edges;
- it acts on internal Clifford factors only at these vertices;
- it is even in total fermion parity, as is $q_0^{-1}B_g$;
- $\|Z_g\|^2 \leq m_g(z)$ and the full derivative obeys the center-only majorant d_g in (5.4).

For a block a outside g , its internal potential charge commutes exactly with the map. The complete internal charge satisfies

$$Q_{a,\alpha}Z_g = Z_gQ_{a,\alpha} + C_{a,\alpha,g},$$

where $C_{a,\alpha,g}$ is bounded multiplication coming only from derivatives of spectator vacua. Its squared norm is bounded by the corresponding derivative majorant, with the finite Clifford contraction factor. Averaging the exact charge squares and applying Young gives

$$\begin{aligned} H_a[Z_g f] &\leq 2 \int m_g(z) [H_a\text{-density of } f] \\ &+ C_N \int d_{a,g}(z) |f|^2. \end{aligned} \tag{6.1}$$

In particular no unrelated $\alpha_a m_g$ loss appears. It would be incorrect to assert exact commutation while differentiating a moving spectator vacuum. For a graph vertex use the elementary uncompressed estimate, with no internal spectral assumptions,

$$\begin{aligned} H_a[Z_g f] &\leq 2 \int m_g(z) [H_a\text{-density of } f] \\ &+ 3c_{n_a} \int \alpha_a m_g(z) |f|^2 \\ &+ 2 \int |\partial_{Aa} Z_g|^2 |f|^2. \end{aligned} \tag{6.2}$$

The notation for a weighted internal density is harmless here because its weight depends only on centers; it is the nonnegative internal form on every fixed-center fiber. The center kinetic estimate has the same derivative-product bound with no Yukawa term.

For pair and triangle dressings from (5.2),

$$\sup m_g \leq C_N r_*^{-3}.$$

At a pair endpoint, $\alpha_a \leq \kappa r_e$ gives

$$\alpha_a m_e \leq C_N \kappa r_e^{-2}.$$

At any triangle vertex, $\alpha_a \leq \kappa \min(\text{incident } r_e) \leq \kappa s_t$, and hence

$$\alpha_a m_t \leq C_N \kappa w_t. \quad (6.3)$$

This remains true when the opposite edge is arbitrarily shorter than the two incident edges. The factor s_t from the virtual denominator is doing essential work.

Combining (6.1)-(6.3), then using positivity of H_s to sum the finitely many graph maps, proves

$$\begin{aligned} & H_s \left[\sum_g Z_g f \right] \\ & \leq C_N r_*^{-3} H_s[f] \\ & + C_N \kappa \int W_2 |f|^2 \\ & + C_N \int W_5 |f|^2. \end{aligned} \quad (6.4)$$

The constants can be chosen from the finite sums of K_g^2 , $K_g'^2$, vertex counts, and c_{n_a} . For instance a direct Cauchy-Schwarz sum multiplies their sum by the number of graphs $\binom{k}{2} + \binom{k}{3}$. This gives an explicit finite-N recipe, without a global ratio parameter.

The full derivative terms in this argument include spectator factors; their product weight is controlled by $W_3 W_2 \leq (\#\text{edges}) W_5$. The same argument applies to a genuinely adapted dressing if its active-graph coefficients and spectator factors satisfy the hypotheses. In particular, (6.4) retains arbitrary internal momenta. No absolute-value inverse expansion of $H_f + H_I$ has been used. It also explains why the coarse estimate $\|Z\|^2 \leq C W_3$ should not first be multiplied by $\sum \alpha_a$: that would incorrectly charge a distant large block against an unrelated short root.

6.1 Direct center-Hardy payment

For each pair, in the mass center metric,

$$\begin{aligned} & \int r_{ab}^{-2} |f|^2 \\ & \leq (4/49) (1/n_a + 1/n_b)^{-1} T_{\text{cent}}[f]. \end{aligned} \quad (6.5)$$

The inequality holds for Hilbert-valued coefficients and unitary connections. Also $W_5 \leq r_*^{-3} W_2$. Hence the last two terms in (6.4) are bounded by an arbitrarily small center kinetic fraction by first choosing κ small at fixed N and then r_* large. The first term is a small fraction of the entire interacting slow form, not a fixed internal kinetic loss.

D Actual adapted source inventory

Technical derivation for review. Equation and subsection numbers in this appendix are local to the source derivation. Historical local labels are retained, including numbering gaps or nonsequential labels. The accompanying source notes retain the original expressions for comparison.

1. Notation and exact origin

Use the kinetic-one normalization and the intrinsic slice in Appendix A. The only internal thinness hypothesis is

$$\alpha_a + \alpha_b \leq \kappa r_{ab},$$

with fixed sufficiently small κ . Set $r_* = \min r_{ab}$. All finite constants below depend on N but not on a ratio of unrelated gaps.

To avoid overloading center C , use these coefficient maps:

$$\begin{aligned} F &= E^* B_{Z+A}, & C &= P_N B_A, \\ L &= E^* B_Y, & N_Y &= P_N B_Y, \\ I_Y &= P_I B_Y, & C_Y &= P_C B_Y, \\ K &= R^{-1} C_Y^*, & J_2 &= K C_Y. \end{aligned}$$

Thus F is pair-block diagonal, C is pair-block diagonal as a map from gauge parameters into normal fast coordinates, and

$$M = F + L - J_2, \quad W = I + K K^*.$$

The maps L and N_Y between distinct fast root pairs have triangle support. They have no same-pair block. The maps I_Y and C_Y use one root and its endpoints. The center metric is $g_C = I + K^* K$; the internal and fast tangent metrics are exactly identity.

The definitive flattened kinetic form is

$$\begin{aligned} &\|(p_H + i\rho)u\|_{g_S^{-1}}^2 + \|Du\|^2, \\ D &= W^{1/2} M^{-*} [S_f - O^* S g_S^{-1} (p_H + i\rho)], \\ \rho &= d_H \log d, \quad d = \sqrt{\det M}. \end{aligned} \tag{1.1}$$

All products below inherit this ordering. Fast adjoints are taken in the flat fiber measure. The normal connection is retained in center derivatives. Nothing is obtained by substituting a momentum constraint into a squared operator.

2. Exact coefficients needed through H2

Use homogeneous weights

$$(Z, A) \rightarrow L_{\text{scale}}(Z, A), \quad Y \rightarrow L_{\text{scale}}^{-1/2} Y,$$

$$p_s \rightarrow L_{\text{scale}}^{-1} p_s, \quad p_F \rightarrow L_{\text{scale}}^{1/2} p_F.$$

Here s comprises centers and all internal coordinates. Keep the internal potential and internal Yukawa terms in their complete internal Hamiltonians throughout.

Write

$$\begin{aligned} \log d &= \log d_0 + \phi_2 + \text{terms of Y-degree} \geq 3, \\ d_0 &= \sqrt{\det F}, \\ \phi_2 &= -(1/2) \text{tr}(F^{-1} J_2) \\ &\quad - (1/4) \text{tr}(F^{-1} L F^{-1} L), \\ \rho_0 &= d_s \log d_0, \quad \rho_{F2} = d_F \phi_2, \\ u_s &= p_s + i \rho_0. \end{aligned} \tag{2.1}$$

There is no degree-one term because $\text{tr}(F^{-1} L) = 0$ for every A , not only at $A = 0$. The triangle term in (2.1) must be retained.

The Gauss derivative stack has coefficients

$$D = D_0 + D_1 + D_2 + \text{higher homogeneous orders},$$

of scaling degrees $1/2$, -1 , and $-5/2$, respectively. They are

$$D_0 = -F^{-1} C^* p_F, \tag{2.2}$$

$$\begin{aligned} D_1 &= F^{-1} (S_f - N_Y^* p_F) \\ &\quad + F^{-1} L^* F^{-1} C^* p_F, \end{aligned} \tag{2.3}$$

and

$$\begin{aligned} D_2 &= -F^{-1} [I_Y^* u_I + (C_Y^* + F K) u_C + i C^* \rho_{F2}] \\ &\quad - F^{-1} L^* F^{-1} (S_f - N_Y^* p_F) \\ &\quad - F^{-1} L^* F^{-1} L^* F^{-1} C^* p_F \\ &\quad - F^{-1} J_2^* F^{-1} C^* p_F \\ &\quad - (1/2) K K^* F^{-1} C^* p_F. \end{aligned} \tag{2.4}$$

These follow by expanding

$$\begin{aligned}
M^{-*} &= F^{-1} - F^{-1}L^*F^{-1} \\
&+ F^{-1}L^*F^{-1}L^*F^{-1} + F^{-1}J_2^*F^{-1} + \dots, \\
W^{1/2} &= I + (1/2)KK^* + \dots, \\
O^*Sg_S^{-1}p_H \\
&= C^*p_F + N_Y^*p_F + I_Y^*p_I + (C_Y^* + FK)p_C + \dots
\end{aligned} \tag{2.5}$$

For example L^*Kp_C first enters the next Gauss order, and $g_C^{-1} - I$ first contributes to direct center kinetic energy at scaling degree -5; neither belongs to H2. The FK in (2.4) is essential and becomes C_Y^* at $A = 0$, so the centered center-orbital coefficient is twice C_Y^* .

The exact leading fast Hamiltonian comprises the direct p_F^2 , $D_0^*D_0$, the full adapted quadratic potential, and the fast fermion mass. Its product vacuum is $U(A)$, with energy $E_{0,\text{total}}$. Define

$$H_{\text{exc}} = H_{\text{fast}} - E_{0,\text{total}}.$$

The vacuum-energy multiplier is handled by Appendix B. It is not inserted into a virtual excitation denominator.

The next two homogeneous form coefficients are exactly

$$H_1 = V_3 + Y_Y + D_0^*D_1 + D_1^*D_0, \tag{2.6}$$

$$\begin{aligned}
H_2 &= \|u_s(\cdot)\|^2 + V_4 \\
&+ 2\text{Re}\langle p_F(\cdot), i\rho_{F2}(\cdot) \rangle \\
&+ \|D_1(\cdot)\|^2 + 2\text{Re}\langle D_0(\cdot), D_2(\cdot) \rangle.
\end{aligned} \tag{2.7}$$

Here Y_Y is the Yukawa term linear in Y , including both slow-fast and fast-fast fermion terms. V_3 is the cross term between $[Z+A, Y]$ and $[Y, Y]$; V_4 is the pure quartic fast potential. The internal potential term paired with $[Y, Y]$ is already in the adapted quadratic potential. Equations (2.6)-(2.7) therefore include the terms which appear as anticommutators of the internal potential charge with higher inverse-Gauss charge coefficients; there is no permission to omit them in a charge derivation. This Hamiltonian derivation fixes them directly from the original scalar potential and the exact kinetic form.

3. Actual H1 source and its graph weights

Define the actual leading source

$$S(A) = H_1U(A). \tag{3.1}$$

It is complementary by boson parity. Each summand is a finite Gaussian-polynomial/Fock map on the full slow Clifford module. It has the graph decomposition

$$S = \sum_e S_e + \sum_t S_t,$$

where an edge source has exactly two excitations on that edge ($N_e = 2$) and a triangle source has exactly one excitation on every triangle edge ($N_e = N_f = N_g = 1$). In the pair case these are one bosonic and one fast-fermionic excitation. Graph subspaces remain orthogonal and invariant under the adapted quadratic oscillator. Parity alone would not be the complete finite-sector statement.

The inventory is explicit:

- $Y_Y U$: a pair has $Y_e \theta_e \theta_s$; a triangle has $Y_e \theta_f \theta_g$.
- $V_3 U$: only three distinct triangle edges, one Y on each.
- $D_0^* F^{-1} S_f U$ and its adjoint: pair terms have one p_e and one fast fermion; triangle terms have $p_e \theta_f \theta_g$.
- $D_0^* F^{-1} N_Y^* p_F U$ and its adjoint: a triangle has $p_e Y_f p_g$.
- The last term of D_1 yields the same triangle support with one additional factor $F_f^{-1} C_f^*$.

A slow internal derivative does not occur in H1. Only the pair source contains internal/center Clifford generators; the triangle source acts trivially on the internal Clifford factors.

For a triangle let $s_t = \sum_{e \in t} r_e$, $w_t = \sum_{e \in t} r_e^{-2}$. Pairwise thinness gives

$$\|F_e^{-1}\| \leq C/r_e, \quad \|F_e^{-1} C_e^*\| \leq C\kappa.$$

On a fixed finite Gaussian degree, Y_e costs $C r_e^{-1/2}$, and p_e costs $C r_e^{1/2}$. Thus

$$\begin{aligned} \|S_e\|^2 &\leq C_N / r_e, \\ \|S_t\|^2 &\leq C_N s_t w_t. \end{aligned} \tag{3.2}$$

For the cubic potential, use

$$\begin{aligned} &s_t / (r_e r_f r_g) \\ &= 1/(r_e r_f) + 1/(r_f r_g) + 1/(r_g r_e) \leq w_t. \end{aligned}$$

For the Gauss triangle $p_e Y_f p_g / r_e$, its norm is bounded by

$$\begin{aligned} &C\kappa \sqrt{r_g / (r_e r_f)} \\ &\leq C\kappa \sqrt{s_t} (r_e r_f)^{-1/2}. \end{aligned}$$

The inverse-L correction has one additional bounded factor κ . These are shape-independent estimates, including when one edge is much shorter than the other two.

Transporting only the finite oscillator coefficient spaces to their centered spaces by the actual pair functional calculus gives the coefficientwise perturbation estimates

$$\begin{aligned}\|S_e(A) - S_e(0)\| &\leq C_N \kappa r_e^{-1/2}, \\ \|S_t(A) - S_t(0)\| &\leq C_N \kappa \sqrt{s_t w_t}.\end{aligned}\tag{3.3}$$

This is an estimate for the actual source (3.1), not a definition by transport. Every new Gauss H1 term has an explicit factor $F^{-1}C^*$, and the remaining actual coefficients have relative $O(\kappa)$ variation in the pair matrices and vacuum factors.

4. Compressed H2 and its actual scalar/Clifford multiplier

Separate the complete slow Berry-covariant form

$$H_s^B = T_{\text{cent}}^B + \sum_a (T_a^B + V_a + Y_a)$$

from the vacuum compression of (2.7). The remainder is a multiplication operator $W(A)$ on the entire slow Clifford fiber. The following procedure is an exact specification of W :

1. In $\|u_s(Uf)\|^2$, retain $\|\nabla_s^B f\|^2$; put the Born-Huang multiplier and $|\rho_0|^2 + \text{div } \rho_0$ into W .
2. Put the vacuum expectations of V_4 , the displayed fast-density cross term, and $D_1^* D_1$ into W .
3. In $2 \text{Re}\langle D_0(Uf), D_2(Uf) \rangle$, keep the written ordering. For the slow-orbital part put

$$\begin{aligned}L_\mu &= \text{component}_\mu \{F^{-1}[I_Y^*, C_Y^* + FK]\}, \\ \langle D_0 U, -L_\mu U \rangle &= i j_\mu, \quad j_\mu \text{ real}.\end{aligned}\tag{4.1}$$

This coefficient is purely imaginary: the bosonic Gaussian and the orbital coefficient maps are real, while D_0 contains one fast momentum. The fermionic factor is its norm. Its first-order coefficient on f therefore integrates to the scalar multiplier $-\text{div } j$. All remaining terms, including derivatives on U and $i\rho_0$, are placed in W . A scalar Berry term in $p_\mu U$ has zero real pairing with $i j_\mu$, so this step is gauge covariant.

An equivalent exact scalar formula removes even the implicit derivative bookkeeping. Write $U = g(Y; s)v(s)$, with g real, put

$$\begin{aligned}A_{\text{op}} &= F^{-1}C^*, \quad a = A_{\text{op}} d_F \log g, \\ B_\mu &= \text{component}_\mu \{F^{-1}[I_Y^*, C_Y^* + FK]\}.\end{aligned}$$

Then $D_0 U = iaU$, and the entire slow-orbital compression is

$$\begin{aligned}V_{\text{orb}} &= - \sum_\mu \langle \partial_\mu (a \cdot B_\mu) \rangle_U \\ &\quad - 2 \sum_\mu \rho_{0,\mu} \langle a \cdot B_\mu \rangle_U.\end{aligned}\tag{4.3}$$

The derivative acts on the displayed fast coefficient; Gaussian-density differentiation has already canceled the derivative of the fiber expectation. Fock/Berry derivative terms have zero real part. The direct fast-density term and the ρ_{F2} term in D_2 combine exactly to

$$V_{\text{fast-density}} = \text{tr}_F[(I + CF^{-2}C^*) \text{Hess}_Y \phi_2]. \quad (4.4)$$

Since ϕ_2 is quadratic in Y , its Hessian is a coefficient matrix independent of Y . These formulas follow by one integration by parts and retain all ordering terms.

No slow derivative has been discarded or bounded by a momentum cutoff. In particular differentiating j , or the coefficient in (4.3), includes internal derivatives at $A = 0$. Those terms participate in the centered cancellation.

4.1 Complete root support and coefficient bounds

Every surviving vacuum contraction in W is supported on one edge or the edges of a triangle. The following list accounts for all terms in (2.7):

- Born-Huang derivatives on two different pair vacua are orthogonal after removing the scalar Berry connection. Each remaining term is bounded by C_N/r_e^2 .
- ρ_0 is a sum of pair gradients. Their inner product vanishes for disjoint endpoint sets. A cross product on two incident edges has weight $1/(r_e r_f)$ and belongs to their triangle. The divergence is pairwise.
- Wick contractions in V_4 involve either one repeated root or two roots sharing an endpoint. Their weights are r_e^{-2} or $(r_e r_f)^{-1}$.
- The first term of ϕ_2 is a sum of Y_e^2/r_e^2 contractions. Its second term follows a root-space path $e \rightarrow f \rightarrow e$ and has $Y_g^2/(r_e r_f)$, where e,f,g form a triangle. A fast derivative and the momentum pairing remove the Y_g scale; the resulting weights are again r_e^{-2} or $(r_e r_f)^{-1}$.
- D_1 contains a pair spin term with weight r_e^{-1} , and triangle terms with weights r_e^{-1} or $r_e^{-1} \sqrt{r_g/r_f}$. Squaring on the vacuum preserves graph orthogonality. The triangle inequality reduces the latter square to Cw_t .
- The slow-orbital part of D_2 paired with D_0 gives $j_e = O(\kappa/r_e)$ on endpoint coordinates. Its derivative is $O(r_e^{-2})$ and varies from its centered value by $O(\kappa/r_e^2)$. Derivatives of spectator vacua are orthogonal in this compression; the products with ρ_0 have pair/triangle support.
- In $D_0^* F^{-1} L^* F^{-1} S_f$, the two bosonic factors lie on distinct roots, so its vacuum compression is zero.
- In $D_0^* F^{-1} L^* F^{-1} N_Y^* p_F$, nonzero Wick contractions force the second root path to close the same triangle. The two possible pairings have weights $\kappa/(r_f r_g)$ and $\kappa/(r_e r_f)$.
- In the two-L term of D_2 the same closure gives the preceding weights with κ squared.
- In the J_2 and KK^* terms, the two orbital center maps vanish against each other for disjoint endpoint sets. Surviving weights are $\kappa^2/(r_e r_f)$, with the repeated-edge case included.
- The ρ_{F2} part of D_2 has the density supports already listed and a prefactor bounded by κ squared after pairing with D_0 .

For estimates on finite excited states, a root path need not close under Wick contraction. It still has a uniform inverse-square coefficient bound. For example a two-L path has

normalized weight

$$\sqrt{r_h}/(\sqrt{r_e}r_f\sqrt{r_g}r_j).$$

The triangle relation $r_h \leq r_f + r_j$ bounds this by a sum of

$$\begin{aligned} 1/\sqrt{r_e r_f r_g r_j}, \\ 1/(\sqrt{r_e}r_f\sqrt{r_g}), \end{aligned}$$

both bounded by a numerical multiple of the sum of inverse-square edge weights by weighted arithmetic-geometric mean. This observation is useful for mixed source terms and prevents an illicit global-frequency comparison.

All of these are finite coefficient expressions in pair matrices with external gaps comparable to their own r_e . Normalize each occurrence of an internal matrix by the incident r_e in that factor. Its norm is $O(\kappa)$. Functional calculus and its required derivatives have uniform convergent resolvent bounds on that fixed pair ball. Termwise product estimates in the above inventory therefore prove

$$\begin{aligned} \|W(A)\| &\leq C_N W_2, \\ \|W(A) - W(0)\| &\leq C_N \kappa W_2, \\ W_2 &= \sum_e r_e^{-2}. \end{aligned} \tag{4.2}$$

The derivative of j is included in the second estimate. It is not estimated merely by the small value of j , which would miss a nonzero centered internal derivative.

5. Actual shifted-fast correction and leading operator defect

On the pair source space the adapted excitation energy is at least cr_e ; on the triangle source space it is at least cs_t . The finite matrix resolvent identity, applied to the actual sources (3.1), gives

$$\begin{aligned} \|S_e^* H_{\text{exc}}^{-1} S_e - S_e(0)^* H_f(0)^{-1} S_e(0)\| \\ \leq C_N \kappa r_e^{-2}, \\ \|S_t^* H_{\text{exc}}^{-1} S_t - S_t(0)^* H_f(0)^{-1} S_t(0)\| \\ \leq C_N \kappa w_t. \end{aligned} \tag{5.1}$$

Cross terms between distinct graph spaces vanish. The centered source lemma, including all internal vacuum derivatives and compression jets, gives

$$W(0) = S(0)^* H_f(0)^{-1} S(0).$$

Consequently the actual leading operator defect satisfies

$$\begin{aligned} & \|W(A) - S(A)^* H_{\text{exc}}(A)^{-1} S(A)\| \\ & \leq C_N \kappa W_2. \end{aligned} \tag{5.2}$$

This is an operator-norm estimate on the complete internal Clifford module. Its actual fast dressing is

$$Z(A) = -H_{\text{exc}}(A)^{-1} S(A), \tag{5.3}$$

and obeys the root-graph bounds and spectator-derivative correction from Appendix C. No interacting internal Hamiltonian occurs in (5.3). The fast energy E0 remains a separate multiplier with its proved commutator-square payment.

6. All-vector use and the remaining reserve requirement

The graph-local lemma now applies to the actual dressing (5.3): its coefficients are the explicit graph-local functions above. Smooth pair functional calculus gives active derivative weights $C r_e^{-5}$ or $C(\sum_t r_e^{-1})^2 w_t/s_t$; spectator derivatives add $C m_g W_2$. Therefore

$$\begin{aligned} H_s[Zf] & \leq C_N r_*^{-3} H_s[f] \\ & + C_N \kappa W_2[f] + C_N W_5[f]. \end{aligned} \tag{6.1}$$

In fact triangle sources have no internal Clifford generator, so outside derivative commutators are the only issue for their internal potential charges. The more general graph bound is valid without using this improvement. All internal momenta in (6.1) are unrestricted.

The identity $H_{\text{exc}}Z + S = 0$ cancels the actual leading complementary source exactly. The complete internal/center form pairs involving Z can be controlled by Cauchy-Schwarz in H_s , using (6.1). The remaining derivative source from U is incident-edge weighted and has the same payment as Section 6 of the geometric lemma. These steps are valid form estimates on arbitrary complementary vectors in the corresponding control norm; they are not a finite oscillator truncation of those vectors.

What is not supplied here is an inequality extracting that control norm from the original complementary form while retaining the same Gauss square. In particular, the expression

$$H_s[v] + H_{\text{exc}}[v] + \|D_{\text{full}}v\|^2$$

cannot be added to the right side of the original Hamiltonian merely because its three terms are nonnegative. H_{exc} already uses the frozen piece of D_{full} . The desired actual all-vector comparison still needs a non-double-counted reserve, compatible with unrestricted slow momenta, and a complete mixed/remainder estimate in that reserve. Finite coefficient cancellation (5.2) cannot prove this coercivity.

Thus Sections 2-5 identify and estimate the actual leading H1/H2 source coefficients. Section 6 supplies their internal-momentum payment and states the precise remaining all-vector obstruction rather than silently assuming it away.

E Rootwise all vector connection estimates

Technical derivation for review. Equation and subsection numbers in this appendix are local to the source derivation. Historical local labels are retained, including numbering gaps or nonsequential labels. The accompanying source notes retain the original expressions for comparison.

1. Assumptions and the excitation energy

Fix the rank and a partition. For every root/pair $e = (a, b)$, let $r_e = |z_a - z_b|$ and assume $\alpha_a + \alpha_b \leq \kappa r_e$, with κ sufficiently small. Write $r_* = \min_e r_e$ and $W_2 = \sum_e r_e^{-2}$. No ratio $r_{\max}/r_*$ is restricted.

Freeze the slow variables. Let

$$\begin{aligned} H_{\text{fast},e} &= H_{B,e} + H_{F,e}, \\ E_{0,e} &= E_{B,e} + E_{F,\min,e}, \\ h_e &= H_{\text{fast},e} - E_{0,e} \\ &= (H_{B,e} - E_{B,e}) + (H_{F,e} - E_{F,\min,e}). \end{aligned}$$

Here $H_{B,e}$ is the exact kinetic-aware bosonic oscillator. The uniform pair-matrix bounds give

$$\begin{aligned} c(p_e^2 + r_e^2 Y_e^2) &\leq H_{B,e} \leq C(p_e^2 + r_e^2 Y_e^2), \\ E_{B,e} &\leq C r_e, \\ h_e &\geq c r_e Q_e, \end{aligned}$$

where P_e is its full boson/fermion vacuum projection and $Q_e = 1 - P_e$. In particular

$$\begin{aligned} \|p_e Q_e v\|^2 &\leq C h_e[v], \\ \|Y_e v\|^2 &\leq C r_e^{-2} h_e[v] + C r_e^{-1} \|v\|^2, \\ \|p_e P_e v\|^2 &\leq C r_e \|P_e v\|^2, \\ \|Y_e p_e P_e v\|^2 &\leq C \|P_e v\|^2. \end{aligned} \tag{1}$$

The last two statements also hold with any uniformly bounded finite color/spatial matrices between the displayed operators. Constants depend only on rank and the fixed small pair-matrix neighborhood. h_e , rather than $H_{B,e}$ or a centered oscillator, is essential.

Let χ be any multiplication cutoff with $|\chi| \leq 1$ supported in $|Y| \leq \sigma r_*$. Derivatives of χ are not used below. Every estimate is a fast-fiber estimate for arbitrary vectors, with arbitrary spectator fast modes and arbitrary slow/Clifford coefficient vectors. It can be integrated over the slow variables. No internal momentum or oscillator-degree cutoff is imposed. No individual internal singlet condition is imposed; the estimates remain valid on the total residual-equivariant space.

2. Same-root and mixed-root lemmas

For any pair e and any positive denominator scale r_g drawn from the retained roots,

$$\begin{aligned} & \|\chi Y_e p_e v / r_g\|^2 \\ & \leq C\sigma^2(r_*^2/r_g^2)h_e[v] \\ & + Cr_g^{-2}\|v\|^2. \end{aligned} \tag{2}$$

Proof: split the INPUT as $P_e v + Q_e v$. On Q_e use $|\chi Y_e| \leq \sigma r_*$ and the first inequality in (1). On P_e use the last inequality in (1), dropping χ . The factor two from the split is harmless. $P_e v$ need not be supported in the tube. No argument asserts that P_e preserves tube support.

For distinct roots e and f ,

$$\begin{aligned} & \|\chi Y_f p_e v / r_g\|^2 \\ & \leq C\sigma^2(r_*^2/r_g^2)h_e[v] \\ & + Cr_e/(r_g^2 r_f^2)h_f[v] \\ & + Cr_e/(r_g^2 r_f)\|v\|^2. \end{aligned} \tag{3}$$

Indeed the Q_e input is as above. On the P_e input, drop χ , commute Y_f with P_e , and use $\|p_e P_e\|^2 \leq Cr_e$ followed by the Y_f bound in (1). Equivalently split the remaining f input into P_f and Q_f . This is the step that cannot be omitted from a mixed-root claim.

If e, f, g are the three edges of a triangle, $r_e \leq r_f + r_g$, so

$$\begin{aligned} r_e/(r_g^2 r_f^2) & \leq 2r_*^{-3}, \\ r_e/(r_g^2 r_f) & \leq r_g^{-2} + (r_g r_f)^{-1} \leq CW_{2,t}. \end{aligned}$$

Consequently

$$\begin{aligned} & \|\chi Y_f p_e v / r_g\|^2 \\ & \leq C(\sigma^2 + r_*^{-3})(h_e[v] + h_f[v]) \\ & + CW_{2,t}\|v\|^2. \end{aligned} \tag{4}$$

For the input-root denominator $g = e$, the same conclusion follows directly, without any triangle hypothesis, with residual $(r_e r_f)^{-1}$. In (2)-(4), Y and p can be contracted with any bounded root-preserving finite matrices. Finite sums of such sources obey the same bounds after enlarging the rank-dependent constant.

A triangle hypothesis (or another restriction on r_e relative to r_f, r_g) is genuinely necessary to replace (3) by (4): the $P_e P_f$ input has squared size comparable to $r_e/(r_g^2 r_f)$. An arbitrary unrelated r_e cannot disappear from this expression.

3. Exact inverse-M factorization for the fast column

Use the notation of the intrinsic geometry and adapted source inventory:

$$\begin{aligned}
M &= F + P, \quad P = L - J_2, \quad J_2 = KC_Y = R^{-1}C_Y^*C_Y, \\
W &= I + KK^*, \quad K = R^{-1}C_Y^*, \\
T &= (I + F^{-1}P^*)^{-1}, \\
M^{-*} &= TF^{-1}, \\
d_0(v) &= F^{-1}C^*p_Fv.
\end{aligned}$$

F is self-adjoint pair-block diagonal; $\|F_e^{-1}C_e^*\| \leq C\kappa$. On the tube,

$$\begin{aligned}
\|PF^{-1}\| + \|F^{-1}P^*\| &\leq C\sigma, \\
\|T\| + \|W^{1/2}\| &\leq C, \\
\|K\| &\leq C\sigma.
\end{aligned}$$

Let $Bp_F = W^{1/2}M^{-*}(C^* + N_Y^*)p_F$ be the exact fast column of the Gauss stack. Algebraically, with the indicated operator ordering,

$$\begin{aligned}
&Bp_F - d_0 \\
&= (W^{1/2} - I)d_0 \\
&\quad + W^{1/2}T[\\
&\quad F^{-1}N_Y^*p_F \\
&\quad - F^{-1}L^*d_0 \\
&\quad + F^{-1}J_2^*d_0].
\end{aligned} \tag{5}$$

There are no derivatives of inverse matrices in this first-order form identity.

The first two bracketed expressions are finite triangle sources Y_{fp_e/r_g} , with respectively bounded and $O(\kappa)$ root-preserving factors. The last expression factors as

$$\begin{aligned}
&F^{-1}J_2^*d_0 \\
&= F^{-1}C_Y^*[C_YR^{-1}d_0],
\end{aligned} \tag{6}$$

where $\|F^{-1}C_Y^*\| \leq C\sigma$ and $C_YR^{-1}d_0$ is a sum of same-root terms Y_{ep_e/r_e} with $O(\kappa)$ coefficients. Also

$$\begin{aligned}
&(W^{1/2} - I)d_0 \\
&= (W^{1/2} + I)^{-1}K[K^*d_0], \\
&K^*d_0 = C_YR^{-1}d_0.
\end{aligned} \tag{7}$$

Thus all potentially troublesome inverse-M matrices remain bounded left factors. Equations (2)-(7) prove

$$\begin{aligned}
& \|\chi(Bp_F - d_0)v\|^2 \\
& \leq C(\sigma^2 + r_*^{-3}) \sum_e h_e[v] \\
& + CW_2\|v\|^2.
\end{aligned} \tag{8}$$

The estimate holds for arbitrary fast vectors. It is not an estimate restricted to vacuum or a finite excited coefficient space.

The spin column obeys separately

$$\|\chi W^{1/2} M^{-*} S_f v\|^2 \leq CW_2\|v\|^2, \tag{9}$$

because the finite fermionic Gauss matrices are bounded and F^{-1} retains the individual output-root denominators.

4. Exact slow-square coefficient and its Schur loss

Let $G_s = \text{diag}(g_C^{-1}, I_I)$, so the direct slow kinetic stack is $w = G_s^{1/2} p_s$. The exact normalized slow column in the Gauss square is

$$\begin{aligned}
A &= W^{1/2} M^{-*} \Psi, \\
\Psi &= [(C_Y^* + (F + L)^* K) g_C^{-1/2}, I_Y^*].
\end{aligned} \tag{10}$$

This follows directly from $O^* S g_s^{-1}$. In particular the FK term must be retained. One has

$$\begin{aligned}
\|\Psi^* F^{-1}\| &\leq C\sigma, \\
\|A\| &\leq C\sigma.
\end{aligned} \tag{11}$$

The adjoint coefficient acting on the frozen fast column has the exact expression

$$A^* d_0 = \Psi^* F^{-1} T^* W^{1/2} d_0. \tag{12}$$

Expand

$$\begin{aligned}
T^* W^{1/2} - I &= (T^* - I) + T^* (W^{1/2} - I), \\
T^* - I &= -T^* P F^{-1}.
\end{aligned} \tag{13}$$

The leading term $\Psi^* F^{-1} d_0$ is the stack of

$$\begin{aligned}
& I_Y F^{-1} d_0, \\
& g_C^{-1/2} [C_Y F^{-1} d_0 + K^* d_0 + K^* L F^{-1} d_0].
\end{aligned} \tag{14}$$

The first three contractions are same-root $Y_e p_e / r_e$. $LF^{-1}d_0$ is a triangle source with INPUT-root denominator r_e , so (3) applies even without using a frequency ratio. The J_2 part of $PF^{-1}d_0$ factors as

$$J_2 F^{-1} d_0 = R^{-1} C_Y^* [C_Y F^{-1} d_0], \quad (15)$$

again a bounded $O(\sigma)$ left factor times a same-root contraction. Equations (7), (11), and (13)-(15) therefore prove the sharper estimate

$$\begin{aligned} & \|\chi A^* d_0(v)\|^2 \\ & \leq C\kappa^2(\sigma^2 + r_*^{-3}) \sum_e h_e[v] \\ & \quad + C\kappa^2 W_2 \|v\|^2. \end{aligned} \quad (16)$$

Combining (8), (9), and $\|A\| \leq C\sigma$ also controls A^* applied to the full fast-plus-spin stack, with a coefficient $C(\kappa^2 + \sigma^2)$ multiplying the right side of (8), and a harmless $C\sigma^2 W_2$ spin term.

For completeness, for any vectors w, b and $\eta > 0$,

$$\begin{aligned} & \eta \|w\|^2 + \|b - Aw\|^2 \\ & = \|(\eta I + A^* A)^{1/2} \\ & \quad [w - (\eta I + A^* A)^{-1} A^* b]\|^2 \\ & \quad + \langle b, (I + \eta^{-1} A A^*)^{-1} b \rangle. \end{aligned}$$

The difference between $\|b\|^2$ and the final term is at most $\eta^{-1} \|A^* b\|^2$. Thus (16) controls the frozen-column Schur loss by exact excitation energies and W_2 , with no $\delta \sum_e r_e$ zero-point cost. This is a first-order form identity: variable coefficients create no omitted ordering terms at this stage.

5. What this lemma does and does not close

It closes the all-vector rootwise CONNECTION-NORM sublemma, including mixed roots, exact inverse-M left factors, and the coefficient occurring after slow-square completion. It remains valid with unrestricted internal momenta because none of its steps estimates those momenta or differentiates a fast projection in a slow direction.

At the connection-norm stage, two additional arguments are necessary. Section 6 below supplies the first for the fast kinetic column:

1. Bounding a squared connection norm is not a bound for the bilinear pairing with d_0 . The direct Young estimate against $\|d_0 v\|^2$ introduces its frozen vacuum energy $C\kappa^2 \sum_e r_e$. The odd H1 source / normal-ordering or graph-local form argument must be used instead; (8) alone does not perform it.
2. Completing all slow squares leaves shifted slow derivatives. Splitting off η times ordinary internal kinetic energy is not the same as splitting η times the colored internal

supersymmetric form: an unweighted Yukawa remainder $\eta C\alpha_a$ may be large for a far block. One must retain a suitable full slow-supercharge form or prove an incident-weighted conversion of the shifted slow squares. Equations (10)-(16) do not silently supply that conversion.

Accordingly these results are compatible with, but do not establish, a near-sharp complementary lower bound in the original Hamiltonian. They provide the rootwise estimate that avoids a global-frequency substitution at the connection stage.

6. Addendum: signed fast kinetic terms without a vacuum-frequency loss

The first obstruction listed in Section 5 can also be removed for the exact fast kinetic column. The following statements concern a smooth core vector v supported in the fast tube; an outer cutoff equal to one near its support may always be used in Sections 2-4. Only that support requirement, not a restriction on the oscillator content of v , is imposed.

6.1 All-vector triangle multiplication

For distinct triangle roots e, f, g , write $s_t = r_e + r_f + r_g$. Any uniformly bounded real coefficient tensor that may depend on the slow variables but is independent of the fast variables Y obeys

$$\begin{aligned} & |\langle v, s_t Y_e Y_f Y_g v \rangle| \\ & \leq C\sigma \sum_{j \in t} h_j[v] \\ & \quad + C\sigma^{-1} W_{2,t} \|v\|^2. \end{aligned} \tag{17}$$

First, splitting inputs into $Q_f, P_f Q_g$, and $P_f P_g$ gives

$$\begin{aligned} & \|\chi Y_f Y_g v\|^2 \\ & \leq C\sigma^2 r_*^2 [r_f^{-2} h_f[v] + r_g^{-2} h_g[v]] \\ & \quad + C/(r_f r_g) \|v\|^2. \end{aligned} \tag{18}$$

On the first component bound χY_g by σr_* and use the excited Y_f oscillator bound; on the second use χY_f and the excited Y_g bound; on the last use both Gaussian widths. The splitting is for estimation only; the projected pieces need not have tube support.

Choose e to be a largest triangle root. If $U_{B,e} = \text{constant} \exp(-Y_e \cdot \Omega_e Y_e / 2)$, put $a_e = \partial_e + \Omega_e Y_e$. The exact bosonic excitation form controls $\|a_e v\|^2$. Since the other two factors do not depend on Y_e , integration by parts gives, componentwise,

$$\begin{aligned} & \langle v, Y_e Y_f Y_g v \rangle \\ & = \text{Re} \langle a_e v, \Omega_e^{-1} (Y_f Y_g) v \rangle. \end{aligned}$$

Using $\|\Omega_e^{-1}\| \leq C/r_e, s_t/r_e \leq 3$, and (18) gives (17) by Young with parameter σ . The proof is valid for arbitrary fermionic and slow vector coefficients.

6.2 Leading triangle differential forms

Let $a(Y)$ be a real symmetric fast coefficient matrix supported only in the off-diagonal root blocks (e, g) and (g, e) , with each entry a bounded linear contraction of Y_f divided by r_e , multiplied by a coefficient $O(\kappa)$ that is independent of the fast variables Y (it may depend on the slow variables). For the product bosonic Gaussian U_B , writing $v = U_B u$ gives the exact identity

$$\begin{aligned} & \int (\partial v)^* a(\partial v) \\ &= \int U_B^2 (\partial u)^* a(\partial u) \\ &+ \int [\text{div}(a\Omega Y) - (\Omega Y) \cdot a(\Omega Y)] |v|^2. \end{aligned} \tag{19}$$

This identity does not project onto the fermionic vacuum. The excitation form satisfies

$$\begin{aligned} & \sum_e (H_{B,e} - E_{B,e})[v] \\ &= \int U_B^2 (\partial u)^* G(\partial u), \end{aligned}$$

with G uniformly positive. On tube-supported v , the first term in (19) is bounded in absolute value by $C\kappa\sigma \sum_e h_e[v]$. The divergence term in (19) vanishes: a connects distinct roots, depends on the third root, and Ω is pair-block diagonal. The remaining scalar is a triangle cubic with coefficient $O(\kappa r_g)$, hence is bounded by (17). Consequently

$$\begin{aligned} & \left| \int (\partial v)^* a(\partial v) \right| \\ & \leq C\kappa\sigma \sum_{j \in t} h_j[v] \\ & + C\kappa\sigma^{-1} W_{2,t} \|v\|^2. \end{aligned} \tag{20}$$

The same proof applies to $O(\kappa^2)$ coefficients. Equation (20) applies to both leading triangle forms from

$$2 \text{Re} \langle d_0, F^{-1} N_Y^* p_F - F^{-1} L^* d_0 \rangle.$$

No derivative cutoff is needed: (19) is applied to the original polynomial coefficient and v itself has tube support. If one instead wants a globally cut-off form on arbitrary unsupported inputs, cutoff derivatives must be included; an even radial cutoff generates additional triangle cubics, with coefficient smaller by $O((\sigma^2 r_*^3)^{-1})$.

6.3 Full inverse-M reduction

There is a short exact reduction avoiding an infinite expansion. Set

$$\begin{aligned}
Z &= F^{-1}P^*, \quad T = (I + Z)^{-1}, \\
n &= F^{-1}N_Y^*p_F, \\
X &= n - Zd_0.
\end{aligned}$$

Then

$$Bp_F = W^{1/2}(d_0 + TX).$$

Because $W \geq I$ and $T - I = -ZT$,

$$\begin{aligned}
&\|Bp_F v\|^2 \\
&\geq \|d_0 v\|^2 + 2\operatorname{Re}\langle d_0 v, Xv \rangle \\
&\quad - 2|\langle Z^* d_0 v, TXv \rangle|.
\end{aligned} \tag{21}$$

Both X and $Z^* d_0 = PF^{-1}d_0$ are covered by the rootwise connection estimates. The last pairing is therefore bounded by

$$C(\sigma^2 + r_*^{-3}) \sum_e h_e[v] + CW_2 \|v\|^2.$$

Inside $\langle d_0, X \rangle$, the J_2 term factors exactly as

$$\begin{aligned}
&\langle d_0, F^{-1}J_2^* d_0 \rangle \\
&= \langle C_Y F^{-1}d_0, C_Y R^{-1}d_0 \rangle,
\end{aligned} \tag{22}$$

and has the same bound. Only the two linear triangle forms in Section 6.2 remain. Combining (20)-(22) gives

$$\begin{aligned}
&\|Bp_F v\|^2 \\
&\geq \|d_0 v\|^2 \\
&\quad - C[\kappa\sigma + \sigma^2 + r_*^{-3}] \sum_e h_e[v] \\
&\quad - C(1 + \kappa/\sigma)W_2 \|v\|^2.
\end{aligned} \tag{23}$$

Thus direct p_F^2 plus the exact Gauss FAST column, the adapted quadratic potential, and its fast fermion mass have a lower bound

$$\begin{aligned}
&E_{0,\text{total}} \|v\|^2 \\
&+ (1 - C[\kappa\sigma + \sigma^2 + r_*^{-3}])H_{\text{exc}}[v] \\
&- C(1 + \kappa/\sigma)W_2 \|v\|^2,
\end{aligned} \tag{24}$$

with the frozen E_0 multiplier understood pointwise in the slow variables. This is a fast-column comparison only; the slow derivatives in the full Gauss square have not been dropped or independently counted.

6.4 Frozen spin-linear term

The leading spin cross is a finite $\sum A_e p_e / r_e$, with Y-independent Hermitian Clifford matrices $\|A_e\| \leq C\kappa$. Since the $\Omega_e Y_e$ part of $a_e = \partial_e + \Omega_e Y_e$ contributes a real expectation against A_e ,

$$\begin{aligned} & |\langle v, A_e p_e v / r_e \rangle| \\ & \leq C\kappa r_e^{-1} \sqrt{h_{B,e}[v]} \|v\| \\ & \leq \epsilon h_e[v] + C\kappa^2 / (\epsilon r_e^2) \|v\|^2. \end{aligned} \tag{25}$$

Here $h_{B,e} = H_{B,e} - E_{B,e}$. This also avoids any bosonic vacuum-frequency cost. The inverse-M corrections to this frozen spin cross pair rootwise connection errors with the bounded spin column (9), and are bounded by $\epsilon H_{\text{exc}} + C_\epsilon W_2$.

7. Final scope after the addendum

Equations (2)-(16) prove the connection and Schur-loss norm statements for arbitrary fast vectors. Equations (17)-(25) additionally prove the signed fast-column comparison on an unrestricted oscillator core supported in the fast tube. These estimates remove the need for a $\delta \sum_e r_e$ loss at the fast-connection stage.

They still do not convert the completed, shifted slow derivative form into a nonnegative colored internal supersymmetric form. One must control its connection curvature/full-supercharge terms or use another incident-weighted argument. In particular a constant loss of internal kinetic energy cannot simply be identified with the same loss of H_a . That slow-form issue, the other physical potential/Yukawa terms, and the final allocation of one non-double-counted reserve belong to the full complementary theorem, not to this helper lemma.

F Pairwise Berry connection comparison

Technical derivation for review. Equation and subsection numbers in this appendix are local to the source derivation. Historical local labels are retained, including numbering gaps or nonsequential labels. The accompanying source notes retain the original expressions for comparison.

4.1 Complete charge comparison

A naive kinetic comparison

$$|(\partial + iA)f|^2 \geq (1 - \epsilon)|\partial f|^2 - \epsilon^{-1}|A|^2|f|^2$$

loses a fixed part of internal kinetic energy. Replacing it by h_I incurs the unwanted $O(\epsilon\kappa r)$ internal Yukawa mass. This is the wrong comparison.

Instead, normalize the sixteen internal charges so h_I is their averaged squared norm on the core. Replace each momentum by its scalar Berry-covariant momentum. Direct Clifford expansion gives

$$\begin{aligned} h_I(A) &= \text{average}_\alpha \|Q_{I,\alpha} + C_\alpha(A)\|^2 \\ &\quad - \text{a finite Clifford contraction of } F_A. \end{aligned}$$

The difference $C_\alpha(A)$ is a bounded multiplication matrix, with norm $\leq C_N|A|$. Consequently

$$\begin{aligned} h_I(A) &\geq (1 - \epsilon)h_I \\ &\quad - C_N[\epsilon^{-1}|A|^2 + |F_A|]. \end{aligned}$$

This retains the complete interacting h_I , rather than its kinetic part, and uses no gap. Spinor averaging eliminates the ordinary gauge-term obstruction to positivity even on the unrestricted internal module; the low coefficient additionally has the physical residual equivariance required by the inductive Hardy input.

On a two-block small-ratio tube the gapped finite matrix is analytic. All determinant curvatures vanish at internal $A = 0$: the two-projector derivative calculation contains a Spin(9) trace of at most three Clifford matrices. Homogeneity and smoothness therefore give

$$|F_A| \leq C_N\kappa/r^2.$$

Use equivariant radial gauge in the internal coordinates, based on the flat center vacuum at $A = 0$. The radial-gauge curvature formula gives $|A_A| \leq C_N\kappa^2/r$ for internal components. Mixed relative components must also be kept: in this gauge they are the flat $A = 0$ base connection plus the integral of $F_{A,b}$ contracted with A . The same vanishing-at- $A = 0$ argument and homogeneity give their bound $C_N\kappa^2/r$. Thus, after this separate mixed-component check,

$$|A_{\text{Berry}}| \leq C_N \kappa^2 / r.$$

It follows that the preceding comparison costs only

$$C_N [\kappa + \epsilon^{-1} \kappa^4] r^{-2},$$

which can be made arbitrarily small after fixing ϵ . A gauge-equivariant radial trivialization preserves ordinary residual SU(n) physicality of the coefficient. Weyl/center flat identifications remain part of the bundle.

This establishes a viable repair of the flatness step provided the local Hamiltonian comparison really identifies the covariant internal form and retains its curvature terms. The following rootwise construction avoids an artificial dependence on the global minimum-gap ratio.

4.2 Pairwise relative multiblock Berry bound

At quadratic order the fast fermionic space is a direct sum over bifundamental block pairs. Its one-particle matrices are

$$D_{ab} = \Gamma \cdot b_{ab} \otimes I + \sum_i \Gamma_i \otimes (A_i^{(a)} \otimes I - I \otimes (A_i^{(b)})^T).$$

If each endpoint internal norm is at most κ times $r_{ab} = |b_{ab}|$, the gap is a positive fixed multiple of r_{ab} after reducing κ by the fixed nine-dimensional norm factor. The adapted determinant line is the tensor product of the occupied pair determinants. On each pair, choose the preceding internal-radial gauge from the flat $A = 0$ center line. The Clifford trace calculation at $A = 0$ is independent of bifundamental dimension, so every pure/mixed curvature component vanishes there.

Smoothness on the compact dimensionless pair tube and homogeneity give, in canonical block/center coordinates,

$$\begin{aligned} |F_{ab}| &\leq C_N \kappa r_{ab}^{-2}, \\ |a_{ab}| &\leq C_N \kappa^2 r_{ab}^{-1}. \end{aligned}$$

The relative-component bound follows from its own mixed-curvature integral, not from simply declaring it gauged away. Product connections add, and Cauchy-Schwarz over the finitely many pairs gives

$$\begin{aligned} |F_{\text{total}}| &\leq C_N \kappa \sum_{a < b} r_{ab}^{-2}, \\ |a_{\text{total}}|^2 &\leq C_N \kappa^4 \sum_{a < b} r_{ab}^{-2}. \end{aligned}$$

The full slow averaged charge stack includes all internal interacting charges and free center charges. Its comparison therefore loses only ϵ times the entire slow supersymmetric form and the inverse-square multiplier

$$C_N(\kappa + \epsilon^{-1}\kappa^4) \sum_{a < b} r_{ab}^{-2}.$$

This multiplier has a direct shape-independent center-kinetic payment. In the mass metric $\sum n_a |dz_a|^2$, the linear map $z \rightarrow z_a - z_b$ has squared norm $l_{ab}^2 = 1/n_a + 1/n_b$. Let T_{ab} be kinetic energy in its nine-dimensional orthogonal relative subspace. Then

$$\begin{aligned} T_{ab} &\geq (49/4) l_{ab}^2 \int r_{ab}^{-2} |f|^2, \\ T_{ab} &\leq T_{\text{centers}}. \end{aligned}$$

Thus $\sum \int r_{ab}^{-2} |f|^2 \leq (4/49) \sum l_{ab}^{-2} T_{\text{centers}}$. The finite rank-dependent coefficient contains no inverse shape gap δ_N . Choose ϵ , then κ , before L and the global radius. Nonflatness therefore causes no parameter-order circle in this Berry comparison.

This is a lemma about the finite adapted vacuum connection and the full slow charge forms. It does not prove that all remaining full-Hamiltonian compression and Schur terms match that covariant slow model, nor does it prove the adapted bosonic/fermionic energy-defect estimate.

G Differential connections and colored forms

Technical derivation for review. Equation and subsection numbers in this appendix are local to the source derivation. Historical local labels are retained, including numbering gaps or nonsequential labels. The accompanying source notes retain the original expressions for comparison.

1. Setting and result

Fix the rank and block partition. Assume only

$$\begin{aligned} \alpha_a + \alpha_b &\leq \kappa r_{ab}, \quad |Y| \leq \sigma r_*, \\ r_* &= \min_e r_e \geq 1, \quad W_2 = \sum_e r_e^{-2}. \end{aligned}$$

Constants depend on fixed rank and fixed sufficiently small upper bounds for κ and σ , never on $r_{\max}/r_*$. Work first on the compactly supported smooth residual-equivariant core in a normal-frame chart. No individual block-singlet assumption is imposed. Let

$$\begin{aligned} H_{\text{exc}} &= \sum_e (H_{\text{fast},e} - E_{0,e}) \geq 0, \\ \lambda(Z) &= \sigma^2 + r_*^{-3}, \\ E[v] &= (\lambda H_{\text{exc}})[v] + W_2[v]. \end{aligned} \tag{1.1}$$

The weights are center-only and the excitation form acts in fast fibers. For the INTERNAL part of the completion connection write

$$\begin{aligned} \beta_j &= b_j(Y, A, Z) \cdot p_F, \\ \beta &= A_s^* B_F p_F, \quad A_s = B_s G_s^{-1/2}. \end{aligned}$$

Here A_s is the normalized Gauss slow coefficient, not the internal matrix tuple. The internal block of G_s is exactly identity. Define the ordered shifted form inherited from the exact kinetic identity:

$$\begin{aligned} H_I^{\text{raw}}(\beta)[v] &= \sum_j \|(p_j + \beta_j)v\|^2 + (V_I + Y_I)[v]. \end{aligned} \tag{1.2}$$

For every ϵ in $(0,1)$,

$$\begin{aligned} H_I^{\text{raw}}(\beta)[v] &\geq (1 - \epsilon) H_I[v] \\ &\quad - C_{N,\epsilon} (\lambda H_{\text{exc}})[v] \\ &\quad - C_{N,\epsilon} W_2[v]. \end{aligned} \tag{1.3}$$

This retains the ENTIRE interacting internal form, including its original potential and Yukawa terms. There is no ϵ times bare internal kinetic loss and no $\epsilon\alpha_a$ mass remainder. No internal gap or physical-sector lower-rank Hardy estimate is used.

The proof gives the actual curvature expectation bound

$$|R_\beta[v]| \leq C_N(\lambda H_{\text{exc}})[v] + C_N W_2[v]. \quad (1.4)$$

The stronger possible coefficient $C_N r_*^{-3}$ with σ^2 absent is not needed and is not claimed. Fixing σ sufficiently small and then taking the exterior radius large makes the excitation coefficient as small as desired. On fast-complement inputs W_2 is also paid by the gap.

2. Exact completion

Remove the finite fermionic Gauss source by the separate estimate chosen for that source. Its estimate is not part of this algebra. Let

$$w = G_s^{1/2} p_s v, \quad a = B_F p_F v, \quad A_s = B_s G_s^{-1/2}.$$

Exactly, as a first-order form identity,

$$\begin{aligned} \|w\|^2 + \|a + A_s w\|^2 \\ = \|w + A_s^* a\|^2 + \|a\|^2 + \|A_s w\|^2 - \|A_s^* a\|^2. \end{aligned} \quad (2.1)$$

Coefficients multiply the differentiated section in their original order. No integration by parts and no coefficient derivative occurs in this completion.

The internal components of $w + A_s^* a$ are $p_j + \beta_j$. Its center components use the exact center metric and NORMAL-BUNDLE covariant momentum. The center square and $\|A_s w\|^2$ may be retained or discarded as positive forms. They are not additional copies of the fast Gauss energy. The negative term is the connection norm already bounded in the rootwise lemma.

If the half-density was separated as in the intrinsic geometry note, its scalar V_d must still be included separately. The symmetrization below neither replaces nor duplicates that calculation.

3. Coefficients and differentiated rootwise bounds

Use intrinsic notation

$$\begin{aligned} F &= E^* B_{Z+A}, \quad C = P_N B_A, \quad N_Y = P_N B_Y, \\ I_Y &= P_I B_Y, \quad M = F + L - J_2, \quad W = I + K K^*. \end{aligned}$$

The internal coefficient is, with the common Gauss sign convention,

$$\beta_I = I_Y M^{-1} W M^{-*} (C + N_Y)^* p_F. \quad (3.1)$$

All coefficients are real in real trace-orthonormal frames. Normal frames depend on centers only; an internal derivative differentiates neither p_F nor the frame. The connection commutes with internal Clifford matrices and every internal potential-supercharge multiplication coefficient.

For a unit internal direction h ,

$$\begin{aligned} \partial_h M &= \partial_h F, \\ \partial_h F \text{ and } \partial_h C &\text{ are pair-block diagonal ,} \\ \|\partial_h F_e\| + \|\partial_h C_e\| &\leq C_N, \\ \partial_h W &= \partial_h I_Y = \partial_h N_Y = 0. \end{aligned} \quad (3.2)$$

Only pairs incident to the differentiated block contribute. The coarse r_* estimate below does not convert any kinetic or Yukawa term, so it creates no unrelated α_a/r_* error.

The cited rootwise connection proof gives

$$\sum_j \|\beta_j v\|^2 \leq C_N E[v]. \quad (3.3)$$

Its explicit first-order decomposition also gives

$$\begin{aligned} \sum_{j,k} \|(\partial_j \beta_k) v\|^2 \\ \leq C_N [(r_*^{-2} \lambda H_{\text{exc}})[v] \\ + (r_*^{-2} W_2)[v]]. \end{aligned} \quad (3.4)$$

Here $\partial_j \beta_k$ differentiates its coefficients only. The following proves (3.4); it is not inferred from the abstract bound (3.3).

3.1 Proof of the differentiated estimate

In the rootwise lemma put

$$d_0 = F^{-1} C^* p_F, \quad T = (I + F^{-1} P^*)^{-1}, \quad P = L - J_2.$$

The exact decompositions (5)-(7) and (12)-(15) in that lemma express β as finite sums of bounded multiplication matrices times the source classes

$$\begin{aligned} &Y_e p_e / r_e, \\ &Y_f p_e / r_g, \quad \text{when } (e, f, g) \text{ is a triangle ,} \\ &Y_f p_e / r_e, \quad \text{with the input-root denominator ,} \end{aligned}$$

with bounded root-preserving factors between the displayed operators. Non-root-preserving inverse matrices remain bounded LEFT factors. Thus no unweighted spectator momentum is hidden.

Differentiate those exact decompositions. $Y, r_e, W, K, I_Y, C_Y, L, J_2$ and g_C have zero internal derivatives. The differentiated factors satisfy

$$\begin{aligned}\|\partial_j F_e^{-1}\| &\leq C_N/r_e^2, \\ \|\partial_j(F_e^{-1}C_e^*)\| &\leq C_N/r_e, \\ \|\partial_j T\| + \|\partial_j T^*\| &\leq C_N/r_*.\end{aligned}$$

The inverse derivative formula and (3.2) prove these inequalities. All other bounded left factors in the displayed decompositions consequently have derivative at most C_N/r_* . Differentiating $F_e^{-1}C_e^*$ replaces its $O(\kappa)$ factor by $O(1/r_e)$; use the source bound with its optional κ gain omitted.

Every resulting term is therefore the SAME source class with one extra factor at most C_N/r_* . Applying the proved same-root and mixed-root inequalities (2)-(4) of the cited lemma, and finite-sum Cauchy, proves (3.4). The argument is all-vector, does not bound internal momenta, and does not require a fast projection to preserve tube support.

3.2 Coarse pointwise derivatives for divergence

Only for the scalar divergence correction we need

$$\begin{aligned}|b| &\leq C_N\sigma(\kappa + \sigma), \\ |\partial_Y b| &\leq C_N/r_*, \\ |\partial_A \partial_Y b| + |\partial_Y^2 b| &\leq C_N/r_*^2.\end{aligned}\tag{3.5}$$

Indeed the internal normalized slow column is $O(\sigma)$, the full fast Gauss column is $O(\kappa + \sigma)$, and $\|M^{-1}\| \leq C_N/r_*$. First derivatives of M, I_Y and $C + N_Y$ are $O(1)$; second derivatives of M are $O(1/r_*)$, coming from J_2 . First and second derivatives of W and its positive square root are $O(1/r_*)$ and $O(1/r_*^2)$. Differentiate the product $B_I^* B_F$ using the inverse derivative formula. The potentially large C is always kept inside $F^{-1}C^*$ or the bounded full fast column, rather than estimated by $\kappa r_{\max}$. This proves (3.5); fixed coordinate dimensions absorb component sums.

4. Exact self-adjoint symmetrization

In flat fast-fiber measure, with $p_F = -i\partial_Y$, put

$$\begin{aligned}d_j &= \text{div}_F b_j, \\ \widehat{\beta}_j &= (\beta_j + \beta_j^*)/2 \\ &= b_j \cdot p_F - (i/2)d_j, \\ P_j &= p_j + \widehat{\beta}_j, \\ X_j &= \partial_j + b_j \cdot \partial_Y.\end{aligned}$$

Then $\beta_j = \widehat{\beta}_j + (i/2)d_j$ and $P_j = -i(X_j + d_j/2)$ is symmetric on the core. Direct integration by parts gives the EXACT identity

$$\begin{aligned} & \|(p_j + \beta_j)v\|^2 \\ &= \|P_j v\|^2 \\ &+ \int [(1/2)X_j d_j + (1/4)d_j^2]|v|^2. \end{aligned} \tag{4.1}$$

The cross term is $(1/2) \int X_j(d_j)|v|^2$; $\text{div } X_j = d_j$. Thus simply treating $b \cdot p_F$ as symmetric would miss a scalar correction.

By (3.5),

$$\begin{aligned} |d_j| &\leq C_N/r_*, \\ |\partial_A d_j| + |\partial_Y d_j| &\leq C_N/r_*^2, \\ \sum_j |(1/2)X_j d_j + (1/4)d_j^2| &\leq C_N W_2. \end{aligned} \tag{4.2}$$

Hence the full correction is inverse-square multiplication. Also

$$\sum_j \|\widehat{\beta}_j v\|^2 \leq C_N E[v], \tag{4.3}$$

and (3.4) holds with $\widehat{\beta}$ replacing β : its extra multiplication derivative is $O(r_*^{-2})$, whose squared norm is bounded by $r_*^{-2}W_2$.

5. Actual curvature expectation

Define the symmetric curvature operator

$$\begin{aligned} F_{jk} &= i[P_j, P_k] \\ &= \partial_j \widehat{\beta}_k - \partial_k \widehat{\beta}_j \\ &\quad + i[\widehat{\beta}_j, \widehat{\beta}_k]. \end{aligned} \tag{5.1}$$

Equivalently F_{jk} is the symmetrized momentum of the fast vector field

$$\begin{aligned} & \partial_j b_k - \partial_k b_j \\ &+ (b_j \cdot \partial_Y) b_k - (b_k \cdot \partial_Y) b_j. \end{aligned}$$

It is first order, not bounded multiplication.

The charge identity contracts (5.1) with fixed bounded internal-Clifford matrices. If L is any such matrix, it commutes with $\widehat{\beta}$ and its coefficient derivatives. Symmetry gives exactly

$$\begin{aligned}
& |\langle v, Li[\widehat{\beta}_j, \widehat{\beta}_k]v \rangle| \\
& \leq 2\|L\|\|\widehat{\beta}_j v\|\|\widehat{\beta}_k v\|.
\end{aligned} \tag{5.2}$$

Thus (4.3) pays all beta-beta curvature terms. There is no need to bound the norm of a fast-differentiated curvature symbol.

For the derivative part, (3.4) and Cauchy give

$$\begin{aligned}
& |\langle v, L(\partial_j \widehat{\beta}_k)v \rangle| \\
& \leq C_N \|r_*^{-1} v\| E[v]^{1/2} \\
& \leq C_N E[v],
\end{aligned} \tag{5.3}$$

since $\|r_*^{-1} v\|^2 \leq W_2[v] \leq E[v]$. To justify the first inequality with varying centers, apply the coefficient derivative bound at each center and then apply weighted Cauchy over centers. Summing the finitely many Clifford components proves (1.4).

This is an all-vector curvature EXPECTATION estimate. It derives the extra coefficient derivative from the actual rootwise expansion, not from a generic relative-norm assertion. It makes no oscillator-degree cutoff and loses no internal kinetic energy.

6. Averaged sixteen-supercharge comparison

Normalize the internal charges by

$$\begin{aligned}
H_I[v] &= (1/16) \sum_{\alpha} \|Q_{\alpha} v\|^2, \\
Q_{\alpha} &= \sum_j c_{\alpha,j} p_j + V_{\alpha}(A).
\end{aligned}$$

The $c_{\alpha,j}$ are fixed internal Clifford matrices. The ordinary colored Gauss terms cancel in the spinor average. An individual Q_{α}^2 must not be identified with H_I on the colored core.

Replace p_j by P_j :

$$\begin{aligned}
Q_{\alpha}^{\beta} &= Q_{\alpha} + C_{\alpha}(\widehat{\beta}), \\
C_{\alpha}(\widehat{\beta}) &= \sum_j c_{\alpha,j} \widehat{\beta}_j.
\end{aligned}$$

The exact averaged expansion is

$$\begin{aligned}
& (1/16) \sum_{\alpha} \|Q_{\alpha}^{\beta} v\|^2 \\
&= \sum_j \|P_j v\|^2 + (V_I + Y_I)[v] + R_{\beta}[v].
\end{aligned} \tag{6.1}$$

Here R_β is a finite-Clifford linear contraction of F_{jk} . The symmetric momentum products give the kinetic form, antisymmetric momentum products give curvature, and

$$[P_j, V_\alpha] = [p_j, V_\alpha]$$

because $\widehat{\beta}$ is fast-bosonic with scalar bosonic coefficients. The original potential and Yukawa terms are consequently unchanged. Spinor-averaged cancellation of the original Gauss term is unchanged too.

Finite-dimensional Clifford bounds and (4.3) imply

$$(1/16) \sum_{\alpha} \|C_{\alpha}(\widehat{\beta})v\|^2 \leq C_N E[v]. \quad (6.2)$$

Young's inequality on the COMPLETE charge stack now gives

$$\begin{aligned} (1/16) \sum_{\alpha} \|Q_{\alpha}^{\beta}v\|^2 \\ \geq (1 - \epsilon)H_I[v] - C_N \epsilon^{-1}E[v]. \end{aligned} \quad (6.3)$$

Subtract curvature using (1.4), then apply (4.1)-(4.2). This proves (1.3). The loss is ϵH_I , not ϵt_I ; that is why no large unrelated internal Yukawa mass is introduced.

7. Normal-bundle center connection and chart changes

The normal fast bundle N_Z depends on centers alone. Its orthogonal connection is already in p_C in the exact intrinsic formula. In a local frame its center generator may contain unbounded $Y\partial_Y$ operators. This argument never estimates those by H_{exc} .

Internal derivatives act in a center-dependent but A-independent frame, so (3.2) has no frame derivative. The internal supercharges contain no center momentum. Neither center normal-bundle curvature nor mixed center/internal curvature occurs in the INTERNAL square (6.1).

The completed center square remains a genuine positive covariant derivative form. Any subsequent center Hardy argument must use that actual connection with its domain justified. It may not replace its unbounded normal-frame generator by a bounded vacuum Berry coefficient. That separate comparison is not claimed here.

All identities intertwine under center-dependent orthogonal changes of fast normal frame and residual gauge changes. β is an actual vector field on the fast fiber, $\widehat{\beta}$ is its half-density action, and curvature is its commutator. The internal comparison is therefore intrinsic on a fixed projector branch.

8. Parameter order and exact scope

Choose the requested internal loss ϵ first, then σ so $C_{N,\epsilon}\sigma^2$ fits the available fast reserve, then r_* large enough for $C_{N,\epsilon}r_*^{-3}$ and the inverse-square multiplier.

On the global fast complement $H_{\text{exc}} \geq c_N r_*$ and $W_2 \leq m_N r_*^{-2}$, so

$$W_2[v] \leq C_N(r_*^{-3} H_{\text{exc}})[v].$$

This final payment is restricted to complementary inputs; (1.3) itself holds on arbitrary colored inputs. No internal Hamiltonian is inverted.

Still separate are the finite spin source, density multiplier, signed fast kinetic remainder, potential/Yukawa remainders, and branch/cutoff compatibility. The present lemma closes the complete internal-form and curvature step only.

H Colored complementary reserve

Technical derivation for review. Equation and subsection numbers in this appendix are local to the source derivation. Historical local labels are retained, including numbering gaps or nonsequential labels. The accompanying source notes retain the original expressions for comparison.

1. Coordinate conventions and exact matrices

Use the intrinsic geometry note and the notation of the actual adapted source inventory:

$$\begin{aligned} F &= E^* B_{Z+A}, C = P_N B_A, L = E^* B_Y, N_Y = P_N B_Y, \\ I_Y &= P_I B_Y, C_Y = P_C B_Y, K = R^{-1} C_Y^*, J_2 = K C_Y, \\ M &= F + L - J_2, W = I + K K^*, g_C = I + K^* K. \end{aligned}$$

Let $p_s = (g_C^{-1/2} p_C, p_I)$, where p_C includes the exact normal connection. Define

$$\begin{aligned} \Psi &= [(C_Y^* + (F + L)^* K) g_C^{-1/2}, I_Y^*], \\ A &= W^{1/2} M^{-*} \Psi, \\ a &= B p_F = W^{1/2} M^{-*} (C + N_Y)^* p_F, \\ s &= -W^{1/2} M^{-*} S_f. \end{aligned}$$

The minus sign in s matches the original Gauss expression S_f —fast—slow after multiplying that whole stack by -1 . The unshifted scalar kinetic stack is exactly

$$|p_s v|^2 + |p_F v|^2 + |a v + A p_s v + s v|^2. \quad (1.1)$$

The scalar half-density term is added after this first-order identity. We never add the same Gauss square a second time to H_{fast} .

Throughout, $T_{\text{cent}}[v] = \|p_C v\|^2$ with the exact normal connection. Its direct induced-metric coefficient obeys $g_C^{-1} \geq (1 + C\sigma^2)^{-1} I$; this small center-only loss is included in the final allocation.

Set $D0 = F^{-1} C^* p_F, P = L - J_2, Z = F^{-1} P^*, T = (I + Z)^{-1}, n = F^{-1} N_Y^* p_F$. Thus

$$a = W^{1/2} T(D0 + n), \quad \|T\| + \|T^{-1}\| \leq C, \quad \|A\| \leq C\sigma. \quad (1.2)$$

The frozen kinetic contribution in H_{fast} is $p_F^2 + \|D0\|^2$.

2. Oscillator estimates with the vacuum kept explicitly

The adapted bosonic Gaussian on edge e is $\exp(-Y_e \cdot \Omega_e Y_e / 2)$, with $cr_e I \leq \Omega_e \leq Cr_e I$. Let $a_e = \partial_e + \Omega_e Y_e$ be its annihilation gradient. Then $\|a_e v\|^2 \leq Ch_e[v]$, including arbitrary fermionic

and spectator components. The shifted fermionic excitation form is nonnegative and only improves this inequality.

The following estimates are consequences of splitting each edge into its adapted full vacuum P_e and $Q_e = 1 - P_e$; see Appendix E for the first two:

$$\|\chi Y_e p_e / r_e v\|^2 \leq C\sigma^2 h_e[v] + Cr_e^{-2} \|v\|^2, \quad (2.1)$$

$$\begin{aligned} \|\chi Y_f p_e / r_g v\|^2 &\leq C(\sigma^2 + r_*^{-3})(h_e + h_f)[v] \\ &+ C \sum_{u \in \text{triangle}} r_u^{-2} \|v\|^2. \end{aligned} \quad (2.2)$$

A useful sharpened two-coordinate version is

$$\begin{aligned} \|\chi Y_f Y_g v\|^2 &\leq C\sigma^2 r_*^2 (h_f[v]/r_f^2 + h_g[v]/r_g^2) \\ &+ C/(r_f r_g) \|v\|^2, \end{aligned} \quad (2.3)$$

for distinct edges f,g. To prove it, decompose into $Q_f, P_f Q_g$ and $P_f P_g$. On the first two pieces use $|\chi Y_g| \leq C\sigma r_*$ or $|\chi Y_f| \leq C\sigma r_*$, and $\|Y_f Q_f v\|^2 \leq Ch_f[v]/r_f^2$ (similarly g). On the last use the exact Gaussian second moments. Finite-sum Cauchy-Schwarz supplies the displayed constant. The cutoff acts only on the left; it need not commute with P_f .

In every triangle e,f,g,

$$r_* r_g / (r_e r_f) \leq 2, \quad r_g / (r_e^2 r_f) \leq C(r_e^{-2} + r_f^{-2}). \quad (2.4)$$

These elementary triangle inequalities are what remove remote zero-point energies.

3. Direct cubic potential, with arbitrary fast vectors

Let e,f,g be a triangle, $s_t = r_e + r_f + r_g$, and let B_t be any fixed bounded trilinear contraction of the three real fast coordinate spaces, also allowing a bounded spectator matrix. Then

$$\begin{aligned} &|\langle v, s_t B_t(Y_e, Y_f, Y_g) v \rangle| \\ &\leq C\sigma(h_e + h_f + h_g)[v] + C\sigma^{-1} \sum_{u \in t} r_u^{-2} \|v\|^2. \end{aligned} \quad (3.1)$$

Choose e with largest r_e . For a scalar component, integration by parts gives exactly

$$\begin{aligned} &\langle v, Y_e Y_f Y_g v \rangle \\ &= \text{Re} \langle a_e v, \Omega_e^{-1} (Y_f Y_g) v \rangle. \end{aligned} \quad (3.2)$$

There is no derivative of $Y_f Y_g$ in the e coordinate. Equation (3.2) applies componentwise to the finite contraction, with Ω_e^{-1} acting on its e index. Use $s_t/r_e \leq 3$, (2.3), $r_* \leq r_f, r_g$ and

Young with parameter σ . The source term is at most $C\sigma^{-1}/(r_f r_g)$, which is bounded by the triangle W_2 . This proves (3.1). No finite oscillator truncation is made.

Every actual cubic potential term has a bounded coefficient times s_t : its coefficient is a commutator with $Z+A$, whose norm is $\leq C(r_e+r_f+r_g)$ under the pairwise-relative internal tube. Therefore (3.1) applies to the full V_3 . V_4 remains positive and is not spent.

4. Signed triangle kinetic forms

For any bounded real coefficient tensor independent of the fast variables Y (but possibly dependent on the slow variables), and three distinct triangle edges,

$$\begin{aligned} & |2 \operatorname{Re} \langle p_e v, (Y_f/r_e) p_g v \rangle| \\ & \leq C\sigma(h_e + h_f + h_g)[v] + C\sigma^{-1} \sum_{u \in t} r_u^{-2} \|v\|^2. \end{aligned} \quad (4.1)$$

The clean proof is the Gaussian ground-state transform. The real symmetric principal coefficient $a_{eg}(Y) = Y_f/r_e$ has no divergence in either active derivative coordinate e.g. For the product Gaussian g_B ,

$$\begin{aligned} & \int (\partial v)^* a \partial v \\ & = \int g_B^2 (\partial(v/g_B))^* a \partial(v/g_B) \\ & + \int [\operatorname{div}(a \Omega Y) - (\Omega Y) \cdot a(\Omega Y)] |v|^2. \end{aligned} \quad (4.2)$$

The first term is $\leq C\sigma \Sigma h_e$ because $|Y_f|/r_e \leq C\sigma$ on the support. The divergence term is zero since the three roots are distinct. The last term is a triangle cubic with coefficient bounded by $C r_g$, hence by $C s_t$, and (3.1) proves (4.1). An outer cutoff equal to one near the support contributes no derivative term there.

One may instead expand $p = -ia + i\Omega Y$. Formula (2.3), together with (2.4), controls the apparently dangerous factor r_g/r_e in the mixed annihilator/quadratic term. This gives the same bound without ever estimating p_e^2 by h_e plus an unassigned r_e vacuum mass.

5. Exact inverse-M reduction: higher paths are products of errors

Let $X = n - ZD0$. The connection bounds imply

$$\begin{aligned} & \|Xv\|^2 + \|Z^*D0v\|^2 \\ & \leq C(\sigma^2 + r_*^{-3})H_{\text{exc}}[v] + CW_2[v]. \end{aligned} \quad (5.1)$$

For X this is the displayed helper lemma. For $Z^*D0 = PF^{-1}D0$ the linear L term is a triangle map $Y_f p_e/r_e$; the J_2 term factors through $C_Y F^{-1}D0$, a same-edge map, followed by bounded K . Thus the identical proof applies. Denominators stay on their actual input or output root.

Since $W \geq I$ and $T - I = -ZT$ exactly,

$$\begin{aligned} \|av\|^2 &\geq \|D0v + TXv\|^2 \\ &\geq \|D0v\|^2 + 2\operatorname{Re}\langle D0v, Xv \rangle - 2|\langle Z^*D0v, TXv \rangle|. \end{aligned} \quad (5.2)$$

The last term is bounded by (5.1). The J_2 part of the remaining cross factors as

$$\langle D0, F^{-1}J_2^*D0 \rangle = \langle C_Y F^{-1}D0, K^*D0 \rangle, \quad (5.3)$$

and both factors satisfy the same-edge bound. The two other terms are precisely

$$2\operatorname{Re}\langle D0, F^{-1}N_Y^*p_F \rangle - 2\operatorname{Re}\langle D0, F^{-1}L^*D0 \rangle. \quad (5.4)$$

Their root supports are triangles; their coefficients are bounded by $C\kappa/r_e$ and $C\kappa^2/r_e$. They obey (4.1). Consequently

$$\begin{aligned} \|av\|^2 &\geq \|D0v\|^2 - C(\kappa\sigma + \sigma^2 + r_*^{-3})H_{\text{exc}}[v] \\ &\quad - C(1 + \kappa/\sigma)W_2[v]. \end{aligned} \quad (5.5)$$

This is an actual all-vector signed-form comparison for the coupled inverse-M fast column. It does not require a separate estimate of every higher Neumann path. Importantly, it never bounds the large $D0$ vacuum norm by the nearest complementary gap.

6. Exact slow completion and finite-spin payments

With s temporarily omitted, the identity

$$\begin{aligned} \|p_s v\|^2 + \|av + Ap_s v\|^2 \\ = \|(p_s + A^*a)v\|^2 + \|av\|^2 + \|Ap_s v\|^2 - \|A^*av\|^2 \end{aligned} \quad (6.1)$$

holds pointwise on the first-order stack. The final negative term is controlled by

$$\|A^*av\|^2 \leq C(\sigma^2 + r_*^{-3})H_{\text{exc}}[v] + CW_2[v], \quad (6.2)$$

because $A^*a = A^*D0 + A^*(a - D0)$, $\|A\| \leq C\sigma$, and both helper norms are established. Equation (6.1) leaves the explicit nonnegative reserve $\|Ap_s v\|^2$. It does not reserve another copy of the full Gauss square.

The center component of the shifted square is compared to the original exact normal-covariant center kinetic by ordinary Young, costing ϵT_{cent} plus ϵ^{-1} times (6.2). Internal components are treated by the complete averaged charges in Section 7; one must not apply this ordinary Young inequality to the internal kinetic square alone.

Restore s . Drop its positive norm square and expand its two cross terms. Put $s_0 = -F^{-1}S_f$, so $\|s_0\|^2 \leq CW_2$. The cross $\langle s, a \rangle$ differs from $\langle s_0, D0 \rangle$ by a quantity bounded by $\epsilon H_{\text{exc}} + C_\epsilon W_2$. To verify this, expand $T^*WT - I$ once: every factor moved onto $D0$ is $ZD0$, Z^*D0 , or KK^*D0 , all controlled by (2.1), (2.2), (5.1). The remaining factors are uniformly bounded. The n terms use (2.2). The leading $\langle s_0, D0 \rangle$ is a finite $\sum p_e$ times a Hermitian fast/slow Clifford matrix of norm $C\kappa/r_e$. Its real form pairs with a_e , since the Gaussian drift has purely imaginary expectation against that Hermitian matrix. Hence it is bounded by $\epsilon h_e + C_\epsilon \kappa^2 r_e^{-2}$. For the internal part of $\langle s, Ap_s \rangle$, define $w_a = \sum_{e \text{ incident } a} r_e^{-3}$. Factoring $M^{-1} = R^{-1}(MR^{-1})^{-1}$ gives, for its coefficient $\beta_{s,a}$,

$$\begin{aligned} & \|\beta_{s,a}v\|^2/w_a \\ & \leq CW_2 \sum_{e \text{ incident } a} r_e \|Y_e v\|^2 \\ & \leq Cr_*^{-3} H_{\text{exc}}[v] + CW_2[v]. \end{aligned} \tag{6.3}$$

Weighted Young loses $\gamma \sum_a w_a t_a$, plus γ^{-1} times the right side. This is a vanishing loss of the complete colored internal forms because

$$\begin{aligned} t_a & \leq H_a + c_a \alpha_a, \\ \sum_a w_a \alpha_a & \leq C\kappa W_2. \end{aligned} \tag{6.4}$$

The weights depend only on centers, so no internal derivative of a weight occurs. Center spin cross terms use ordinary center Young. The direct linear-Y Yukawa terms obey $\epsilon H_{\text{exc}} + C_\epsilon W_2$ by (3.2) with one Y and a bounded Hermitian Clifford matrix.

All spin estimates remain on the whole colored internal module. They make no physical-sector assumption.

7. Complete differential-connection curvature payment

Write the internal component of $\beta = A^*a$ as $\beta_j = b_j(Y; Z, A) \cdot p_F$. This is real and fast-bosonic, hence commutes with internal coordinates and internal Clifford matrices. Its symmetric version is

$$\widehat{\beta}_j = \beta_j - (i/2) \text{div}_F b_j.$$

The averaged sixteen internal charges can therefore be used with $p_A + \widehat{\beta}$. There is no additional internal-potential anticommutator; the only algebraic correction is a finite Clifford contraction of the fast-bosonic curvature

$$\begin{aligned} F_{jk} & = i[P_j, P_k] = p_{\text{sym}}(\partial_j b_k - \partial_k b_j + [b_j, b_k]), \\ P_j & = p_{A_j} + \widehat{\beta}_j. \end{aligned} \tag{7.1}$$

The nonsymmetric-to-symmetric kinetic conversion also has the exact scalar correction

$$\begin{aligned} & (1/2)X_j(\operatorname{div}_F b_j) + (1/4)(\operatorname{div}_F b_j)^2, \\ & X_j = \partial_{A_j} + b_j \cdot \partial Y. \end{aligned} \tag{7.2}$$

The companion proof Appendix *G* establishes the actual rootwise bound

$$\begin{aligned} & |\langle v, F_{jk}v \rangle| + |\langle (7.2)v, v \rangle| \\ & \leq C(\sigma^2 + r_*^{-3})H_{\text{exc}}[v] + CW_2[v]. \end{aligned} \tag{7.3}$$

The proof does not infer (7.3) from a norm bound alone. Internal differentiation of the explicit rootwise source decomposition preserves the same-edge and triangle source classes and adds one factor at most C/r_* . Thus $\|r_*\partial A\hat{\beta}v\|^2 \leq C[(\sigma^2 + r_*^{-3})H_{\text{exc}}[v] + W_2[v]]$. Weighted Cauchy with $\|r_*^{-1}v\|^2 \leq W_2[v]$ pays the derivative curvature. Symmetry bounds each commutator-curvature expectation by $2\|\hat{\beta}_jv\|\|\hat{\beta}_kv\|$. The scalar divergence correction is bounded by CW_2 from the actual first and second coefficient derivatives. Averaged-charge Cauchy-Schwarz therefore yields

$$H_I(p + \beta) \geq (1 - \epsilon)H_I - C_\epsilon(\sigma^2 + r_*^{-3})H_{\text{exc}} - C_\epsilon W_2, \tag{7.4}$$

retaining the entire interacting H_I , not a fixed fraction of t_A alone.

8. Density, vacuum defect, and the final allocation

The density term is bounded by CW_2 uniformly on the pairwise tube. Indeed MR^{-1} and its inverse are uniformly bounded; each slow or fast derivative of their explicit normalized entries is $\leq C/r_*$, and a second derivative is $\leq C/r_*^2$. The principal co-metric and the required normal-covariant coefficient derivatives obey the same bounds. In $d^{-1}\operatorname{div}(a\partial d)$, these give $C/r_*^2 \leq CW_2$. This argument uses actual normalized matrices and does not compactify a set of gap ratios.

The frozen $E_{0,\text{total}}$ is paid separately by the incident commutator-square result Appendix *B*. Directly, $|E_{0,\text{total}}| \leq C\sum_a w_a V_a \leq Cr_*^{-3}H_I + C\kappa W_2$, with fixed endpoint multiplicities absorbed in C . This uses $V_a \leq H_a + c_a\alpha_a$ and the incident bound $w_a\alpha_a \leq C\kappa W_2$. The first term is a vanishing complete-internal-form loss; the second remains in the allowed inverse-square remainder. Thus this local reserve does not require a separate center-Hardy argument. $E_{0,\text{total}}$ is never included in H_{exc} or a virtual excitation denominator.

Equations (7.3), (3.1), (5.5), (6.1)–(6.4), the density bound, and the exact Hamiltonian potential/Yukawa inventory give, for every desired $\delta > 0$, constants $\kappa_\delta, \sigma_\delta, R_\delta$ and C_δ , independent of all gap ratios,

$$H_{\text{full}}[v] \geq (1 - \delta)(H_I[v] + T_{\text{cent}}[v] + H_{\text{exc}}[v]) - C_\delta W_2[v], \tag{8.1}$$

on the tube with $\kappa \leq \kappa_\delta, \sigma \leq \sigma_\delta$ and $r_* \geq R_\delta$. Uniformity for nested smaller fast tubes follows by proving the estimate at the fixed positive width σ_δ and restricting its core; C_δ is not obtained by substituting a varying σ into σ^{-1} . A nonnegative fraction of $\|Ap_s v\|^2$ and

V_4 may also be retained by allocating their positive terms explicitly. This is not a reserve of the original full Gauss square.

For v fiberwise orthogonal to the adapted product vacuum, $H_{\text{exc}}[v] \geq cr_* \|v\|^2$. Since $W_2 \leq (\#\text{edges})r_*^{-2}$, increasing R_δ absorbs $C_\delta W_2$ into δH_{exc} . The resulting complementary reserve is near-unit in the full colored H_I , center kinetic, and shifted excitation form. Starting (8.1) with $\delta/2$ gives a final displayed coefficient $1 - \delta$ after this absorption.

For a projection-stable tube realization first fix a positive cutoff width $\sigma_0 \leq \sigma_\delta$, then choose the exterior radius so $\sigma_0^2 r_*^3$ is large. Choose a smooth χ_0 with support strictly inside the validity tube and equal to one on a smaller tube, using the smooth harmonic minimum of pair separations. Put $U_\sigma = \chi_0 U / \|\chi_0 U\|$. The fixed-rank Gaussian tail gives $\|U_\sigma - U\| \leq \tau \leq C \exp(-c\sigma_0^2 r_*^3)$, uniformly in all gap ratios. For v fiberwise orthogonal to U_σ , $|\langle U, v \rangle| \leq \tau \|v\|$, hence $H_{\text{exc}}[v] \geq cr_*(1 - \tau^2)\|v\|^2$. Both $U_\sigma f$ and $v = u - U_\sigma f$ have Dirichlet form support. Thus the same complementary reserve holds for this cutoff projection once the exterior radius is enlarged. Gaussian derivative tails needed in a subsequent mixed-source calculation are separate estimates.

The reserve retains the actual normal-bundle center derivative, with no replacement by a bounded Berry coefficient. On the low vacuum line the scalar pure-internal and mixed center/internal Berry comparison is the rootwise full-charge lemma in source Section 4.2 in Appendix F: its costs are $C_N(\kappa + \epsilon^{-1}\kappa^4)W_2$. This does not require a flat determinant line.

No claim about mixed low/complement source estimates follows merely from (8.1), especially if those estimates spend the full original Gauss square instead of the actual reserve left by (6.1). The full inverse-square Feshbach conclusion remains separate from this complementary theorem.

I Local Schur comparison on the form domain

Technical derivation for review. Equation and subsection numbers in this appendix are local to the source derivation. Historical local labels are retained, including numbering gaps or nonsequential labels. The accompanying source notes retain the original expressions for comparison.

1. Statement and inputs

Fix a partition of a finite N . Put

$$r_* = \min_e r_e, \quad W_p = \sum_e r_e^{-p},$$

$$H_s = H_I + T_C, \quad H_{\text{exc}} = \sum_e (H_{\text{fast},e} - E_{0,e}).$$

The internal tube is $\alpha_a + \alpha_b \leq \kappa r_{ab}$. The fast tube has $|Y| < \sigma r_*$. There is no bound on ratios of unrelated r_e and no internal spectral-gap assumption. H_I is the complete interacting colored internal form. T_C on the full fiber retains the intrinsic normal connection. On the slow bundle, H_s denotes the center-plus-internal model identified by internal radial transport with the $A = 0$ vacuum line. Its unperturbed internal forms are the colored ones; no lower-rank physical-sector Hardy theorem is used.

For every $\epsilon > 0$, there are positive κ, σ, R and a smooth even normalized cutoff vacuum U_σ supported strictly inside the fast tube such that, for $r_* \geq R$,

$$\begin{aligned} (1 - \epsilon)H_s[f] - \epsilon W_2[f] \\ \leq F_\sigma[f] \\ \leq (1 + \epsilon)H_s[f] + \epsilon W_2[f], \end{aligned} \tag{1.1}$$

where the actual zero-energy variational Hamiltonian Schur form is

$$\begin{aligned} F_\sigma[f] = \inf_{v \in Q(H), U_\sigma^* v = 0} \\ H[U_\sigma f + v]. \end{aligned} \tag{1.2}$$

Here $Q(H)$ denotes the CLOSED QUADRATIC-FORM DOMAIN of the Hamiltonian, not its operator domain $D(H)$. Orthogonality is fiberwise over the slow variables. For every f in $Q(H_s)$, the infimum is attained by a unique v_f in $Q(H)$ with $U_\sigma^* v_f = 0$. The form is closed on the slow model's form domain, interpreted with Dirichlet support on the selected branch. The constants depend on the fixed partition and ϵ , never on $r_{\max}/r_*$.

Inputs proved in the companion notes are:

1. The exact intrinsic kinetic identity and normalized matrices from Appendix A.
2. The rootwise all-vector connection estimates in Appendix E and differential-curvature identities in Appendix G.

3. The actual H1/H2 inventory, finite source graph grading, and operator identity

$$\|W - S^* H_{\text{exc}}^{-1} S\| \leq C_N \kappa W_2, \quad (1.3)$$

from Appendix *D*. Here $S = H1U$ is the actual source, not a transported replacement.

4. The graph-local complete-slow-form bound for $Z = -H_{\text{exc}}^{-1}S$:

$$H_s[Zf] \leq CR^{-3}H_s[f] + C\kappa W_2[f] + CW_5[f], \quad (1.4)$$

including spectator-vacuum derivatives, from Appendix *C* and the actual-source continuation.

5. The colored complementary reserve. A fixed positive part of the actual square $J[v] = \|Ap_s v\|^2$ can be retained:

$$H[v] \geq (1 - \delta)(H_s[v] + H_{\text{exc}}[v]) + c_0 J[v] \quad (1.5)$$

on the cutoff complement, after choosing the parameters and exterior radius. For example $c_0 = 1/2$ is allowed. This is NOT another copy of the original full Gauss square.

This note supplies the mixed and diagonal-remainder estimates needed to assemble these inputs. It never inverts H_I or expands a resolvent in internal momenta.

2. Exact error-column factorization, including spin

Use the common positive-fast/positive-slow convention

$$\begin{aligned} H_{\text{kin}} &= \|p_s u\|^2 + \|p_F u\|^2 + \|au + Ap_s u + su\|^2, \\ p_s &= (g_C^{-1/2} p_C, p_I), \\ a &= W^{1/2} M^{-*} (C + N_Y)^* p_F, \\ s &= -W^{1/2} M^{-*} S_f. \end{aligned}$$

The scalar density term is added separately. Set

$$\begin{aligned} d_0 &= F^{-1} C^* p_F, \quad P = L - J_2, \quad Z_m = F^{-1} P^*, \\ T &= (I + Z_m)^{-1}, \quad n = F^{-1} N_Y^* p_F, \\ X &= n - Z_m d_0, \quad s_0 = -F^{-1} S_f, \\ q &= X + s_0, \quad X_1 = n - F^{-1} L^* d_0, \quad X_2 = F^{-1} J_2^* d_0. \end{aligned}$$

The symbol Z_m is an inverse-matrix error, distinct from the dressing Z . Exactly,

$$a + s = W^{1/2} (d_0 + Tq),$$

and, as ordered first-order forms,

$$\begin{aligned}
& \|(a + s)u\|^2 - \|d_0 u\|^2 \\
&= 2 \operatorname{Re} \langle d_0 u, (X_1 + s_0)u \rangle + 2 \operatorname{Re} \langle d_0 u, X_2 u \rangle \\
&\quad - 2 \operatorname{Re} \langle Z_m^* d_0 u, T q u \rangle + \|T q u\|^2 \\
&\quad + \|K^*(d_0 + T q)u\|^2.
\end{aligned} \tag{2.1}$$

The first line's first term is exactly the fast-orbital/spin part of H1. No derivative has been moved through an inverse matrix. The remaining apparently dangerous $d_0 - X_2$ product factors exactly:

$$\begin{aligned}
& \langle d_0 u, X_2 u \rangle \\
&= \langle C_Y F^{-1} d_0 u, C_Y R^{-1} d_0 u \rangle.
\end{aligned} \tag{2.2}$$

Every factor on the right of (2.1)-(2.2), after removing the displayed H1 term, belongs to the finite error-column list

$$\begin{aligned}
& X_1, X_2, Z_m^* d_0, K^* d_0, \\
& C_Y F^{-1} d_0, C_Y R^{-1} d_0, s_0,
\end{aligned}$$

followed by uniformly bounded left multiplication matrices. Their all-vector estimates are

$$\begin{aligned}
& \|Eu\|^2 \leq C[(\lambda H_{\text{exc}})[u] + W_2[u]], \\
& \lambda = \sigma^2 + r_*^{-3}.
\end{aligned} \tag{2.3}$$

For the spin column the stronger bound $\|s_0 u\|^2 \leq C W_2[u]$ holds. On the uncut vacuum, and on each fixed-degree Gaussian/Fock coefficient vector normalized in its own roots,

$$\|E U f\|^2 \leq C W_2[f]. \tag{2.4}$$

For (2.4), a same-edge $Y_e p_e / r_e$ costs C / r_e ; a triangle $Y_f p_e / r_g$ costs $C \sqrt{r_e / r_f} / r_g$, whose square is bounded by $C W_{2,t}$ by $r_e \leq r_f + r_g$. Root-preserving pair matrices have uniformly bounded normalized coefficients. All inverse matrices in the exact expressions remain bounded LEFT factors. The same proof therefore works on a fixed finite excited degree, with a degree-dependent constant, and does not introduce r_{max} .

For a complementary v , $W_2[v] \leq C R^{-3} H_{\text{exc}}[v]$. Polarization of each product in (2.1)-(2.2) consequently gives, for any $\eta > 0$,

$$\begin{aligned}
& |R_{\text{fast}}[U f, v]| \leq \eta H_{\text{exc}}[v] \\
& \quad + C_\eta (\sigma^2 + R^{-3}) W_2[f].
\end{aligned} \tag{2.5}$$

This includes spin-fast and spin-square remainders. It is the central reason a finite inverse-square constant is not left unpaid: no Taylor remainder is ever paired directly with an unassigned d_0 vacuum norm.

For diagonal matching, retain the quadratic terms in (2.1), replace T by I , q by $X_1 + s_0$, Z_m by $F^{-1}L^*$, and retain (2.2) and $\|K^*d_0\|^2$. These are precisely the corresponding H2 terms. Each discarded product has one additional uniformly $O(\sigma)$ factor and two low error columns. Hence

$$|R_{\text{fast}}[Uf] - R_{\text{fast},H2}[Uf]| \leq C\sigma W_2[f]. \quad (2.6)$$

One may obtain an additional inverse-radius improvement from Gaussian moments, but it is unnecessary. The order of the two operations matters: first perform the exact error-column factorization, then bound the inverse difference. Doing them in reverse would expose the forbidden remote vacuum mass.

3. Other nondifferential fast terms

The actual V_3 and linear-Y Yukawa are exactly the remaining H1 terms; there is no omitted higher polynomial potential. They are boson-odd, so their diagonal expectation on U or an even cutoff U_σ is zero.

The quartic potential is a sum of squares of quadratic fast-coordinate contractions. In addition to the distinct-root Q/P estimate already proved, the repeated-root estimate is

$$\|\chi Y_e^2 u\|^2 \leq C\sigma^2 h_e[u] + Cr_e^{-2} \|u\|^2. \quad (3.1)$$

Split the input into Q_e and P_e , bound one Y_e by σr_* on Q_e , use $\|Y_e Q_e u\|^2 \leq Cr_e^{-2} h_e[u]$, and use the exact fourth Gaussian moment on P_e . This gives

$$\begin{aligned} 0 \leq V_4[u] &\leq C\sigma^2 H_{\text{exc}}[u] + CW_2[u], \\ V_4[Uf] &\leq CW_2[f]. \end{aligned} \quad (3.2)$$

Thus its low/complement cross obeys the same bound as (2.5). Its full diagonal expectation belongs to W and is retained there.

The density multiplier satisfies $|V_d| \leq CW_2$. Its mixed term is paid by the complementary L2 gap:

$$|\langle Uf, V_d v \rangle| \leq \eta H_{\text{exc}}[v] + C_\eta R^{-3} W_2[f]. \quad (3.3)$$

The leading coefficient $V_{d,2}$ is the one specified by the exact H2 inventory, including the slow density, fast Hessian, and orbital ordering terms. Differentiating the normalized matrices in the exact density formula and subtracting their zero-fast-coordinate jets gives

$$|V_d - V_{d,2}| \leq C\sigma W_2 \quad (3.4)$$

on the tube. Each normalized inverse and its required derivatives are bounded in its own root scale. More explicitly, V_d uses two coefficient derivatives, and one additional fast derivative of that finite rational expression is bounded by C/r_*^3 ; integrating it from $Y = 0$ gives $C|Y|/r_*^3 \leq C\sigma W_2$. The leading zero-fast-coordinate coefficient includes the quadratic determinant jet before fast differentiation. This is a density-coefficient estimate, not a bound on d_0^2 . Equations (3.3)-(3.4) settle both density uses.

The frozen E_0 is a separate scalar multiplier. It has zero U-v cross because it commutes with the vacuum projection. Its incident payment is

$$|E_0[u]| \leq CR^{-3}H_I[u] + C\kappa W_2[u]. \quad (3.5)$$

It is never included in H_{exc} or in a virtual denominator.

4. Scalar fast-slow orbit cross: complete charges instead of bare t_I

Write $\beta = A^*a$, with internal components $\beta_j = b_j(Y; s) \cdot p_F$. The b_j are real, scalar in internal Clifford factors, and commute with all internal potential-charge coefficients. Let

$$\begin{aligned} d_j &= \text{div}_F b_j, \quad \widehat{\beta}_j = \beta_j - (i/2)d_j, \\ C_\alpha(\widehat{\beta}) &= \sum_j c_{\alpha,j} \widehat{\beta}_j. \end{aligned}$$

The differentiated connection lemma and its explicit rootwise decomposition imply

$$\begin{aligned} \sum_j \|\widehat{\beta}_j v\|^2 &\leq C(\lambda H_{\text{exc}}[v] + W_2[v]), \\ \sum_j \|\widehat{\beta}_j Uf\|^2 &\leq C(\kappa^2 + \sigma^2)W_2[f], \end{aligned} \quad (4.1)$$

and

$$\begin{aligned} \sum_{jk} \|(\partial_{A_j} \widehat{\beta}_k) Uf\|^2 \\ \leq Cr_*^{-2} W_2[f]. \end{aligned} \quad (4.2)$$

The SMALL low-leg divergence estimate in (4.1) is essential. Directly, $b = A^*B$ with $A = O(\sigma)$, $B = O(\kappa + \sigma)$, $\partial_Y A = O(r_*^{-1})$ and $\partial_Y B = O(r_*^{-1})$. Hence

$$|\text{div}_F b| \leq C(\kappa + \sigma)/r_*.$$

Together with the stronger rootwise estimate for A^*d_0 and the $O(\sigma)$ factor multiplying $a - d_0$, this proves the second line of (4.1). It is not inferred from a coarse $O(1/r_*)$ divergence bound.

4.1 Exact linear charge identity

Polarize the following identity, initially on the smooth core:

$$\begin{aligned}
& 2 \operatorname{Re} \sum_j \langle p_j u, \beta_j u \rangle \\
&= 2 \operatorname{Re}(1/16) \sum_\alpha \langle Q_\alpha u, C_\alpha(\hat{\beta}) u \rangle \\
&\quad - R_{\text{der}}[u] + (1/2) \sum_j \langle u, (\partial_{A_j} d_j) u \rangle.
\end{aligned} \tag{4.3}$$

Here R_{der} is the fixed bounded-Clifford contraction of $\partial_j \hat{\beta}_k - \partial_k \hat{\beta}_j$. To verify it, expand the exact shifted averaged-charge identity and subtract the beta-square. The beta-beta curvature in $C_\alpha(\hat{\beta})^2$ cancels the beta-beta curvature in the full shifted charge identity. Only the derivative curvature remains. The raw-versus-symmetric cross contributes $+(1/2)\partial_A d$, with this sign. No anticommutator with the internal potential survives the spinor average. The first term in the polarized identity contains

$$\langle Q(Uf), C(\hat{\beta})v \rangle \quad \text{and} \quad \langle C(\hat{\beta})Uf, Qv \rangle.$$

The first is bounded using (4.1), the complementary gap, and

$$(1/16) \sum \|Q_\alpha(Uf)\|^2 \leq C(H_s[f] + W_2[f]).$$

The second uses the full $H_I[v]$ reserve and the small low-leg estimate in (4.1). No constant fraction of bare t_I is spent. Since each derivative-curvature operator is symmetric, it can be moved onto U in a mixed pairing. Equation (4.2) and the complementary gap then give a $C_\eta R^{-3} W_2$ cost. The scalar term has the same payment.

Consequently the complete internal scalar-orbit mixed form obeys

$$\begin{aligned}
& |R_{\text{orb},I}[Uf, v]| \leq \eta(H_I[v] + H_{\text{exc}}[v]) \\
& \quad + C_\eta \lambda H_s[f] \\
& \quad + C_\eta(\kappa^2 + \sigma^2 + R^{-3})W_2[f].
\end{aligned} \tag{4.4}$$

This is uniform over unrestricted internal momenta and colored internal states. The center component obeys the same inequality by ordinary center-kinetic Cauchy-Schwarz, with its actual normal connection retained. Normal-frame generators are not estimated by a vacuum Berry scalar on arbitrary v .

4.2 Independent incident-row verification

There is a second direct check of the high-momentum issue. With $U = g$ times its fermion vacuum and real Gaussian g ,

$$\begin{aligned}
& (\beta_j + \beta_j^*)U \\
& = i(2b_j \cdot \Omega Y - \operatorname{div}_F b_j)U.
\end{aligned}$$

Ground-state integration by parts therefore gives

$$\begin{aligned}
& \|H_{\text{exc}}^{-1/2}(\beta_j + \beta_j^*)U\|^2 \\
& \leq C\|b_j U\|^2.
\end{aligned} \tag{4.5}$$

For an internal row belonging to block a , factor the exact coefficient as

$$b_a = I_{Y,a} F^{-1} [T^* W M^{-*} (C + N_Y)^*].$$

The bracket is uniformly $O(\kappa + \sigma)$, while $I_{Y,a} F^{-1}$ contains only Y_e/r_e on edges incident to a . Thus

$$\begin{aligned}
& \|b_a U\|^2 \leq C(\kappa + \sigma)^2 w_a, \\
& w_a = \sum_{e \text{ incident } a} r_e^{-3}.
\end{aligned} \tag{4.6}$$

This proves directly that integrating the single slow derivative in the orbit cross cannot create an unrelated internal-radius loss. It also accounts for the adjoint fast divergence, rather than dropping it. Either (4.3) or (4.5)-(4.6) supplies the needed first-derivative estimate; the proof above uses (4.3).

4.3 Diagonal orbit remainder

Let β_1 be the first homogeneous slow-orbit coefficient: use the linear slow column with F^{-1} , the leading $C_Y^* + FK$ center term, and d_0 . Its raw diagonal compression, including derivatives of its coefficient at $A = 0$, is retained in W . Together with the corresponding $V_{d,2}$ orbital-density term, it equals the V_{orb} in the density-shifted inventory. In particular the inventory term $-2 \sum_{\mu} \rho_{0,\mu} j_{\mu}$ belongs to $V_{d,2}$ in the present unshifted-kinetic convention; it is counted exactly once, not inserted a second time in the raw β_1 compression. Only the SUM of these contributions is identified with the H2 scalar. The raw β_1 contribution is NOT discarded on the grounds that β_1 vanishes at $A = 0$.

The exact identities for $A^* d_0$ in the rootwise helper express $\beta - \beta_1$ as a sum with one extra $O(\sigma)$ factor multiplying a controlled error column. They give

$$\begin{aligned}
& \|(\beta - \beta_1)U\|^2 + \|(\operatorname{div}(b - b_1))U\|^2 \leq C\sigma^2 W_2, \\
& \|\partial_A(\beta - \beta_1)U\| \leq C\sigma r_*^{-1} W_2^{1/2}.
\end{aligned}$$

Differentiate those identities, not an abstract norm bound: the extra fast-degree factor remains when F or $F^{-1}C^*$ is differentiated. Applying (4.3) to this difference gives, for any $\eta > 0$,

$$|R_{\text{orb,diag}}[f]| \leq \eta H_s[f] + C_\eta(\sigma + \sigma^2)W_2[f]. \quad (4.7)$$

This retains the exact leading coefficient while paying only its true higher-order remainder.

5. Spin-slow, slow-orbit square, and induced center metric

5.1 Spin-slow cross

Put $c = A^*s$. Its internal row a satisfies the exact incident coefficient bound

$$|c_a|^2 \leq CW_2 \sum_{e \text{ incident } a} |Y_e|^2/r_e^2, \quad (5.1)$$

$$\|c_a U\|^2 \leq Cw_a W_2.$$

For the mixed form, integrate its single slow derivative onto the smooth coefficient cU . This is legitimate for the complete matrix c , with no commutation claim about internal Clifford matrices. The coefficient of $\partial_{Aa}f$ has norm squared bounded by $Cw_a W_2$. The complementary gap therefore gives the weighted derivative cost

$$C(W_2/r_*) \sum_a w_a t_a[f] \leq CR^{-3}[R^{-3}H_I[f] + \kappa W_2[f]]. \quad (5.2)$$

The exact inverse derivative formula, incident left factor $I_{Y,a}F^{-1}$, and finite Gaussian derivative moments give

$$\begin{aligned} & \sum_a \|\partial_{Aa}(c_a U)\|^2 \\ & \leq Cr_*^{-2}(\sum_a w_a)W_2 + C(\sum_a w_a)W_2^2 \\ & \leq CR^{-5}W_2. \end{aligned} \quad (5.3)$$

Spectator-vacuum derivatives are included in the W_2^2 term. Center derivatives have the same or better bound and are paid by T_C . After the complementary gap, (5.2)-(5.3) are smaller than the mixed budget in (4.4).

For the DIAGONAL, (5.1) alone would leave a fixed W_2 constant and is insufficient. The leading coefficient c_1 is boson-odd: its internal part is $-I_Y F^{-2}S_f$, with the corresponding linear center row. U , its slow derivatives, and the even cutoff are boson-even. Its diagonal cross is exactly zero. Every term of $c - c_1$ has an additional inverse/metric difference of size $O(\sigma)$, so

$$\|(c_a - c_{1,a})U\|^2 \leq C\sigma^2 w_a W_2.$$

Weighted Young and $\sum_a w_a \alpha_a \leq C\kappa W_2$ yield only

$$\eta R^{-3} H_s[f] + C_\eta(\kappa + \sigma^2) W_2[f]. \quad (5.4)$$

This is the specific parity use that prevents an unpaid diagonal constant.

5.2 The positive square actually retained by completion

Let $J[u] = \|Ap_s u\|^2$. Its low-leg estimate is

$$J[Uf] \leq CR^{-3} H_s[f] + C\kappa W_2[f] + CW_5[f]. \quad (5.5)$$

For derivatives on f , the exact incident factorization gives $\sum_a w_a t_a$ and the analogous center coefficient. Convert only these incident weights using

$$\sum_a w_a t_a \leq CR^{-3} H_I + C\kappa W_2.$$

Derivatives on U are finite degree-two Gaussian/Fock vectors. Their rootwise moments give W_5 , including spectator products $W_3 W_2 \leq C_N W_5$. The Berry connection adds its already proved smaller incident contribution. Thus

$$\begin{aligned} & |\langle Ap_s Uf, Ap_s v \rangle| \\ & \leq \eta J[v] + C_\eta [R^{-3} H_s[f] + \kappa W_2[f] + W_5[f]]. \end{aligned} \quad (5.6)$$

Equation (5.6) uses only the true postcompletion reserve J . No original full Gauss square is used.

5.3 Center metric

The direct center coefficient differs from identity by

$$g_C^{-1} - I = -K^*(I + KK^*)^{-1}K.$$

It is $O(\sigma^2)$ on arbitrary tube vectors. On a Gaussian low leg its squared coefficient norm is $O(R^{-6})$; derivatives on U have the corresponding W_5 -or-smaller moments. Ordinary center-form Young controls the mixed term by $\eta T_C[v] + C_\eta R^{-6} T_C[f] + C_\eta W_5[f]$. Its diagonal has bound $CR^{-3} T_C[f] + CW_5[f]$. No internal derivative is involved.

5.4 Direct slow kinetic cross

The preceding orbit estimates do not replace the cross from the DIRECT center and internal kinetic forms. For v perpendicular to U , differentiate $U^*v = 0$ and integrate the remaining slow derivative once. In a covariant vacuum-line frame the complementary source is

$$\begin{aligned} & -2 \sum_j (1-P)(\nabla_j U) \nabla_j f \\ & - (1-P) \sum_j \nabla_j^2 U f. \end{aligned} \tag{5.7}$$

This is a form identity and contains no second derivative of f . Internal potential and Yukawa terms commute with the vacuum injection and have zero mixed compression. The same is true of E_0 . Normal-bundle derivatives are kept covariantly in (5.7).

An internal derivative in block a differentiates only incident pair vacua. After removing its scalar Berry part, each term has a fixed excited sector on that pair, norm C/r_e , and gap at least cr_e . Its squared H_{exc} -dual norm is therefore at most Cw_a . The center derivative satisfies the analogous sum of pair weights. Second derivatives have active-pair and two-pair finite excitation terms, of norm at most C/r_e^2 or $C/(r_e r_f)$. Their dual squared norm is bounded by CW_5 ; spectator products are included using $W_3 W_2 \leq C_N W_5$. Thus ordinary source/gap Young gives

$$\begin{aligned} 2|H_s[Uf, v]| & \leq \eta H_{\text{exc}}[v] \\ & + C_\eta [R^{-3} H_s[f] + \kappa W_2[f] + W_5[f]]. \end{aligned} \tag{5.8}$$

The conversion on the right is incident: $\sum_a w_a t_a \leq CR^{-3} H_I + C\kappa W_2$. It is performed before replacing w_a by its maximum. The cutoff version gains only the source-specific exponentially small errors in Section 7. This explicitly includes the direct slow derivative source in the mixed ledger.

6. Summary of the mixed and diagonal estimates

Let $B[v] = H_s[v] + H_{\text{exc}}[v] + J[v]$. After a fixed finite allocation of η to the preceding terms, for every $\eta > 0$ the exact mixed form has the decomposition

$$H[Uf, v] = \langle Sf, v \rangle + R_{\text{mix}}[f, v],$$

with

$$\begin{aligned} & 2|R_{\text{mix}}[f, v]| \\ & \leq \eta B[v] + a_\eta(\kappa, \sigma, R) H_s[f] \\ & + b_\eta(\kappa, \sigma, R) W_2[f], \end{aligned} \tag{6.1}$$

where one can choose finite constants such that

$$\begin{aligned} a_\eta &\leq C_\eta(\sigma^2 + R^{-3}), \\ b_\eta &\leq C_\eta(\kappa + \kappa^2 + \sigma^2 + R^{-3}). \end{aligned} \tag{6.2}$$

The exact diagonal is

$$H[Uf] = H_s^B[f] + E_0[f] + W[f] + R_{\text{diag}}[f], \tag{6.3}$$

and for an independently chosen $\theta > 0$,

$$\begin{aligned} |R_{\text{diag}}[f]| &\leq \theta H_s[f] \\ &+ C_\theta[\kappa + \sigma + \sigma^2 + R^{-3}]W_2[f] \\ &+ CR^{-3}H_s[f]. \end{aligned} \tag{6.4}$$

The H_s^B comparison is the full averaged-charge Berry comparison from source Section 4.2 of Appendix F:

$$\begin{aligned} |H_s^B[f] - H_s[f]| \\ \leq \theta H_s[f] + C(\theta^{-1}\kappa^4 + \kappa)W_2[f]. \end{aligned} \tag{6.5}$$

Equations (3.5), (6.3)-(6.5) keep every internal potential and Yukawa term. No frozen slow-momentum resolvent has appeared.

7. Cutoff vacuum, derivative tails, and source parity

Fix $\sigma > 0$ before increasing R . Let $h = (\sum_e r_e^{-2})^{-1/2}$, and choose a smooth even radial $\chi = \chi_0(|Y|^2/(\sigma_0^2 h^2))$, with $0 < \sigma_0 < c_N \sigma$ fixed, equal to one on an inner tube and supported strictly inside the validity tube. Put

$$U_\sigma = \chi U / n, \quad n = \|\chi U\|.$$

This is intrinsic and residual-equivariant. Its support is projection-stable. The normalized Gaussian coordinates $\xi_e = r_e^{1/2} Y_e$ have uniformly positive bounded covariance matrices. On the cutoff transition region,

$$\sum_e |\xi_e|^2 \geq c\sigma_0^2 r_*^3.$$

Consequently any fixed-degree root-normalized Gaussian polynomial has a tail bounded by

$$\tau_R = C_{N,\sigma_0} \exp(-c_N \sigma_0^2 R^3), \quad (7.1)$$

after absorbing fixed powers of R into the exponential. These estimates do not contain $r_{\max}$. They apply after the rootwise error-column factorization, and to finite graph source/dressing coefficients with their own graph weights.

One must not claim that $\|p_F(U_\sigma - U)\|$ has a root-independent bound: a far vacuum derivative can indeed carry an arbitrarily large zero-point norm. Instead use the excitation ground-state transform. Since

$$\begin{aligned} H_{\text{exc}} U &= 0, \\ H_{\text{exc}}[\chi U] &= \int |\nabla_F \chi|_G^2 |U|^2, \end{aligned}$$

no spectator vacuum energy occurs. In the operator cutoff commutator, $\nabla_e \chi$ is a scalar multiple of Y_e/h^2 ; its product with $\Omega_e Y_e$ has a root-normalized quadratic coefficient of order h^{-2} , because Ω_e/r_e is bounded. The same observation applies to the active finite graph states in Z and to H1: replacing a p_e by $\nabla_e \chi$ adds a factor bounded by $C/(r_e h^2)$, with a Gaussian tail. Thus the only cutoff errors required here satisfy, uniformly in all root ratios,

$$\begin{aligned} |\delta \text{ diagonal}| &\leq \tau_R(H_s[f] + W_2[f]), \\ |\delta \text{ mixed}| &\leq \eta B[v] + C_\eta \tau_R(H_s[f] + W_2[f]), \\ H_{\text{exc}}[\chi Z f/n] &= H_{\text{exc}}[Z f] + O(\tau_R W_2[f]), \end{aligned} \quad (7.2)$$

with the analogous graph-local H_s and J estimates. Slow cutoff derivatives use $|\partial h|/h \leq C/r_*$; internal derivatives of h vanish. Derivatives of n are controlled by the same normalized tail calculation. The normal connection preserves $|Y|^2$, so it creates no extra radial-cutoff generator.

S and Z are boson-odd. The cutoff is even, hence $\chi Z/n$ is EXACTLY orthogonal to U_σ . The leading cutoff H1 source is also odd and is orthogonal to the uncut U , so its shifted-fast inverse never encounters the vacuum kernel. Replacing it by S in the mixed identity costs only (7.2). In the lower bound the source completion with the uncut H_{exc} is valid for every v , whether or not v is exactly orthogonal to U : S is orthogonal to its kernel.

Finally v perp U_σ has $\|Pv\| \leq \tau_R \|v\|$ and therefore

$$H_{\text{exc}}[v] \geq cr_*(1 - \tau_R^2) \|v\|^2. \quad (7.3)$$

All estimates (6.1)-(6.4) consequently hold for U_σ and its genuine Dirichlet complement after adding the vanishing τ_R terms.

8. Lower Schur comparison

Choose the reserve loss δ and the mixed allocation η small. A fixed positive fraction of J is available, so η can be chosen below c_0 . Equations (1.5) and (6.1) leave

$$\begin{aligned}
& H[U_\sigma f + v] \\
& \geq H[U_\sigma f] + c_s H_s[v] + c_J J[v] \\
& \quad + (1 - \delta - \eta) H_{\text{exc}}[v] + 2 \operatorname{Re} \langle Sf, v \rangle \\
& \quad - a_\eta H_s[f] - b_\eta W_2[f],
\end{aligned} \tag{8.1}$$

where c_s and c_J are positive after the finite allocation. Completing only the shifted fast form gives

$$\begin{aligned}
& (1 - \delta - \eta) H_{\text{exc}}[v] + 2 \operatorname{Re} \langle Sf, v \rangle \\
& \geq -(1 - \delta - \eta)^{-1} \langle f, S^* H_{\text{exc}}^{-1} S f \rangle.
\end{aligned} \tag{8.2}$$

The virtual multiplier is $\leq C W_2$. Therefore replacing its coefficient by one costs only $C(\delta + \eta) W_2$. Use (1.3), (3.5), and (6.3)-(6.5). Their total error is made smaller than $\epsilon(H_s + W_2)$ in the parameter order below. Taking the infimum proves the lower half of (1.1). One may retain a further small positive fraction of $H_s[v] + H_{\text{exc}}[v] + J[v]$ for the form-domain argument, at an additional $O(\eta W_2)$ cost.

9. Upper trial and the complete internal-source graph bound

Use the genuine complementary trial

$$v_f = \chi Z f / n, \quad Z = -H_{\text{exc}}^{-1} S.$$

The actual graph components obey

$$\begin{aligned}
\|Z_e\|^2 & \leq C r_e^{-3}, \\
\|Z_t\|^2 & \leq C w_t / s_t, \\
w_t & = \sum_{e \in t} r_e^{-2}, \quad s_t = \sum_{e \in t} r_e,
\end{aligned}$$

and their full derivatives include the spectator term $m_g W_2$. The complete interacting internal estimate is (1.4). At a graph vertex, $\alpha_a m_g \leq C \kappa W_{2,g}$; outside the graph the potential charge commutes with the even graph map, and only differentiated spectator vacua remain. This is why no α_a from an unrelated far block can multiply a nearest-root dressing norm. The rootwise graph proof covers arbitrary momenta of f .

There is also a direct incident estimate for the retained orbit square:

$$\begin{aligned}
J[Zf] & \leq C R^{-6} H_s[f] \\
& \quad + C \kappa R^{-3} W_2[f] + C \sigma^2 W_5[f].
\end{aligned} \tag{9.1}$$

Indeed each fixed-degree graph state has the same Y_e second-moment scale C/r_e on active and spectator roots. The coefficient of $t_a[f]$ is at most $C w_a \sum_g m_g$. Convert $w_a t_a$ BEFORE

using $\sum_g m_g \leq CR^{-3}$. Derivatives on Z use the full active-plus-spectator derivative bound and the pointwise coefficient $\|A\| \leq C\sigma$, giving $C\sigma^2 W_5$. This proves (9.1) without a global internal kinetic loss.

The same absolute estimates used in the reserve proof, with the upper Young inequality for the complete shifted charges, give

$$\begin{aligned} H[w] &\leq (1 + \delta)(H_s[w] + H_{\text{exc}}[w]) \\ &\quad + C_\delta W_2[w] + CJ[w] + V_4[w]. \end{aligned} \tag{9.2}$$

For the fast part this is immediate from the exact identity (2.1), whose leading signed terms have two-sided bounds. The internal shifted-charge comparison is also two-sided; density, E0 and spin estimates are absolute. If desired V_4 can be absorbed using (3.2). Thus (9.2) uses no unproved upper control of the full original Gauss square.

Apply (9.2) to v_f . Since $H_{\text{exc}}[Zf] = \langle f, S^* H_{\text{exc}}^{-1} S f \rangle$, (1.4), (9.1), (3.2), and the cutoff estimates give

$$\begin{aligned} H[v_f] &\leq (1 + \delta + C\sigma^2)H_{\text{exc}}[Zf] \\ &\quad + CR^{-3}H_s[f] + C\kappa W_2[f] \\ &\quad + C_\delta R^{-3}W_2[f] + O(\tau_R(H_s + W_2)). \end{aligned} \tag{9.3}$$

The mixed remainder (6.1) applied to this trial is small by the same graph estimates. Meanwhile

$$2 \operatorname{Re} \langle Sf, Zf \rangle = -2 \langle f, S^* H_{\text{exc}}^{-1} S f \rangle.$$

Together with the diagonal expansion, this leaves $W - S^* H_{\text{exc}}^{-1} S$ and only the displayed arbitrarily small errors. Equation (1.3) therefore proves the upper half of (1.1). The graph trial is used only for this upper bound; arbitrary complementary vectors remain unrestricted in the lower bound.

10. Simultaneous parameter order

There is no circular order and no unattached constant CW_2 :

1. Fix the partition and desired ϵ . Choose the finite Young allocations θ, η and reserve loss δ so the coefficients they alone contribute, including $C(\delta + \eta)$ from (8.2), use less than epsilon/10 each.
2. Choose κ small enough for the fixed pair functional-calculus neighborhood, $C_\eta \kappa, C\kappa$ from the source defect and E0 payment, and $C\theta^{-1}\kappa^4$ to fit their assigned W_2 budgets.
3. Choose a FIXED positive σ small enough for the reserve, $C_\theta \sigma, C_\eta \sigma^2$ and the complete-charge mixed costs to fit their budgets. Choose a fixed smaller σ_0 for the radial cutoff.
4. Increase R until every constant now fixed times R^{-3} , all complementary W_2 absorptions, and all cutoff-tail quantities fit the remaining budgets.

Constants such as C_δ or $1/\sigma$ are finite before R is chosen. No estimate uses a later inverse power of a parameter to undo an earlier smallness requirement without allowing that earlier parameter to be chosen in the stated order. All estimates are uniform on nested smaller supports at the selected fixed widths.

11. Closed domains and variational meaning

First work on compact slow supports strictly inside the chosen branch and tube. The exact first-order coefficients and density are smooth there, with uniformly positive principal kinetic form and bounded local matrix potentials. The associated Dirichlet form is closed after a scalar shift, with its ordinary local H1 core. U_σ has smooth compact fast support; P_σ and Q_σ map this core into the same form domain. No assertion is made that the uncut P preserves Dirichlet support.

The estimates above imply, after adding a bounded scalar multiple of $\|u\|^2$,

$$\begin{aligned} H[u] + C_R\|u\|^2 &\geq cH_s[f] + cB[v], \\ u &= U_\sigma f + v, \end{aligned}$$

with $c > 0$; $W_2 \leq CR^{-2}$ is bounded. Conversely the diagonal upper estimate bounds $H[U_\sigma f]$ by $C(H_s[f] + \|f\|^2)$. Hence P_σ extends boundedly in the closed quadratic-form norm, equivalently the graph norm of the averaged supercharge stack for the stated supersymmetric realization. This proves $P_\sigma Q(H) \subset Q(H)$; preservation of the Hamiltonian operator domain $D(H)$, or boundedness in its operator graph norm, is not asserted. Exhaustion by compact slow supports gives the same identification on an unbounded selected branch, using these UNIFORM estimates rather than local ellipticity constants that might depend on $r_{\max}$.

The restriction of the closed quadratic form of H to $Q(H) \cap \ker(U_\sigma^*)$ is closed and strictly L2-coercive by (1.5) and (7.3). For f in the slow form domain, the mixed functional is continuous in that complementary form norm by (6.1) and the exact source dual estimate. Riesz representation gives a unique minimizing v_f in $Q(H) \cap \ker(U_\sigma^*)$ for each f in $Q(H_s)$. This is form-domain attainment only; no operator-domain regularity of v_f is inferred. The resulting Schur form is continuous in the shifted H_s form norm, and (1.1) makes the shifted quadratic-form norms equivalent after a bounded scalar shift. It is therefore closed.

These are form statements. They do not claim a bounded operator B or a second-slow-derivative operator identity on arbitrary f , and do not assume an inverse for H_I . All cutoffs, projections, graph dressings, and estimates intertwine under the same-branch residual/color/normal-frame unitaries. Different cluster branches and off-cone selector transport remain outside this local theorem.

13. Explicit module and differentiated-tail checks

The following coefficient identities make the retained-module and differentiated-tail steps explicit.

13.1 Internal potential/Yukawa commute with the actual injection

At a fixed intrinsic branch, the orthogonal Lie-algebra splitting into off-block and retained block/center directions gives the graded tensor product of the fast and slow Clifford modules. The injection is

$$U(A, Z)f = g_{A,Z}(Y)v_F(A, Z) \otimes f,$$

with the identity map on the ENTIRE slow Clifford module. There is no projection onto an internal singlet or internal eigenstate. Each complex bifundamental fast mass has $8n_a n_b$ occupied negative modes, an even number. This remains constant under the gapped continuation, so the product determinant v_F is even. In particular its graded tensor injection intertwines the internal odd Clifford generators with the same retained representation. The internal Hamiltonian's Yukawa term is even in any case: it is a sum of bilinears in retained internal fermions only.

Pointwise in (A, Z) , V_I is scalar multiplication and $Y_I(A)$ acts only on those retained Clifford factors. Therefore, as actual maps,

$$\begin{aligned} V_I U &= U V_I, & Y_I U &= U Y_I, \\ U^* Y_I v &= Y_I U^* v. \end{aligned}$$

This proves their zero mixed compression for $U^* v = 0$. The cutoff is a scalar bosonic multiplier, so the identities hold exactly for U_σ as well.

The pair negative-projector radial transport is implemented on the fast Fock module by a fast-fermion even unitary; its bosonic covariance transport also acts only on the fast module. Both commute with the retained Clifford action. The line may carry a nontrivial scalar Berry connection, but this does not rotate the slow Clifford representation. Differentiation gives

$$p_A(Uf) = U \nabla_A^B f + (1 - P)(p_A U)f.$$

The transverse term is retained in (5.7); the Berry term is retained in H_s^B and compared by the full charge estimate (6.5). Nothing in the radial identification sets the full derivative of U to zero. Residual color transformations act simultaneously on the fast determinant line and the retained module, and U is the corresponding equivariant intertwiner. These pointwise tensor identities therefore descend to the residual-equivariant/physical subspace without requiring individual block singlets.

13.2 Exact beta-tail decomposition before internal differentiation

Use the notation of Section 2, and set

$$\begin{aligned} B &= \Psi^* F^{-1}, \\ B_1 &= [C_Y F^{-1} + K^*; I_Y F^{-1}], \\ R_m &= T^* W T - I. \end{aligned}$$

The center row of B is

$$B_C = g_C^{-1/2} [C_Y + K^*(F + L)] F^{-1},$$

and the internal row is $B_I = I_Y F^{-1}$. Direct substitution into $\beta = A^* a$ gives the exact identities

$$\begin{aligned} \beta &= B T^* W T (d_0 + n), \quad \beta_1 = B_1 d_0, \\ \beta - \beta_1 &= (B - B_1) d_0 + B R_m d_0 + B T^* W T n. \end{aligned} \quad (13.1)$$

The internal part of $B - B_1$ is zero, and its center part has the exact factorization

$$\begin{aligned} (B - B_1)_C d_0 &= (g_C^{-1/2} - I) (C_Y F^{-1} d_0 + K^* d_0) \\ &\quad + g_C^{-1/2} K^* L F^{-1} d_0. \end{aligned} \quad (13.2)$$

The metric difference is $O(\sigma^2)$, and the last term has the explicit $O(\sigma)$ left factor K^* . The other factors in (13.2) are same-root or input-root-denominator triangle columns.

To expand the middle term of (13.1) without exposing a bare frozen vacuum norm, use $T - I = -Z_m T$ and the fact that T commutes with Z_m . Exactly,

$$\begin{aligned} R_m d_0 &= -T^* Z_m^* d_0 - T^* T Z_m d_0 \\ &\quad + T^* K (K^* d_0) - T^* K K^* T (Z_m d_0). \end{aligned} \quad (13.3)$$

Thus it contains only uniformly bounded left factors multiplying $Z_m^* d_0$, $Z_m d_0$, or $K^* d_0$. These are the controlled rootwise columns. For example

$$\begin{aligned} Z_m d_0 &= F^{-1} L^* d_0 \\ &\quad - F^{-1} C_Y^* [C_Y R^{-1} d_0], \\ Z_m^* d_0 &= L F^{-1} d_0 - K [C_Y F^{-1} d_0], \\ K^* d_0 &= C_Y R^{-1} d_0. \end{aligned}$$

The last term of (13.1) has the same explicit $O(\sigma)$ outer B multiplying the triangle column n . This proves the extra small factor by an exact finite identity, not by a formal infinite series.

For an INTERNAL derivative $\partial_j, Y, K, L, N_Y, I_Y, C_Y, W$ and g_C are independent of A . Consequently

$$\begin{aligned} \|\partial_j B\| &\leq C\sigma/r_*, \\ \partial_j Z_m &= -F^{-1}(\partial_j F) Z_m, \\ \|\partial_j T\| &\leq C\sigma/r_*. \end{aligned}$$

Every derivative of a root-preserving coefficient F_e^{-1} adds $1/r_e$. A derivative of $F_e^{-1}C_e^*$ can remove its optional $O(\kappa)$ factor, but replaces it by $O(1/r_e)$. The same-root or triangle column therefore still has an extra $1/r_e \leq 1/r_*$ in its low Gaussian norm. Crucially the outer B, K^* , or $g_C^{-1/2} - I$ displayed in (13.1)-(13.3) remains, or its derivative has the SAME σ gain and another $1/r_*$.

Applying the finite Gaussian rootwise bounds to these differentiated identities gives

$$\begin{aligned} \|(\beta - \beta_1)U\| &\leq C\sigma W_2^{1/2}, \\ \|[\partial_A(\beta - \beta_1)]U\| & \\ &\leq C\sigma r_*^{-1} W_2^{1/2}. \end{aligned} \tag{13.4}$$

The derivative in the second line acts on the operator coefficient, exactly as required by the curvature term, not on U . Derivatives of U are already present in the complete-charge cross. At $A = 0$, $\partial_A d_0$ need not vanish; (13.1)-(13.3) explicitly retain its contribution with the same extra fast factor. The non-small derivative of β_1 itself is not included in the tail estimate and remains in the leading H2 scalar.

Finally the coefficients of $\beta - \beta_1$ have at least two fast-coordinate factors before p_F . A fast divergence removes at most one. Differentiating the displayed normalized coefficients therefore gives $|\operatorname{div}_F(b - b_1)| \leq C\sigma/r_*$; an additional internal derivative costs at most $1/r_*$ and leaves this σ factor. This proves the divergence portions of Section 4.3 with the required gain as well.

13.3 Form domain convention

The infimum, projection bound, and Riesz minimizer are statements about $Q(H)$, the closed quadratic-form domain. They do not establish invariance of the Hamiltonian operator domain $D(H)$. The minimizing complement is asserted only in $Q(H)$, for f in $Q(H_s)$. The global lower bound does not require attainment.

J Off cone incidence localization

Technical derivation for review. Equation and subsection numbers in this appendix are local to the source derivation. Historical local labels are retained, including numbering gaps or nonsequential labels. The accompanying source notes retain the original expressions for comparison.

1. A slightly regularized commuting tree

Write a traceless ordered commuting configuration as $\lambda = (\lambda_1, \dots, \lambda_N)$, with $\rho^2 = \sum |\lambda_i|^2$. For a proper subset B with at least two members put

$$\begin{aligned} c_B &= |B|^{-1} \sum_{i \in B} \lambda_i, \\ a_B^2 &= \sum_{i \in B} |\lambda_i - c_B|^2, \\ d_B^{-2} &= \sum_{j \text{ outside } B} (|\lambda_j - c_B|^2 + a_B^2)^{-1}. \end{aligned}$$

The extra a_B^2 in the denominator is deliberate. It removes inverse-matrix singularities from the off-cone extension whenever $a_B > 0$, without changing the clustering mechanism. For $0 < \kappa < 1/20$, $L > 1$ and $\delta = \kappa \exp(-L)$, use

$$\begin{aligned} \theta_B &= \Theta(\log(a_B/(\delta d_B))/L), \\ C_B &= \cos(\theta_B), S_B = \sin(\theta_B), \end{aligned}$$

where Θ is a fixed smooth nondecreasing function, 0 on $(-\infty, 0]$ and $\pi/2$ on $[1, \infty)$. Every derivative of the trigonometric factors is flat at the constant-tail boundaries. If $a_B = 0$ and all exterior points are separated, $C_B = 1$ locally. Coincident unused nodes are handled by the same identically zero-prefix mechanism as in the decision tree.

The regularization retains a uniform derivative bound

$$\begin{aligned} |da_B| &\leq 1, |dd_B| \leq C_N, \\ |d\theta_B|^2 &\leq C_N \|\Theta'\|_\infty^2 L^{-2} a_B^{-2} \end{aligned}$$

on $\delta d_B < a_B < \kappa d_B$. The constant is independent of δ and L . For example a conservative C_N can be obtained by taking $|dd_B| \leq 4N$ and then squaring $1 + 4N\kappa$.

For crossing subsets B, C , the same three-point argument gives

$$d_B \leq 2a_B + 2a_C, d_C \leq 2a_B + 2a_C.$$

Thus $C_B C_C = 0$ if $\kappa < 1/4$. All simultaneously selected clusters are laminar. The decreasing-cardinality decision tree and its exact pathwise IMS identity therefore still apply.

Symmetric raw leaf formula

For an unordered partition π into at least two final blocks, its tree leaf multiplier is exactly

$$q_\pi = \prod_{B \in \pi, |B| \geq 2} C_B \prod_{B \text{ proper union of } \geq 2\pi\text{-blocks}} S_B. \quad (1)$$

This is not a claim that all visited nodes are unions on their entire domain. Rather, any omitted rejected crossing node has $S_B = 1$ on a neighborhood of the nonzero final-block selection support, by laminarity. Nodes strictly inside a selected final block are skipped. Thus (1) equals the original leaf product as a smooth function, including its derivatives. In particular

$$\sum_{\pi} q_\pi^2 = 1, \quad \sum_{\pi} |dq_\pi|^2 = \sum_{\pi} q_\pi^2 E_\pi, \quad (2)$$

where E_π sums $|d\theta_B|^2$ for the final nontrivial blocks and the proper unions in (1). Terms that were skipped or crossing contribute zero on that leaf. No arbitrary ordering of equal-cardinality clusters remains in (1).

2. Intrinsic matrix extension on one stationary flag

Use the notation and associated-bundle coordinates of the intrinsic multiblock geometry note. On an ordered flag of ranks $n_1, \dots, n_k$, write

$$X = Z + A + Y, B_Z^* Y = 0, \alpha_a^2 = \sum_i \|A_i^{(a)}\|_F^2, \\ r_{ab} = |z_a - z_b|, r_* = \min r_{ab}.$$

For a union I of flag blocks define

$$n_I = \sum_{a \in I} n_a, \\ c_I = n_I^{-1} \sum_{a \in I} n_a z_a, \\ a_I^2 = \sum_{a \in I} \alpha_a^2 + \sum_{a \in I} n_a |z_a - c_I|^2, \\ H_{b,I} = \sum_i (A_i^{(b)} + (z_b - c_I)_i I)^2 + a_I^2 I, \\ d_I^{-2} = \sum_{b \text{ outside } I} \text{Tr } H_{b,I}^{-1}. \quad (3)$$

For a single nontrivial final block, $a_I = \alpha_a$. Formula (3) ignores Y , as it must for the desired leading slow localization. It uses traces, positive matrix inverses, and projector compressions; it chooses no internal eigenbasis. If $a_I > 0$ every H is positive. If $a_I = 0$ for a selected final block, exterior separation keeps every H positive and its angle has a constant select tail. A union of at least two blocks has $a_I > 0$ when $r_* > 0$.

The inverse-trace differentiation estimate remains uniform. If $H = \sum_i T_i^2 + a^2 \mathbf{I}$, then Cauchy-Schwarz and $\sum_i \|T_i H^{-2}\|_F^2 \leq \text{Tr } H^{-3}$ bound the derivative of $\text{Tr } H^{-1}$. The inequality $\sum \text{eigenvalues}(H)^{-3} \leq (\sum \text{eigenvalues}(H)^{-1})^3$ gives a dimension-controlled Lipschitz bound for d . Center subtraction only introduces the fixed factors from n_I and N . Hence the θ derivative estimate of Section 1 holds for the canonical (Z, A) metric.

For any gauge-invariant scalar $f(Z, A)$, the exact co-metric identity gives

$$|\nabla_X f|^2 \leq (1 + C_N \sigma^2) |d_{Z, A} f|^2 \quad (4)$$

on a thin fast tube. Indeed put $p = (d_Z f, d_A f, 0)$. Solving the full coordinate differential and scalar gauge constraint gives the ambient covector $(p - K^* \beta, \beta)$, where $\beta = -M^{-*} O_H^* p$ and $O_H^* p = O_C^* d_Z f + O_A^* d_A f$. Both O_C and O_A are projections of B_Y . Factoring M^{-*} with F^{-*} on the right gives $|\beta| \leq C|Y||p|/r_*$. Since $\|K\| \leq 2|Y|/r_*$, the full squared covector norm is at most $(1 + C_N \sigma^2) |p|^2$. This argument includes the full orbital contribution and has no unrelated internal-radius loss.

3. Support margins and an explicit center-gap bound

If q_π is nonzero, all nontrivial final blocks satisfy $\alpha_a < \kappa d_a$. From

$$d_a^2 \leq r_{ab}^2 + \alpha_b^2 + \alpha_a^2$$

and the analogous inequality with a,b interchanged, obtain

$$\begin{aligned} \alpha_a + \alpha_b &\leq \beta r_{ab}, \\ \beta &= 2\kappa / \sqrt{1 - 2\kappa^2}. \end{aligned} \quad (5)$$

Singleton radii are zero. Choose the tree κ so β is strictly smaller, for example by a factor four, than the endpoint constant allowed by the geometric lemma.

Every proper union I of at least two final blocks has been rejected, so $a_I \geq \delta d_I$ on q_π 's support. Suppose the currently included centers have diameter D . Equation (5) implies

$$a_I \leq \sqrt{N(1 + \beta^2)} D.$$

From $d_I \leq a_I/\delta$, at least one eigenvalue of some exterior $H_{b,I}$ is $\leq N a_I^2 / \delta^2$. Evaluating its unit eigenvector and using the triangle inequality yields

$$|z_b - c_I| \leq \sqrt{N} a_I / \delta + \alpha_b.$$

Using $\alpha_b \leq \beta|z_b - z_a|$ for an included a and adjoining this center grows D by at most M_N/δ , where

$$M_N = [N\sqrt{1 + \beta^2} + 1]/(1 - \beta).$$

Starting from a closest pair and making at most $k-2$ additions proves

$$\begin{aligned} r_* &\geq \Delta_N R, \\ R^2 &= r_\pi^2 + \sum_a \alpha_a^2, \\ \Delta_N &= [N(1 + \beta^2)]^{-1/2} (\delta/M_N)^{N-2}. \end{aligned} \tag{6}$$

Since $\rho_X^2 = R^2 + |Y|^2$ and $|Y| \leq \sigma r_*$, this also gives

$$r_* \geq \Delta_N \rho_X / \sqrt{1 + 2\sigma^2}. \tag{7}$$

All nonzero raw weights, and the closures of their derivative supports, have this margin. This is the key quantitative substitute for an unspecified finite cover.

For a selected-block angle transition,

$$\begin{aligned} d_a &\geq (1 - \beta)r_*/\sqrt{N}, \\ \alpha_a &\geq \delta(1 - \beta)r_*/\sqrt{N}. \end{aligned} \tag{8}$$

For a proper-union angle, a_I dominates its center-relative radius. Thus all nonconstant cutoff factors live at a fixed positive multiple of ρ , once N, κ, L have been fixed. Internal coincidences never cause an unbounded derivative of a surviving multiplier.

4. All branches, finite count, and smooth zero extension

For every ordered composition $N = n_1 + \dots + n_k, k \geq 2$, take the incidence set of stationary ordered orthogonal projector flags for Φ_X . Admit branches inside a padded domain with:

- endpoint bound strictly larger than β , but still inside the stated geometric regime
- center gap strictly smaller than the lower bound (7), for example one fourth of that bound
- fast width strictly larger than the support of an invariant smooth fast cutoff

Use a smooth cutoff $\chi_Y = 1$ for $|Y| \leq \sigma h/4$ and $\chi_Y = 0$ for $|Y| \geq \sigma h/2$, with $h = (\sum_{a < b} r_{ab}^{-2})^{-1/2}$. The geometric domain can allow $|Y| < \sigma r_*$. Put

$$b_j = (k!)^{-1/2} \chi_Y q_\pi \tag{9}$$

on each stationary flag sheet j . Extend it by zero outside the padded support. Equations (5)-(7), the flat cutoff tails, and the strict fast margin make this a smooth finite-cover section. There is no cutoff associated with a choice of coordinate chart or block frame.

The Hessian estimate from the geometry note makes each admitted stationary flag a nondegenerate critical point. In particular it is locally a smooth branch, and no branch can be born or disappear inside the nonzero support. Branches can have monodromy; chart transitions merely permute direct-sum components.

Explicit N-only multiplicity bound

A crude bound is sufficient. A flag of type (n_a) has real dimension $m = N^2 - \sum n_a^2 \leq N^2$. It is covered by at most $N!$ pivot charts. In a complex block-lower-unipotent chart, every cumulative orthogonal projector is

$$V(V^*V)^{-1}V^*.$$

A common denominator for all projectors has degree at most $2N^2$ in real chart coordinates. Φ has numerator and denominator degree at most $4N^2$; clearing denominators from each first derivative gives polynomial equations of degree at most $8N^2$. Nondegenerate real critical points are isolated complex zeros as well. The elementary Bezout isolated-zero bound therefore gives at most $(8N^2)^m$ per chart. Summing the at most 2^{N-1} ordered compositions gives the conservative bound

$$D_N = 2^{N-1}N!(8N^2)^{N^2}. \quad (10)$$

The bound is deliberately huge and only used to ensure constants do not secretly depend on the logarithmic inner scale. It counts ordered sheets; (9) corrects their actual multiplicity.

Why commuting admitted sheets are exactly cluster partitions

Let $B(X) = \sum_{i < j} \| [X_i, X_j] \|_F^2$. On a stationary thin flag, the transverse linear commutator has norm

$$\|L_Z Y\|^2 = \sum_{a < b} r_{ab}^2 |Y_{ab}|^2$$

with the corresponding fixed real-block normalization. Endpoint perturbations are bounded by C times the padded geometric endpoint constant times this norm (and by $C\beta$ on the raw-weight support), and the quadratic Y term by $C\sigma$ times it. With the fixed small geometric constants,

$$B(X)^{1/2} \geq c_N (\sum r_{ab}^2 |Y_{ab}|^2)^{1/2}. \quad (11)$$

Consequently a commuting admitted sheet has $Y = 0$. Its projectors group common eigenspaces; a repeated joint eigenvalue cannot be split between two selected thin blocks, by (5). Each unordered particle partition appears exactly $k!$ times among all ordered compositions and flags. Thus on the commuting cone

$$Q = \sum_j b_j^2 = 1. \quad (12)$$

No extra $N!$ multiplier belongs in (9).

Equation (11) also ensures that on a sufficiently small near-commuting region the fast cutoff is identically one on every contributing sheet. Using (7) and $h \geq r_*/\sqrt{k(k-1)}/2$, it suffices to take

$$t = B(X)/\rho_X^4 \leq c_N \sigma^2 \Delta_N^4. \quad (13)$$

The constant in (13) can be reduced to include every fixed geometric margin. There is then no fast-cutoff derivative in the near-cone localization; outside it, its usual cost could instead be paid by a retained fast reserve.

5. The exact cone identity and the small off-cone defect

Where $Q > 0$ define $w_j = b_j/\sqrt{Q}$. Then $\sum_j w_j^2 = 1$. Let

$$\begin{aligned} E(X) &= \sum_j |\nabla_X w_j|^2, \\ R(X) &= \sum_j w_j^2 E_j(X), \end{aligned} \quad (14)$$

where E_j is the sum of the ambient $|\nabla \theta_I|^2$ for the single-block select factors and proper-union reject factors of that sheet. All expressions are taken before any slow/fast projection.

At a commuting X on an admitted sheet, $Y = 0$. A pointwise common diagonalization verifies that the first derivative of every scalar in (3) in an off-diagonal internal direction is zero. This is a calculation of an invariant Frechet derivative, not a choice of smoothly varying eigenvectors. It remains true at repeated joint eigenvalues: trace derivatives of inverse matrices against off-diagonal perturbations vanish, and the radius-zero select factor is constant on a neighborhood. Further, $K = O_C = O_A = 0$ at $Y = 0$ in the exact kinetic identity. Therefore the ambient gradients of all raw weights equal their commuting-base gradients, including at singular strata. The same statement holds for the angles wherever their derivatives matter.

The whole symmetric commuting identity (2) therefore implies

$$Q = 1, dQ = 0, E = R \quad (15)$$

at every nonzero commuting tuple. This is stronger than just equality of the weight values.

On the unit sphere, all supported branch sums and their first two derivatives are bounded: the incidence Hessian is uniformly invertible after (7), the flag multiplicity is bounded by (10), all nonconstant cutoff transition scales a_I obey fixed positive lower bounds, and the support stays strictly inside the padded branch domain. Accordingly Q and E-R are smooth scalar functions in a neighborhood of the compact commuting unit locus. There is a finite explicitly defined derivative bound $K_{N,\kappa,L,\sigma,\Theta}$, obtained as the supremum of their first derivatives on a closed smaller padded neighborhood, such that

$$|Q(X) - 1| + |E(X) - R(X)| \leq K \operatorname{dist}(X, K_{\text{comm}}) \quad (16)$$

for unit X close enough to that locus. This constant can grow with Δ_N^{-1} ; it is not asserted uniform in the shape gap. Its role is to determine a later near-cone thickness, never to appear as an unpaid inverse-square coefficient.

For every $\epsilon > 0$ choose a neighborhood thickness $\eta > 0$ small enough that $Q \geq 1/2$ and $K\eta \leq \epsilon$. Homogeneity then proves

$$E(X) \leq R(X) + \epsilon \rho_X^{-2} \quad (17)$$

on that conical neighborhood. If desired, the elementary estimate of Section 6 makes the choice in terms of t : take $t_1 \leq (\eta/(2C_N))^4$, together with (13) and the branch-continuation radius. This is a concrete order of choices, not a simultaneous assumption $\kappa = \kappa(\Delta_N)$.

Why direct leafwise normalization was insufficient

For a single select factor, $|d \cos \theta|^2 = \sin^2 \theta |d\theta|^2$ cannot be bounded by $\cos^2 \theta$ times the same cost near its zero. Hence one cannot bound $|dw_j|^2$ by $w_j^2 E_j$ separately. Nor may one charge every nested transition to every final leaf: a tiny subcluster transition inside a larger selected block is not controlled with a shape-independent coefficient by the latter's total-radius Hardy estimate. Only the summed defect argument (15)-(17) repairs this issue.

6. An elementary almost-commuting distance certificate

There is no need to assume a smooth retraction onto the singular commuting cone. For a traceless Hermitian nine-tuple X of norm one, put $\epsilon_0 = \max_{i < j} \|[X_i, X_j]\|_F \leq \sqrt{t}$. An elementary spectral-block construction produces a commuting traceless tuple Z with

$$\begin{aligned} \|X - Z\| &\leq C_N \sqrt{\epsilon_0} \leq C_N t^{1/4}, \\ C_N &= 729 \sqrt{N^3 + 9} \end{aligned} \quad (18)$$

as a conservative choice.

To see this, process one component at a time inside the already constructed invariant blocks. In each current block, group consecutive eigenvalues across gaps at most $\eta = \sqrt{\epsilon_j}$.

Distinct groups have spectral distance greater than η . Replace the processed component on each new spectral block by its trace-average; its tuple error is at most $N^{3/2}\eta$. Delete interblock entries of all remaining components. The Frobenius norm of each deleted part is at most ϵ_j/η , by its commutator with the processed component. The new block-diagonal commutator differs from the compressed old commutator only by two products of deleted off-block parts. Hence $\epsilon_{j+1} \leq \epsilon_j + 2(\epsilon_j/\eta)^2 \leq 3\epsilon_j$. At most nine steps give (18); all changes preserve traces. Previously processed components are scalar on their blocks and stay commuting with subsequent steps.

This proof uses spectral subspaces only to establish a distance inequality. The actual localization never uses this approximate tuple, its ordering, or any internal eigenbasis. If the right side of (18) is at most $1/2$, normalize Z to norm one; its distance from X is at most $2C_N t^{1/4}$. Thus (16) can be converted into an explicit $t^{1/4}$ modulus with the finite derivative constant K .

7. Payment and physical measure

The first term R in (17) has the original leafwise payments:

1. A selected-block transition costs at most $C_N L^{-2} \alpha_a^{-2}$, with $\alpha_a \geq c_N \delta \Delta_N r_\pi$ by (8). On the physical low coefficient the lower-rank soft Hardy estimate pays this from an arbitrarily small internal-form fraction, with faster-than-inverse-square exterior defect.
2. A proper-union transition has $a_I^2 = \sum \alpha_a^2 + s_I^2$, where s_I is the canonical relative-center radius of that union. Therefore $a_I^{-2} \leq s_I^{-2}$, paid by ordinary Hilbert-valued relative-center Hardy in dimension $9(1-1)$. There are at most 2^N such terms per sheet. Their coefficient C_N/L^2 can be made arbitrarily small before δ and Δ_N are fixed.
3. On the colored complementary coefficient all scalar costs are bounded by $C_{N,\kappa,L} r_\pi^{-2}$, using (6)-(8), and are paid by a retained fast gap at sufficiently large radius. The physical lower-rank Hardy theorem is not applied to colored vectors.
4. The additional $\epsilon \rho_X^{-2}$ in (17) is at most ϵr_π^{-2} . On the low coefficient it uses an arbitrarily small fraction of the center Hardy reserve; on the complement it is absorbed by the fast reserve at large radius. If the near-cone region is set by t , the existing outer cutoff has its separately paid commutator-potential IMS cost.

The only form inputs in this payment are the previously identified internal soft-Hardy estimates and the separately supplied all-vector multiblock comparison with sufficient retained reserves. This construction does not manufacture those reserves.

On each flag sheet the physical density is the exact $J = \det M$ from the intrinsic geometry note. Same-sheet frame changes are unitary residual-block/normal-frame transitions and introduce no scalar chart partition. Distinct sheets remain distinct summands. Pull back the physical section to the incidence cover and multiply by w_j . Then

$$\begin{aligned} \|u\|^2 &= \sum_j \|w_j u\|^2, \\ h[u] &= \sum_j h[w_j u] - \int E(X) |u|^2 \end{aligned} \tag{19}$$

with sheet integration against the pulled-back physical measure, equivalently $J \, dZ \, dA \, dY$ after gauge integration. The $k!$ factor in (9) and normalization, rather than discarding

branches, enforce single counting. Local form estimates can therefore be integrated chart-free on every sheet. No globally smooth projector, angular gauge-fixing multiplier, or internal bound-state projector is used.

8. Noncircular parameter order

For each fixed N and target loss ϵ :

1. Fix the necessary fractions of center/internal/fast reserves in the conditional local form theorem.
2. Choose the geometric endpoint and fast constants; choose tree κ so β in (5) lies strictly inside the endpoint bound. Any analytic requirement on these constants must itself be shape-independent, as required by the local comparison.
3. Choose L so the sum of at most 2^N logarithmic costs is paid by the allocated slow reserves.
4. Set $\delta = \kappa e^{-L}, \Delta_N$ from (6), padded incidence domains, and the finite derivative/continuation constants K .
5. Choose the near-commuting threshold t_1 using (13), (16)-(18), so the added IMS defect is at most $\epsilon \rho^{-2}$.
6. Choose the outer cutoff parameters and finally the exterior radius large enough to pay all fixed-shape constants, complementary inverse-square costs, and soft-Hardy remainders.

No rank-uniform statement or effective small radius is asserted. The point is that every large delta-dependent constant is either multiplied by a later freely chosen cone thickness or absorbed at a later exterior radius.

K Corrected rotation algebra and support estimates

This appendix records the algebra, normalization and domain extension used in Section 9. The rotation anticommutator and kernel-invariance theorem are due to the cited prior work; the quantitative use of a bounded internal radius is the estimate needed here.

Two charge conventions and two summation scopes

Retain the kinetic-one Hamiltonian h . Let $\tilde{q}$ denote the convention

$$\{\tilde{q}_\alpha, \tilde{q}_\beta\} = \delta_{\alpha\beta} h, \quad \sum_{\alpha=1}^{16} \|\tilde{q}_\alpha g\|^2 = 8\mathfrak{h}[g] \quad (\text{K.1})$$

on the physical core. The manuscript's baseline charges are $q = \sqrt{2}\tilde{q}$ and have stack coefficient 16. For a fixed $j = (u, v)$ the primitive of [3, Lemma 2(c)] is

$$\tilde{a}_{j,\alpha} = \frac{1}{16}(X_u \Gamma_v - X_v \Gamma_u)_{\alpha\epsilon} \theta_\epsilon + \frac{1}{56} \sum_{w \notin \{u,v\}} X_w (\Gamma_w \Gamma_{uv})_{\alpha\epsilon} \theta_\epsilon. \quad (\text{K.2})$$

The color index is contracted in each summand. This follows from $(5b + 2c)/7$ because $\Gamma_u \Gamma_{uv} = \Gamma_v$ and $\Gamma_v \Gamma_{uv} = -\Gamma_u$.

All coefficient matrices are real. The first two are symmetric grade-one matrices, the others antisymmetric grade-three matrices. They are mutually orthogonal in the Frobenius inner product and each has squared Frobenius norm 16. Each $\tilde{a}_{j,\alpha}$ is a real linear combination of Majoranas, so its square is scalar by the CAR. Consequently

$$\sum_{\alpha} \tilde{a}_{j,\alpha}^2 = \frac{1}{2} \left[\frac{16}{16^2} (|X_u|^2 + |X_v|^2) + \frac{16}{56^2} \sum_{w \notin \{u,v\}} |X_w|^2 \right] I \quad (\text{K.3})$$

$$= \left[\frac{|X_u|^2 + |X_v|^2}{32} + \frac{\sum_{w \notin \{u,v\}} |X_w|^2}{392} \right] I. \quad (\text{K.4})$$

For several blocks, concatenate color coordinates; no color-dimension factor occurs. A fixed coordinate belongs to eight of the 36 pairs and is outside the other 28, giving

$$\sum_{u < v} \sum_{\alpha} \tilde{a}_{(u,v),\alpha}^2 = \left(\frac{8}{32} + \frac{28}{392} \right) \alpha^2 I = \frac{9}{28} \alpha^2 I. \quad (\text{K.5})$$

Thus $1/32$ is the bound for one generator, while $9/28$ is the exact coefficient after summing all generators. They are not alternative estimates for the same sum.

Rescale reciprocally:

$$q_\alpha = \sqrt{2}\tilde{q}_\alpha, \quad a_{j,\alpha} = \tilde{a}_{j,\alpha}/\sqrt{2}, \quad J_j = \sum_{\alpha} \{q_\alpha, a_{j,\alpha}\}. \quad (\text{K.6})$$

The fixed-generator bound becomes $\alpha^2/64$, and (K.5) becomes $9\alpha^2/56$. The products $8(1/32)$ and $16(1/64)$ agree. Rescaling the charge alone would change the generator identity and is not allowed.

Direct Clifford verification of the identity

Write $\tilde{a}_\alpha = X_w(M_w)_{\alpha\epsilon}\theta_\epsilon$ for a fixed pair (u, v) . In the real gamma convention the coefficient matrices obey

$$\text{Tr}(\Gamma_t^T M_w) = \delta_{wu}\delta_{tv} - \delta_{wv}\delta_{tu}, \quad (\text{K.7})$$

$$\sum_w \Gamma_w^T M_w = \frac{1}{4}\Gamma_{uv}, \quad (\text{K.8})$$

$$\text{Tr}(M_w^T \Gamma_{st}) = 0. \quad (\text{K.9})$$

The first identity yields the orbital rotation, the second the spin contribution $-i\theta\Gamma_{uv}\theta/4$, and the last removes the interaction anticommutator. These are exact finite Clifford identities. The included integer-matrix program checks them for all 36 pairs and all indicated indices. It uses a self-contained real symmetric Spin(9) representation constructed from the octonion multiplication table. This test verifies the finite identities only; integration by parts and closed-domain statements require the arguments in the text.

A stronger Casimir variant

The weaker coefficient $1/4$ in Lemma 9.2 is sufficient for the manuscript. For comparison, the all-generator identity proves

$$H_I[g] \geq \frac{7}{9R^2} \|(1 - P_I)g\|^2 \quad \text{when } \text{supp } g \subset \{\alpha \leq R\}. \quad (\text{K.10})$$

Use the graded sum of block supercharges, or equivalently a blockwise charge index, with stack coefficient 8. Its squares add to the complete H_I . Let $C_I = \sum_j (J_j^I)^2$ and initially take a physical smooth vector in a Casimir eigenspace $C_I g = \lambda g$, $\lambda > 0$. Put $E = H_I[g]$ and $K = 9/28$. Rotational invariance of the complete form gives

$$\sum_j \|J_j^I g\|^2 = \lambda \|g\|^2, \quad \sum_j H_I[J_j^I g] = \lambda E. \quad (\text{K.11})$$

Individual supercharges need not commute with rotations; only their averaged square is used. Pair the rotation identity with $J_j^I g$, sum over j , and integrate by parts on the core. The first resulting term is bounded by $\sqrt{8\lambda E} \sqrt{KR^2 \|g\|^2}$. For the other, pointwise Cauchy-Schwarz and the fact that each primitive square is scalar give

$$\sum_\alpha \left\| \sum_j \tilde{a}_{j,\alpha}(X) (J_j^I g)(X) \right\|^2 \leq K\alpha(X)^2 \sum_j \|(J_j^I g)(X)\|^2. \quad (\text{K.12})$$

That term has the same bound. Therefore

$$\lambda \|g\|^2 \leq 2\sqrt{8K} R \sqrt{\lambda E} \|g\|, \quad E \geq \frac{7\lambda}{72R^2} \|g\|^2. \quad (\text{K.13})$$

In the generator convention used here, a dominant B_4 weight has Casimir

$$\lambda_C = \sum_{i=1}^4 \lambda_i(\lambda_i + 9 - 2i), \quad \lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4 \geq 0, \quad (\text{K.14})$$

with all coordinates integral or all half-integral. The smallest positive integer case $(1, 0, 0, 0)$ has Casimir 8, and the smallest half-integral case $(1/2, 1/2, 1/2, 1/2)$ has Casimir 9. Summing positive Casimir sectors proves (K.10).

The rotation-invariant ball preserves all compact-group projections. At finite Casimir cutoff the generators are bounded angular operators, and the projected smooth physical core remains radially compact. Invariant nonnegative forms split orthogonally over the Casimir decomposition. Finite cutoff, approximation in the form norm, and a limit therefore extend the result to the closed form domain. For support in a closed ball use a slightly larger open radius and decrease it to R . Spectator Hilbert spaces pass through the same argument by finite sums and closure. Neither $\tilde{a}g$ nor Jg is asserted to lie in the Hamiltonian operator domain for a general form vector.

Kernel invariance without an assumed positive moment

For completeness, the domain mechanism behind [3, Theorem 1(a)] is as follows. The physical kernel is invariant under the compact rotation group. Decompose it into isotypes so that J_j is bounded on each finite-dimensional representation factor. Choose a smooth radial cutoff χ_R that is one in a ball and whose derivative is supported where $R < \rho < 3R$. In the generator identity, test with two kernel vectors of finite isotypic type and move the charge through χ_R . The surviving shell terms have bounds of the form

$$C \|\phi\|_{\{R < \rho < 3R\}} \|\Psi\|_{\{R < \rho < 3R\}}, \quad (\text{K.15})$$

since $[\tilde{q}, \chi_R] = O(R^{-1})$ and $\tilde{a} = O(R)$ on that shell. The compatible maximal/self-adjoint realization and physical-core approximation are the ones in the cited theorem. The bound tends to zero by L^2 integrability alone. Taking $\phi = J_j \Psi$ within each isotype gives $J_j \Psi = 0$. Completing the compact-group decomposition proves invariance of the full kernel. This is an explanation of the cited prior theorem, not a new existence or domain theorem.

L Equivariant vacuum and reference form domains

This appendix expands the geometric identifications used in Section 10. It addresses the added rotation argument on the local structures supplied by Appendices A–J; it does not replace their collision-uniform estimates.

Spatial rotations preserve actual stationary flags

For an ordered flag $P = (P_1, \dots, P_k)$, the functional

$$\Phi_X(P) = \sum_{a,i} n_a^{-1} (\text{Tr}(P_a X_i))^2 \quad (\text{L.1})$$

is spatially invariant at the same flag. Thus Φ_{RX} and Φ_X are identical functions on the flag manifold, including all their flag derivatives. The map $(X, P) \mapsto (RX, P)$ preserves stationarity and every Hessian admission test. It fixes the flag along each spatial orbit; apparent permutations in a local numbering of sheets are only chart changes.

Center gaps, internal radii, fast norms and commutator norms are spatial scalars. The inverse-trace regularized cluster scales use spatial contractions and are likewise invariant. Raw select/reject weights and their square-sum normalization therefore preserve singlets. No angular cutoff is inserted. Color frames may be chosen as functions of the flag alone, so their transition maps are gauge changes independent of the spatial rotation. Actual normal-bundle frame changes are accounted for by the fast unitaries of Appendix A. Density flattening is scalar and cannot supply an extra angular character.

Global two-block anchor including the central element

At $A = 0$, fix a reference center direction $n_0 \in S^8$. The fast ground line is a one-dimensional genuine unitary representation of its stabilizer $\text{Spin}(8)$. Every continuous character of a connected compact semisimple group is trivial. The statement includes the central element -1 , rather than only the infinitesimal Lie algebra.

The fast action is obtained from the physical spatial and Clifford actions, restricted to the fast factor. Its occupied determinant contains $8n_1n_2$ modes and has even parity. There is therefore no hidden projective character or extra fermionic sign on the line. Choose a unit vector v_* in the line over n_0 and define

$$v_0(gn_0) = U_{\text{fast}}(g)v_*, \quad g \in \text{Spin}(9). \quad (\text{L.2})$$

Stabilizer invariance makes this independent of the representative g , smooth on the sphere, and exactly equivariant. It is a global normalized section, not only a local choice of phase.

Its Berry one-form is group invariant. The tangent representation of $\text{Spin}(8)$ on S^8 has no invariant covector, so the angular one-form is zero. The occupied determinant is independent of the radial center coordinate. The positive normalized Gaussian has zero scalar Berry one-form, because its norm is one and its entries are real. Hence the unperturbed center connection is trivial and its kinetic form is the ordinary T_x .

This argument is restricted to the anchor. At nonzero internal coordinates the vacuum may have nonzero Berry curvature; the finite- A comparison is precisely the paid one in

Appendices E–G and I. Gauge and Weyl identifications are retained, rather than being discarded by the trivialization.

Covariant Kato transport and retained fermions

The path tA , $0 \leq t \leq 1$, remains in the local endpoint tube. Let $P_t(x, A)$ be the gapped negative fast spectral projector. Covariance gives

$$P_t(Rx, RA) = U_{\text{fast}}(R)P_t(x, A)U_{\text{fast}}(R)^*. \quad (\text{L.3})$$

Differentiating in t shows the same covariance for the anti-Hermitian Kato generator $K_t = [\dot{P}_t, P_t]$. The solution of $\dot{W}_t = K_t W_t$, $W_0 = I$, is unique, so W_t is covariant. The induced fast Fock transport is even and acts only on fast fermions. Gaussian covariance transport and the scalar normalized radial cutoff are also equivariant. All these maps commute with the retained Clifford action.

Applying these maps to v_0 gives the actual injection U_σ as an intertwiner. The retained relative module is exactly the 256-dimensional module of sixteen Majoranas, and the slow generator on the smooth core is $L_x + S_{\text{rel}} + J_I$. This does not make the spatial derivatives of U_σ vanish. Their transverse source and scalar Berry contributions stay in the complete comparison already charged in the local ledger; there is no extra Schur dressing or a second completion square.

The model form and its extension map

For the two physical internal Hilbert spaces, the reference model is (10.3). The internal forms are the complete nonnegative supersymmetric forms, with arbitrary energy and rotation type. The baseline bounded projection and padded support give an actual coefficient f in the Dirichlet form closure of the padded branch. Extend each approximating compactly supported smooth coefficient by zero; the baseline form bounds yield a Cauchy sequence in $\mathcal{Q}(T_x) \cap \mathcal{Q}(H_I)$. Its limit is the zero extension of f . This establishes the extension for this domain. It asserts no boundary-free extension for an arbitrary section or preservation of $D(h_N)$.

The extended coefficient is supported in $\mathcal{C}_\beta$ from (10.6). A detailed leaf need not survive internal-only rotations: those may alter extra direction-dependent tests. This is why every original local comparison and leaf-multiplier payment is completed before extension and averaging.

Let $V_I(g)$ rotate both internal blocks, including their fermions, while fixing x and $\mathcal{F}_{\text{rel}}$. It preserves each α_a , commutes with H_I and T_x , and preserves the radial tube. Individual $V_I(g)$ need not commute with every diagonal rotation. Their Haar average

$$P_I = \int_{\text{Spin}(9)} V_I(g) \, dg \quad (\text{L.4})$$

does commute, by conjugation invariance of Haar measure. It is a contraction for the norm and the model form norm. The corresponding invariant projection reduces each of T_x and H_I , giving

$$T_x[f] = T_x[P_I f] + T_x[(1 - P_I)f], \quad H_I[f] = H_I[P_I f] + H_I[(1 - P_I)f]. \quad (\text{L.5})$$

It commutes with center radial weights, so the s^{-2} weighted norm splits as well. Both summands of a diagonal singlet remain diagonal singlets.

Apply these identities only to the already-paid model form. The enlarged support remains in $\mathcal{C}_\beta$, so for almost every fixed center x the internal vector has support in the ball of radius $\beta_{\text{pad}}\nu|x|$. The direct-integral form description gives internal form-domain membership almost everywhere. Lemma 9.2 then applies fiberwise; Fubini and nonnegativity integrate the inequality. Approximation first on compact center annuli, then in the closed form norm, removes any smoothness restriction. No derivative of a radius-dependent internal spectral projection occurs; P_I is independent of x .

Representation types and residual constraints

The relative module has highest weights

$$\mathbf{44} : (2, 0, 0, 0), \quad \mathbf{84} : (1, 1, 1, 0), \quad \mathbf{128} : (3/2, 1/2, 1/2, 1/2). \quad (\text{L.6})$$

A scalar harmonic on S^8 of degree l has highest weight $(l, 0, 0, 0)$. Only $\mathbf{44}$ pairs with such a harmonic into a singlet, and it does so at $l = 2$. Internal invariants contribute arbitrary multiplicity spaces. The argument does not require each block to be separately invariant: nontrivial block representations can couple to a simultaneous internal singlet.

The relative $\mathbf{128}$ cannot give a singlet with a scalar harmonic and an internal trivial representation. It can contribute when paired with nontrivial spinorial internal representations; these states belong to $(1 - P_I)\mathcal{K}$ and are controlled by the nonsinglet uncertainty bound. They are not omitted. Equal-block exchange and residual physical conditions only reduce the admissible ordered-cover state space and cannot weaken the lower bound.

For three or more final blocks, radial center Hardy already gives at least 64 in kinetic-one units, so none of these angular identifications is needed for their coercive constant. Their geometry, cutoff errors, colored complements and soft internal payments remain those of the baseline proof. The only additional two-block restriction is the initial padded width $\beta_{\text{pad}} \leq 1/12$.